Carlsen vs Caruana, World Championship 2018

Generated by Stockfish 16

October 18, 2026

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Chapter 1

Caruana, Fabiano vs Carlsen, Magnus

1.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.09
Round	1
White	Caruana, Fabiano
Black	Carlsen, Magnus
Result	1/2-1/2
Opening	??? – Unknown

1.2 Player Statistics

1.2.1 Caruana, Fabiano (White)

- Total moves: 115
- Accuracy: 95.7%
- Average centipawn loss: 8.8
- Best/Excellent moves: 84 / 13
- Good moves: 10
- Inaccuracies: 4
- Mistakes: 0
- Blunders: 4

1.2.2 Carlsen, Magnus (Black)

- Total moves: 114
- Accuracy: 95.5%
- Average centipawn loss: 9.2

- Best/Excellent moves: 90 / 12
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 1
- Blunders: 4

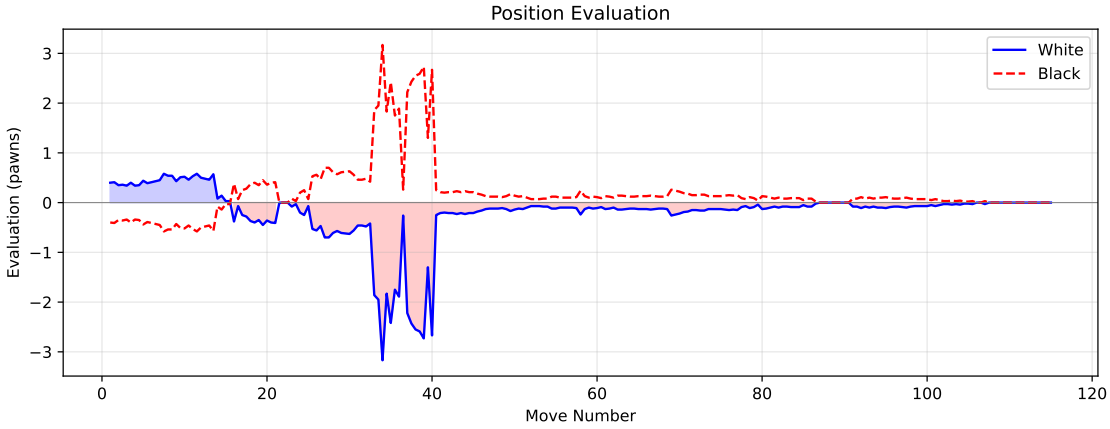
1.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

1.3.1 Position Evaluation Over Time

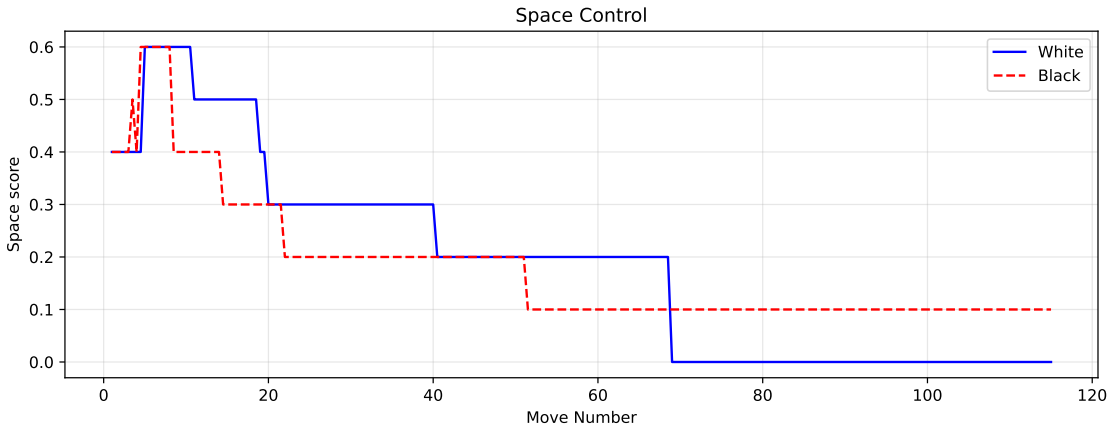
Game Character Classification. Evaluation range: minimum $m_1 = -3.17$, maximum $m_2 = +0.58$, spread $d = m_2 - m_1 = 3.75$.

This game is classified as **tactical** (normal). A tactical game with moderate swings.



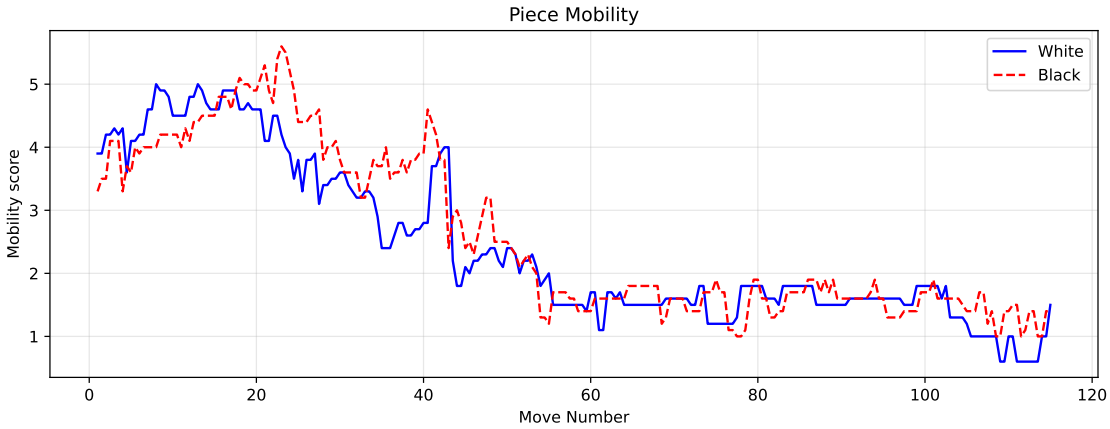
1.3.2 Space Control

- White average: 0.19 (range: 0.00 to 0.60)
- Black average: 0.18 (range: 0.10 to 0.60)



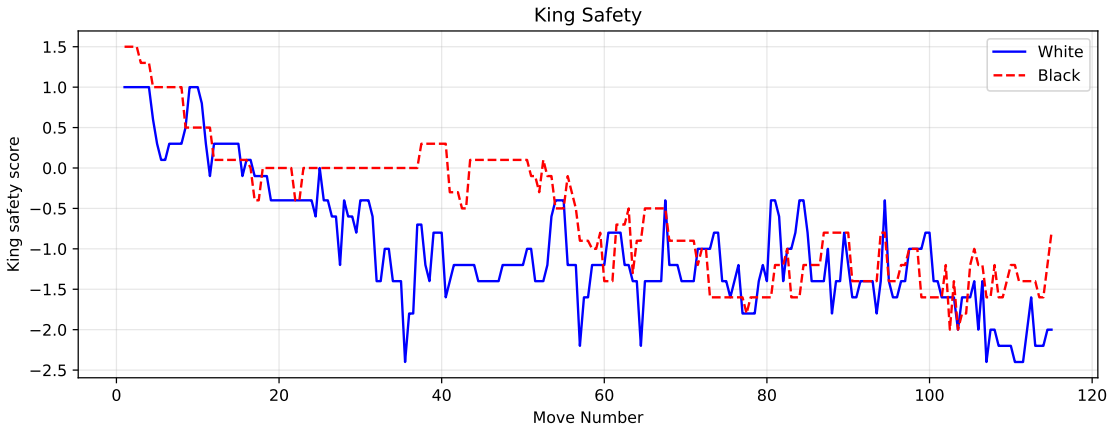
1.3.3 Mobility (Piece Activity)

- White average: 2.44 (range: 0.60 to 5.00)
- Black average: 2.61 (range: 1.00 to 5.60)



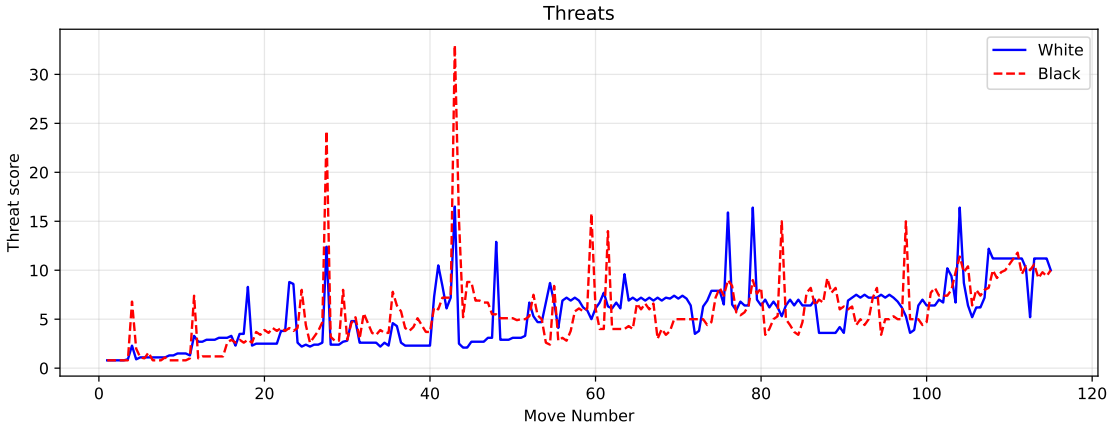
1.3.4 King Safety

- White average: -0.97
- Black average: -0.54



1.3.5 Threats

- White average: 5.45 (range: 0.80 to 16.50)
- Black average: 5.61 (range: 0.80 to 33.00)
- Average advantage: -0.16 (Black maintained more threats)



1.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. Bb5 g6 4. Bxc6 dxc6 5. d3 Bg7 6. h3 Nf6 7. Nc3 Nd7 8. Be3! e5 9. O-O b6 10. Nh2! Nf8 11. f4! exf4! 12. Rxf4 Be6 13. Rf2! h6 14. Qd2?! g5! 15. Raf1 Qd6! 16. Ng4?! O-O-O 17. Nf6 Nd7 18. Nh5 Be5! 19. g4! f6! 20. b3! Bf7! 21. Nd1 Nf8?! 22. Nxf6! Ne6! 23. Nh5 Bxh5 24. gxh5 Nf4! 25. Bxf4! gxf4! 26. Rg2 Rhg8! 27. Qe2 Rxg2+ 28. Qxg2! Qe6! 29. Nf2 Rg8! 30. Ng4! Qe8! 31. Qf3 Qxh5! 32. Kf2! Bc7! 33. Ke2?? *White loses 126cp. Consider instead: e5, Rg1.*

33...Qg5! 34. Nh2?? *White loses 125cp. Consider instead: Qf2, Re1, Rf2.*

34...h5?? *Black loses 147cp. Consider instead: Qe5.*

35. Rf2?! Qg1? *Black loses 68cp. Consider instead: Qg7, Kb7.*

36. Nf1 h4?? *Black loses 192cp. Consider instead: Qg7, Kb7, Qg5.*

37. Kd2?? *White loses 191cp. Consider instead: e5.*

37...Kb7 38. c3 Be5 39. Kc2?! Qg7?? *Black loses 140cp. Consider instead: b5.*

40. Nh2?? *White loses 158cp. Consider instead: Nd2, a4, Qh5.*

40...Bxc3?? *Black loses 254cp. Consider instead: Qg1.*

41. Qxf4! Bd4! 42. Qf7+! Ka6! 43. Qxg7 Rxg7! 44. Re2! Rg3! 45. Ng4! Rxh3! 46. e5! Rf3 47. e6! Rf8! 48. e7! Re8! 49. Nh6! h3 50. Nf5! Bf6! 51. a3! b5 52. b4! cxb4! 53. axb4! Bxe7 54. Nxe7! h2 55. Rxh2! Rxe7! 56. Rh6 Kb6 57. Kc3 Rd7! 58. Rg6 Kc7 59. Rh6! Rd6! 60. Rh8! Rg6 61. Ra8 Kb7 62. Rh8! Rg5 63. Rh7+ Kb6 64. Rh6! Rg1 65. Kc2 Rf1 66. Rg6 Rh1 67. Rf6! Rh8 68. Kc3 Ra8 69. d4 Rd8! 70. Rh6! Rd7 71. Rg6 Kc7! 72. Rg5 Rd6! 73. Rg8! Rh6! 74. Ra8! Rh3+! 75. Kc2! Ra3 76. Kb2 Ra4 77. Kc3! a6 78. Rh8! Ra3+! 79. Kb2! Rg3! 80. Kc2 Rg5 81. Rh6 Rd5 82. Kc3! Rd6 83. Rh8! Rg6! 84. Kc2 Kb7! 85. Kc3 Rg3+! 86. Kc2 Rg1 87. Rh5! Rg2+! 88. Kc3! Rg3+ 89. Kc2! Rg4 90. Kc3! Kb6 91. Rh6 Rg5! 92. Rf6! Rh5 93. Rg6 Rh3+! 94. Kc2 Rh5 95. Kc3 Rd5! 96. Rh6! Kc7 97. Rh7+ Rd7 98. Rh5 Rd6! 99. Rh8! Rg6 100. Rf8 Rg3+ 101. Kc2! Ra3 102. Rf7+! Kd6! 103. Ra7! Kd5! 104. Kb2! Rd3! 105. Rxa6! Rxd4! 106. Kb3 Re4! 107. Kc3 Rc4+ 108. Kb3! Kd4 109. Rb6! Kd3! 110. Ra6! Rc2 111. Rb6! Rc3+! 112. Kb2! Rc4 113. Kb3! Kd4 114. Ra6! Kd5 115. Ra8

1/2-1/2

1.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

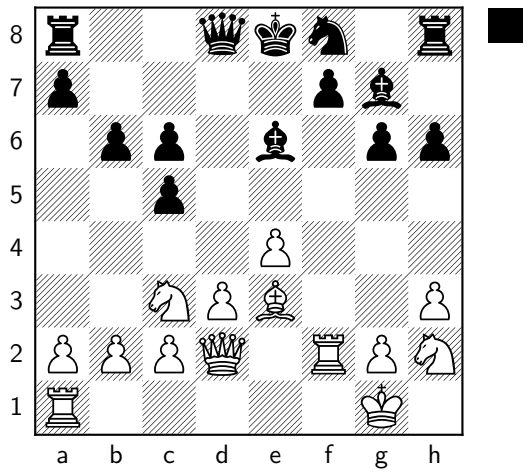
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 14. Qd2 ★

Biggest evaluation swing(s) (49cp)

Evaluation: +0.08



Metric	White	Black
Space	0.50	0.40
Mobility	4.70	4.50
King Safety	0.30	0.10
Threats	2.90	1.20

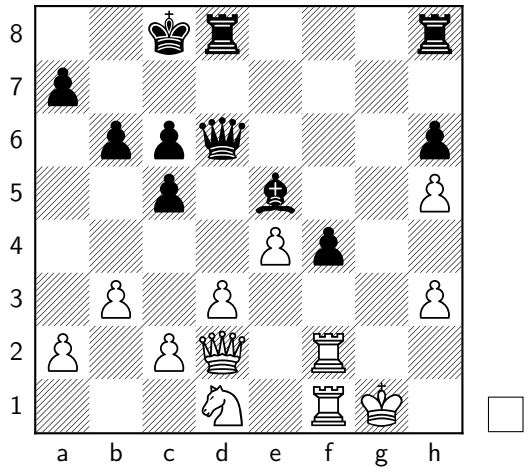
Instead of Qd2: a4, Nf3, Qe2

Best continuation: g5 Nf3 Ng6

Position 2: After 25...gxf4 ★

Biggest evaluation swing(s) (46cp)

Evaluation: -0.53



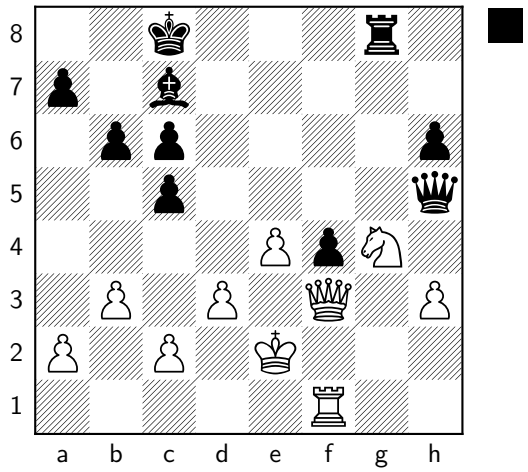
Metric	White	Black
Space	0.30	0.20
Mobility	3.30	4.40
King Safety	-0.40	0.00
Threats	2.20	2.60

Best continuation: Rxf4 Bxf4 Qxf4

Position 3: After 33. Ke2

White blunders (-144cp)

Evaluation: -1.86



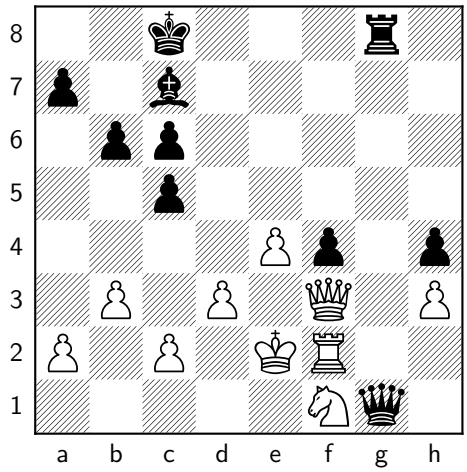
Metric	White	Black
Space	0.30	0.20
Mobility	3.30	3.20
King Safety	-1.00	0.00
Threats	2.60	3.60

Instead of Ke2: e5, Rg1 Best continuation: Qg5 e5 h5

Position 4: After 36...h4

Black blunders (+163cp)

Evaluation: -0.26



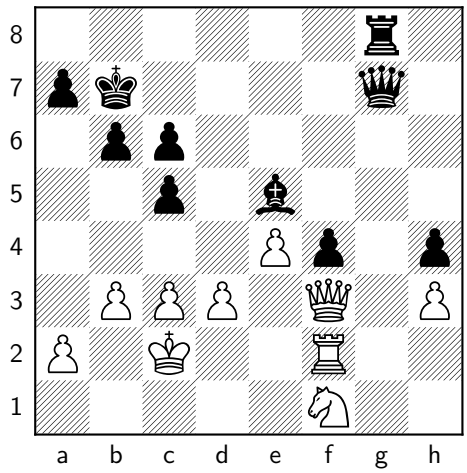
Metric	White	Black
Space	0.30	0.20
Mobility	2.60	3.60
King Safety	-1.80	0.00
Threats	2.60	5.80

Instead of h4: Qg7, Kb7, Qg5 Best continuation: e5 Rg5 Kd2

Position 5: After 39...Qg7

Black blunders (+143cp)

Evaluation: -1.30



Metric	White	Black
Space	0.30	0.20
Mobility	2.70	3.90
King Safety	-0.80	0.30
Threats	2.30	3.70

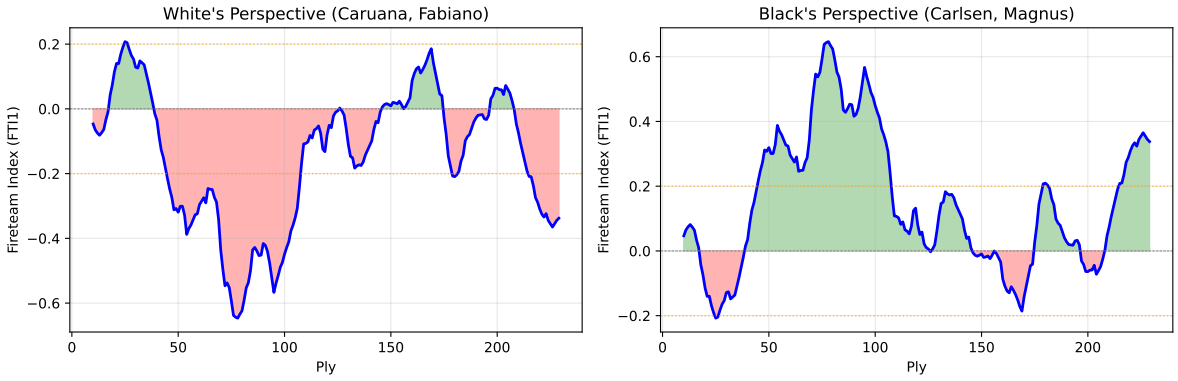
Instead of Qg7: b5 Best continuation: a4 Qg1 Rd2

1.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

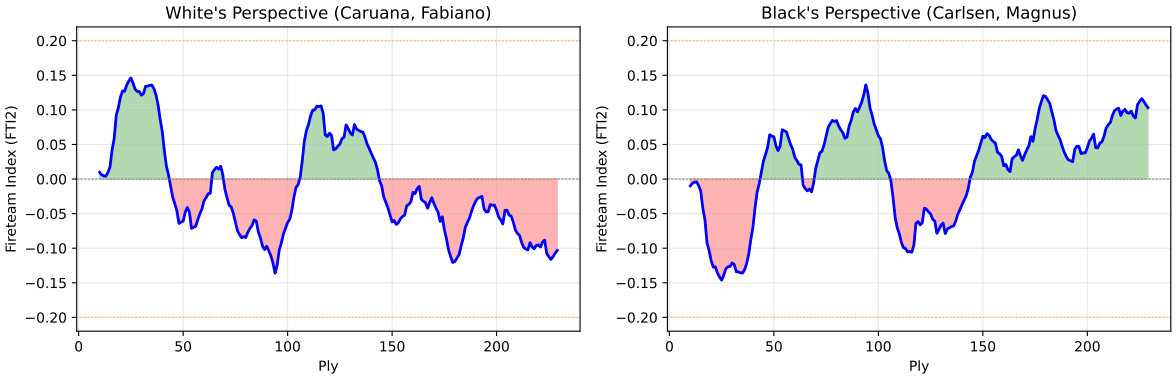
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	2 plies
Carlsen, Magnus	Per-ply	WIN	22 plies
Carlsen, Magnus	Windowed	WIN	63 plies



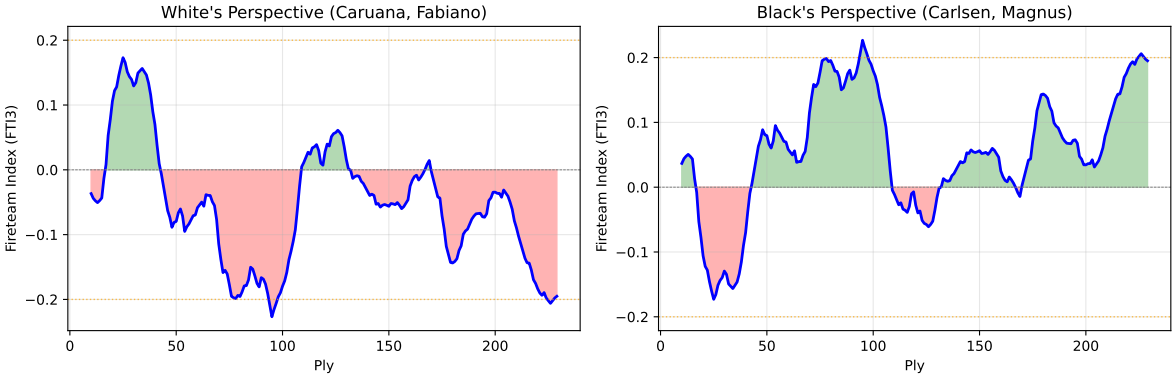
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



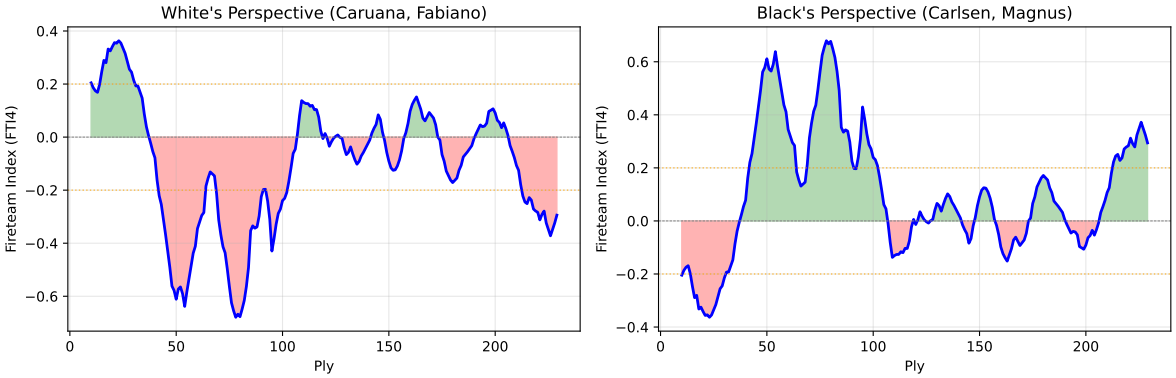
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	0 plies
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	DRAW	4 plies



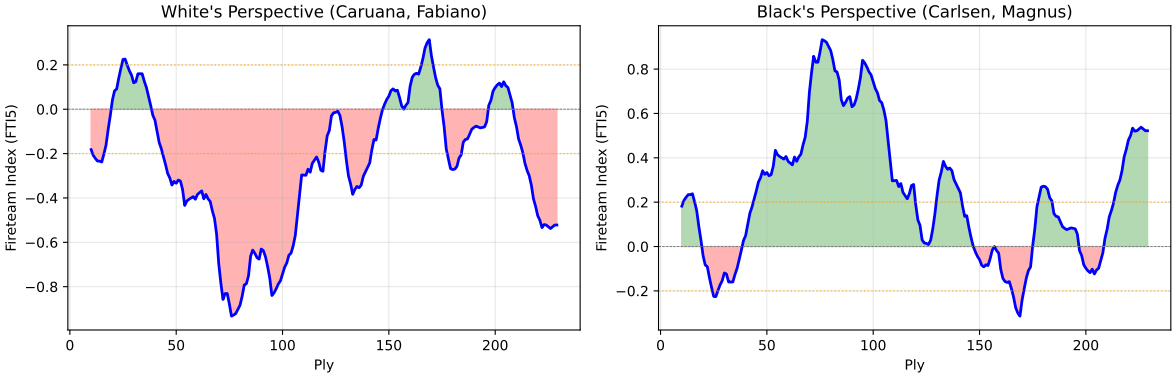
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	5 plies
Caruana, Fabiano	Windowed	WIN	14 plies
Carlsen, Magnus	Per-ply	WIN	16 plies
Carlsen, Magnus	Windowed	WIN	22 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	5 plies
Carlsen, Magnus	Per-ply	WIN	24 plies
Carlsen, Magnus	Windowed	WIN	76 plies



Interpretation. Black (Carlsen, Magnus) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting White successfully defended or Black failed to convert the advantage.

1.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 272.7 seconds

Chapter 2

Carlsen, Magnus vs Caruana, Fabiano

2.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.10
Round	2
White	Carlsen, Magnus
Black	Caruana, Fabiano
Result	1/2-1/2
Opening	??? – Unknown

2.2 Player Statistics

2.2.1 Carlsen, Magnus (White)

- Total moves: 49
- Accuracy: 98.2%
- Average centipawn loss: 3.6
- Best/Excellent moves: 31 / 10
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

2.2.2 Caruana, Fabiano (Black)

- Total moves: 48
- Accuracy: 98.9%
- Average centipawn loss: 2.3

- Best/Excellent moves: 36 / 5
- Good moves: 7
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

2.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

2.3.1 Position Evaluation Over Time

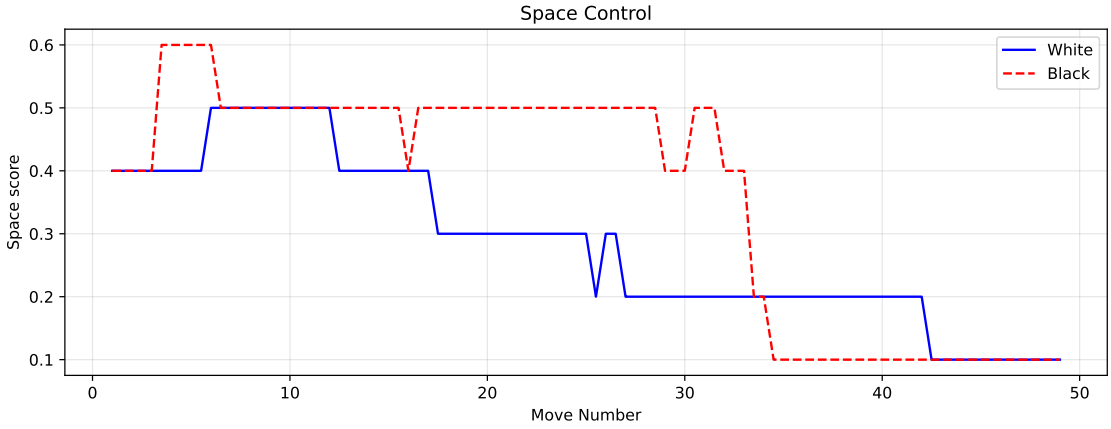
Game Character Classification. Evaluation range: minimum $m_1 = -0.27$, maximum $m_2 = +0.43$, spread $d = m_2 - m_1 = 0.70$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



2.3.2 Space Control

- White average: 0.29 (range: 0.10 to 0.50)
- Black average: 0.36 (range: 0.10 to 0.60)



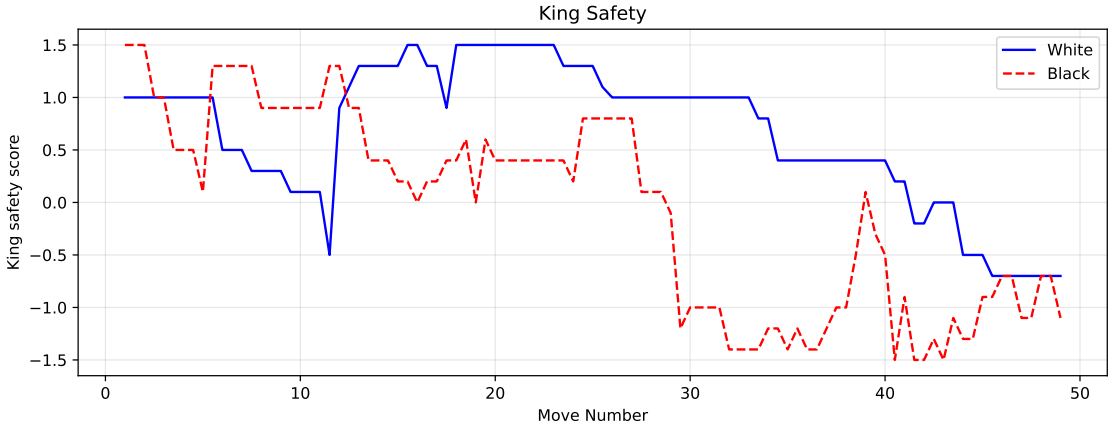
2.3.3 Mobility (Piece Activity)

- White average: 3.25 (range: 1.20 to 5.90)
- Black average: 2.84 (range: 1.30 to 5.10)



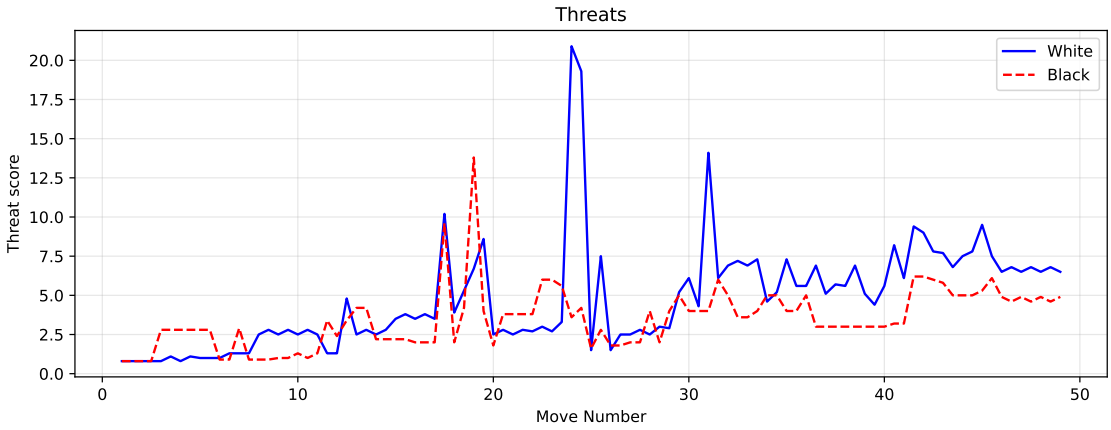
2.3.4 King Safety

- White average: 0.67
- Black average: -0.05



2.3.5 Threats

- White average: 4.78 (range: 0.80 to 20.90)
- Black average: 3.51 (range: 0.80 to 13.80)
- Average advantage: 1.28 (White maintained more threats)



2.4 Annotated Game

1. d4 Nf6 2. Nf3 d5 3. c4 e6 4. Nc3 Be7 5. Bf4 O-O 6. e3 c5 7. dxc5 Bxc5! 8. Qc2! Nc6! 9. a3! Qa5! 10. Rd1 Rd8 11. Be2 Ne4 12. O-O! Nxc3! 13. bxc3 h6 14. a4 Ne7! 15. Ne5 Bd6! 16. cxd5! Nxd5 17. Bf3 Nxf4 18. exf4! Bxe5! 19. Rxd8+! Qxd8! 20. fxe5! Qc7 21. Rb1 Rb8! 22. Qd3 Bd7 23. a5 Bc6! 24. Qd6! Qxd6! 25. exd6! Bxf3! 26. gxf3! Kf8! 27. c4! Ke8! 28. a6 b6! 29. c5! Kd7! 30. cxb6! axb6! 31. a7 Ra8! 32. Rxb6! Rxa7! 33. Kg2 e5 34. Rb4 f5 35. Rb6! Ke6! 36. d7+! Kxd7! 37. Rb5 Ke6! 38. Rb6+ Kf7 39. Rb5! Kf6! 40. Rb6+ Kg5 41. Rb5 Kf4 42. Rb4+ e4 43. fxe4 fxe4! 44. h3 Ra5 45. Rb7! Rg5+ 46. Kf1! Rg6 47. Rb4 Rg5 48. Rb7! Rg6! 49. Rb4

1/2-1/2

2.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

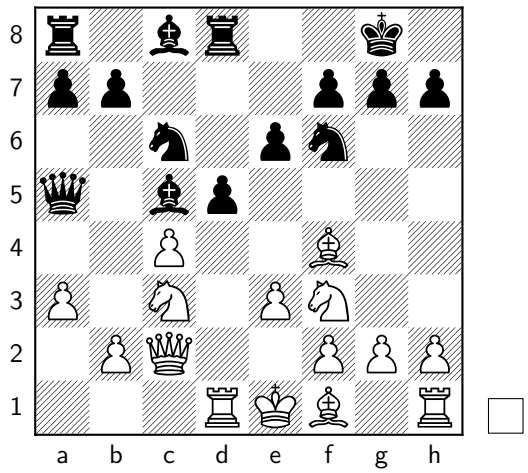
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 10...Rd8 ★

Biggest evaluation swing(s) (23cp)

Evaluation: +0.43



Metric	White	Black
Space	0.50	0.50
Mobility	5.80	4.80
King Safety	0.10	0.90
Threats	2.80	1.00

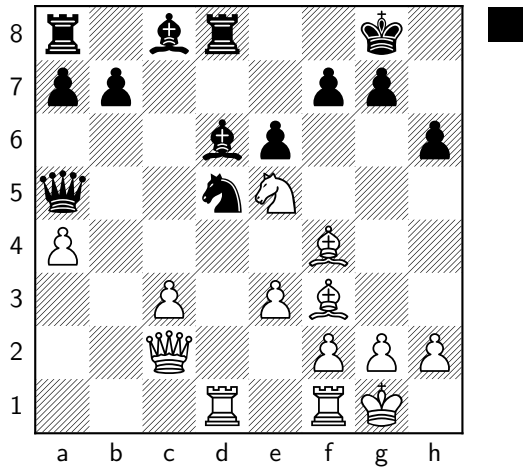
Instead of Rd8: Be7, h6

Best continuation: Nd2 dxc4 Nxc4

Position 2: After 17. Bf3 ★

Biggest evaluation swing(s) (27cp)

Evaluation: -0.27



Metric	White	Black
Space	0.40	0.50
Mobility	5.20	4.30
King Safety	1.30	0.20
Threats	3.50	2.00

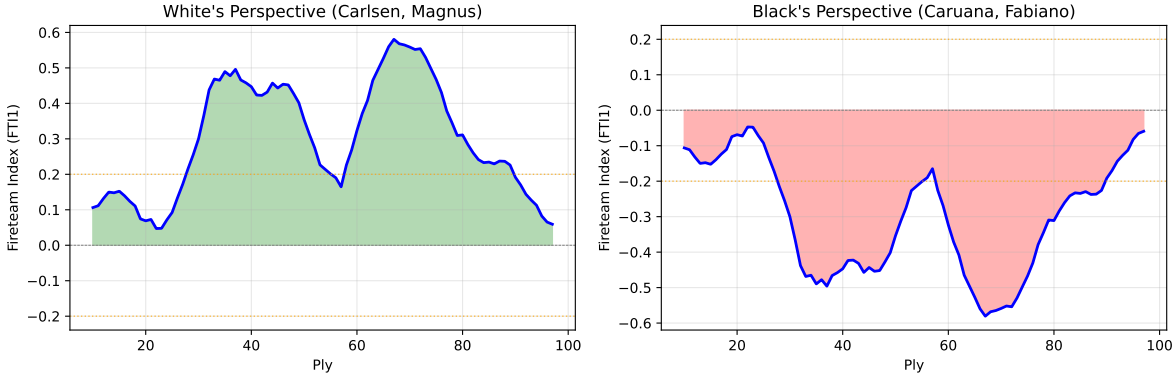
Instead of Bf3: Bg3, Nxf7 Best continuation: Nxf4 exf4 Bxe5

2.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

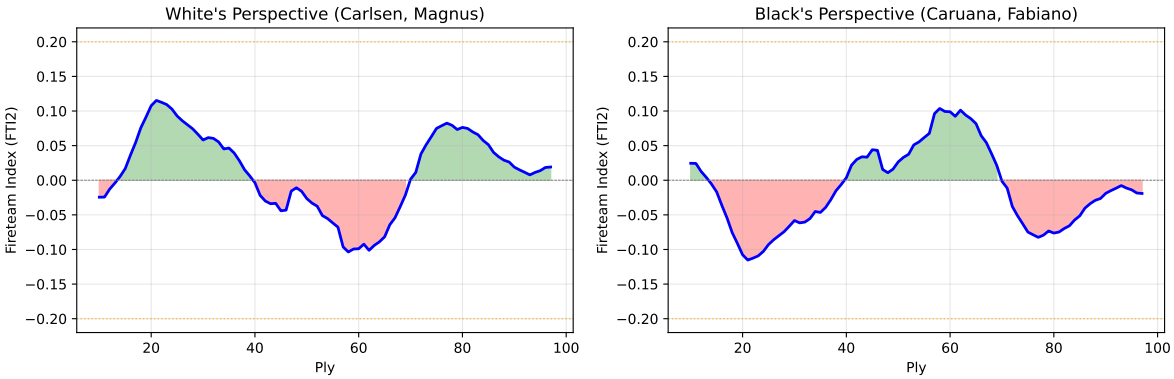
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	23 plies
Carlsen, Magnus	Windowed	WIN	32 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



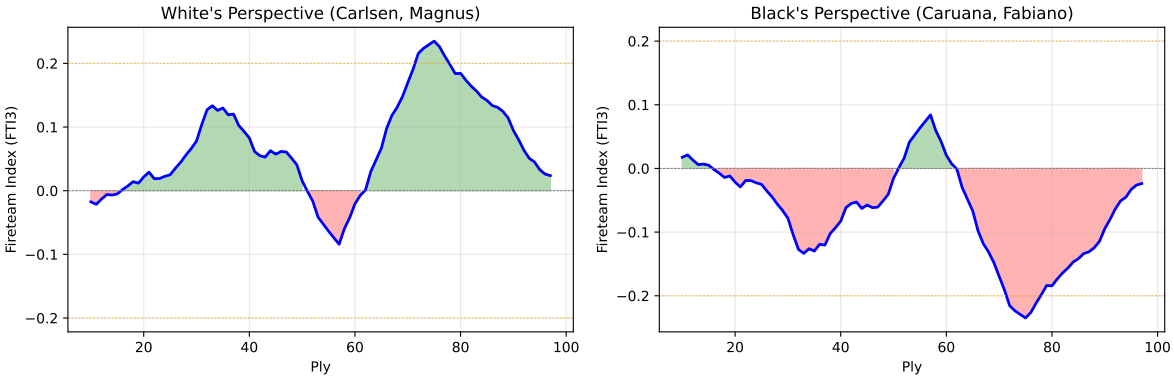
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



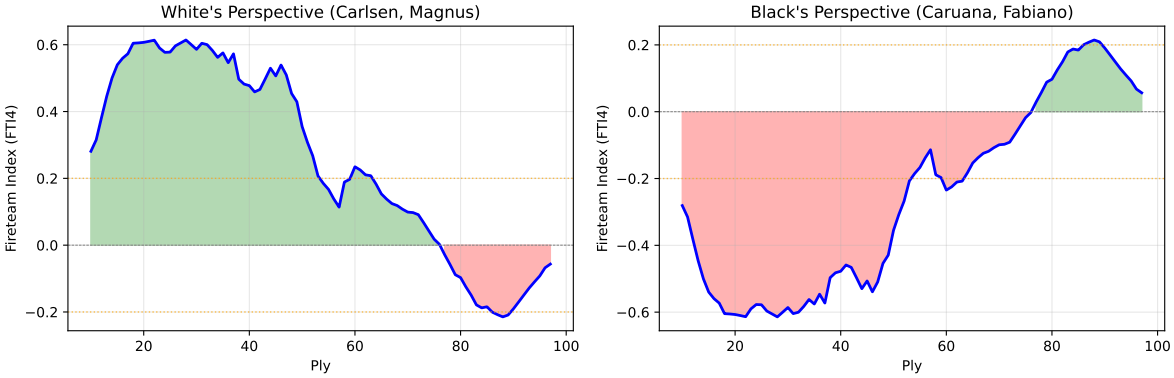
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	8 plies
Carlsen, Magnus	Windowed	DRAW	6 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



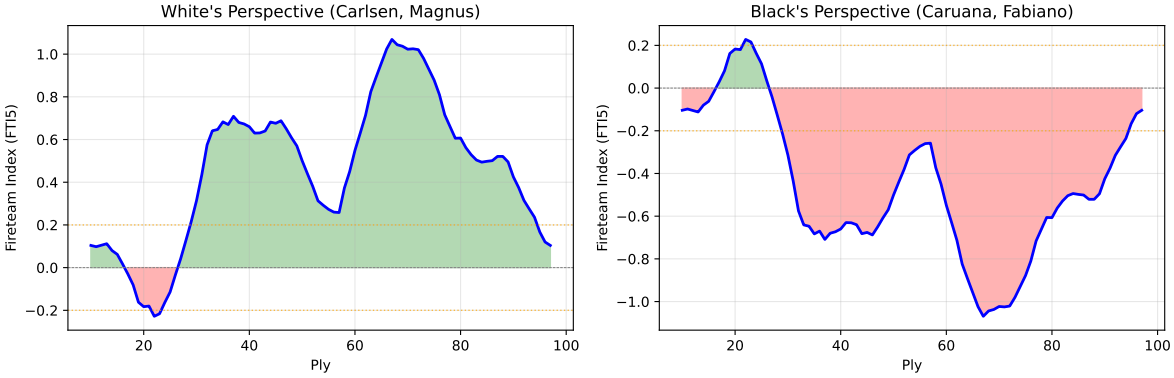
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	21 plies
Carlsen, Magnus	Windowed	WIN	37 plies
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	4 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	24 plies
Carlsen, Magnus	Windowed	WIN	66 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	2 plies



Interpretation. White (Carlsen, Magnus) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

2.7 Analysis Information

- Engine: Stockfish 16

- Depth: 20
- Analysis time: 130.4 seconds

Chapter 3

Caruana, Fabiano vs Carlsen, Magnus

3.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.12
Round	3
White	Caruana, Fabiano
Black	Carlsen, Magnus
Result	1/2-1/2
Opening	??? – Unknown

3.2 Player Statistics

3.2.1 Caruana, Fabiano (White)

- Total moves: 49
- Accuracy: 96.7%
- Average centipawn loss: 6.8
- Best/Excellent moves: 25 / 10
- Good moves: 13
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

3.2.2 Carlsen, Magnus (Black)

- Total moves: 48
- Accuracy: 97.0%
- Average centipawn loss: 6.1

- Best/Excellent moves: 27 / 11
- Good moves: 7
- Inaccuracies: 3
- Mistakes: 0
- Blunders: 0

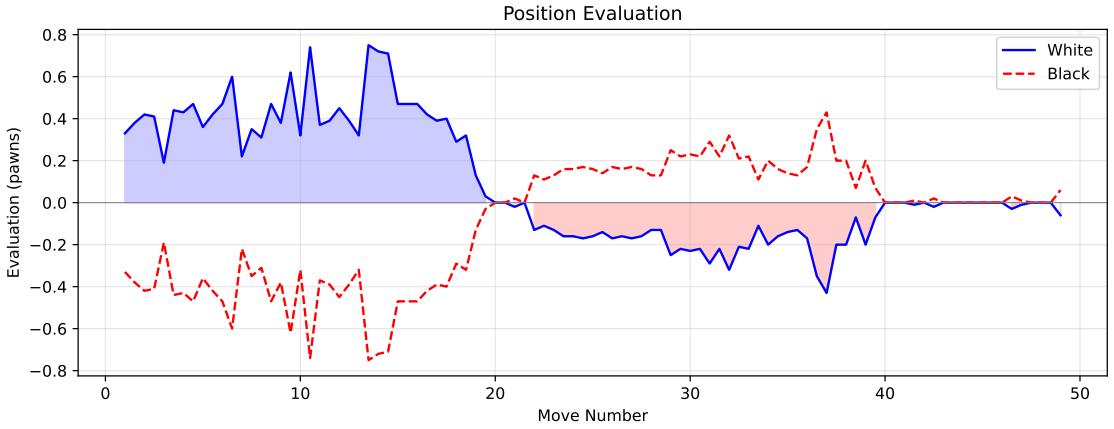
3.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

3.3.1 Position Evaluation Over Time

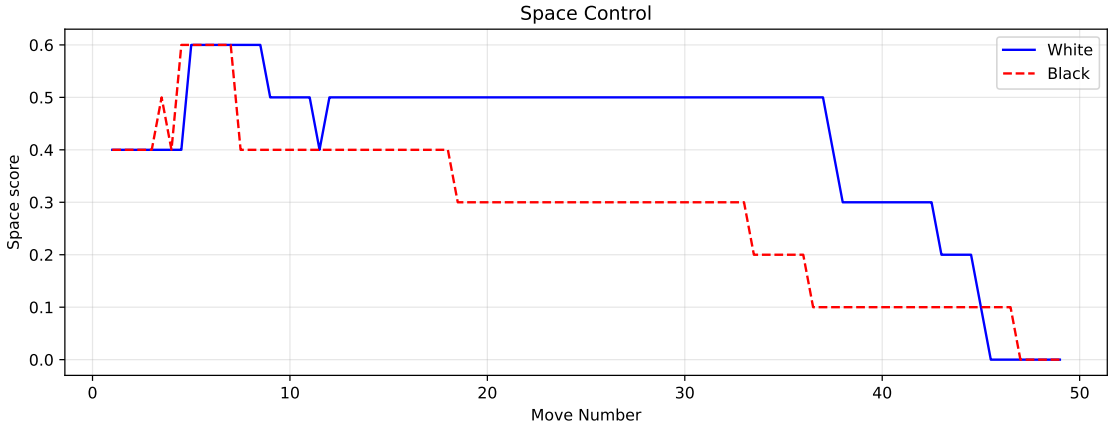
Game Character Classification. Evaluation range: minimum $m_1 = -0.43$, maximum $m_2 = +0.75$, spread $d = m_2 - m_1 = 1.18$.

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.



3.3.2 Space Control

- White average: 0.42 (range: 0.00 to 0.60)
- Black average: 0.28 (range: 0.00 to 0.60)



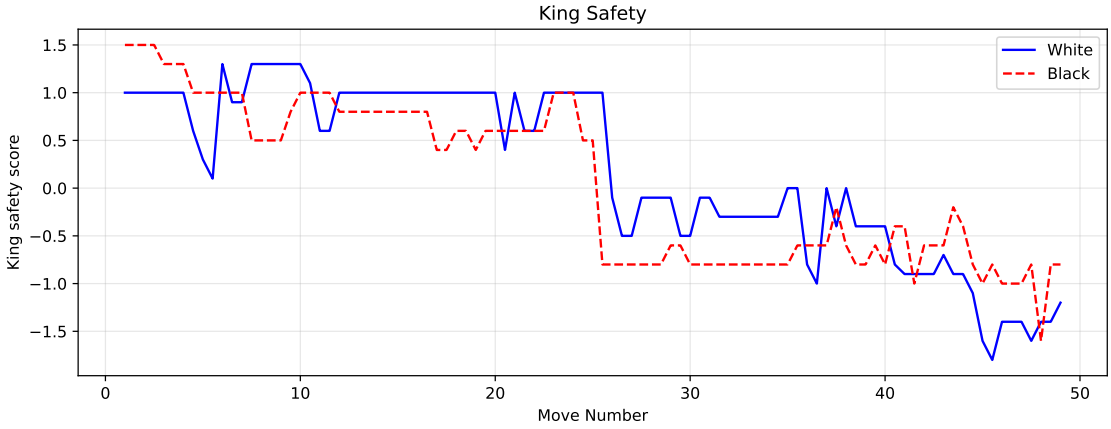
3.3.3 Mobility (Piece Activity)

- White average: 2.72 (range: 0.30 to 5.80)
- Black average: 2.78 (range: 0.80 to 5.00)



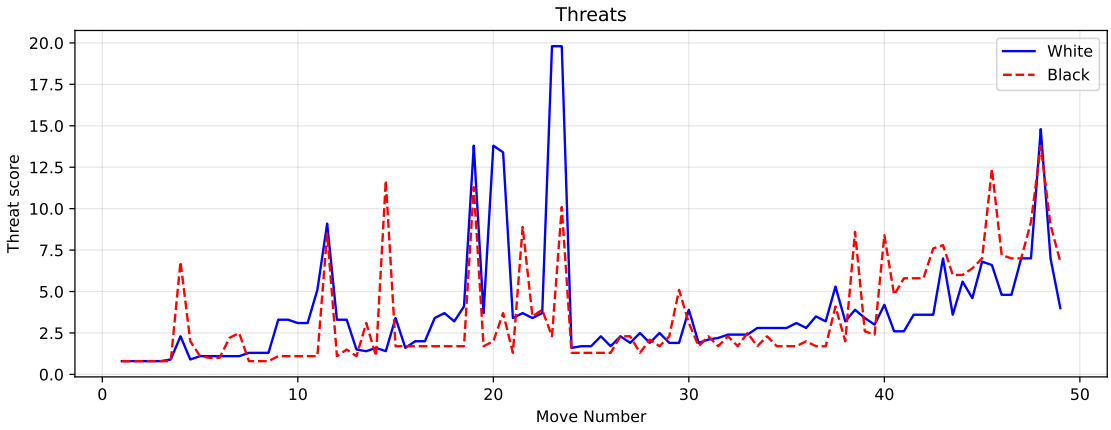
3.3.4 King Safety

- White average: 0.19
- Black average: 0.06



3.3.5 Threats

- White average: 3.78 (range: 0.80 to 19.80)
- Black average: 3.47 (range: 0.80 to 13.80)
- Average advantage: 0.31 (White maintained more threats)



3.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. Bb5 g6 4. Bxc6 dxc6 5. d3 Bg7 6. O-O Qc7 7. Re1 e5 8. a3! Nf6 9. b4 O-O?! 10. Nbd2 Bg4?! 11. h3?! Bxf3 12. Nxf3! cxb4 13. axb4! a5?! 14. bxa5 Rxa5! 15. Bd2 Raa8! 16. Qb1! Nd7 17. Qb4 Rfe8 18. Bc3 b5 19. Rxa8 Rxa8! 20. Ra1 Rxa1+! 21. Bxa1! Qa7! 22. Bc3 Qa2 23. Qb2 Qxb2! 24. Bxb2! f6! 25. Kf1! Kf7 26. Ke2! Nc5! 27. Bc3 Ne6! 28. g3! Bf8! 29. Nd2 Ng5! 30. h4! Ne6! 31. Nb3 h5 32. Bd2 Bd6 33. c3! c5 34. Be3! Ke7 35. Kd1! Kd7 36. Kc2 f5! 37. Kd1 fxe4 38. dxe4! c4 39. Nd2 Nc5 40. Bxc5! Bxc5! 41. Ke2! Kc6 42. Nf1! b4! 43. cxb4! Bxb4! 44. Ne3! Kc5! 45. f4! exf4 46. gxf4! Ba5 47. f5! gxf5! 48. Nxc4 Kxc4! 49. exf5!

1/2-1/2

3.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

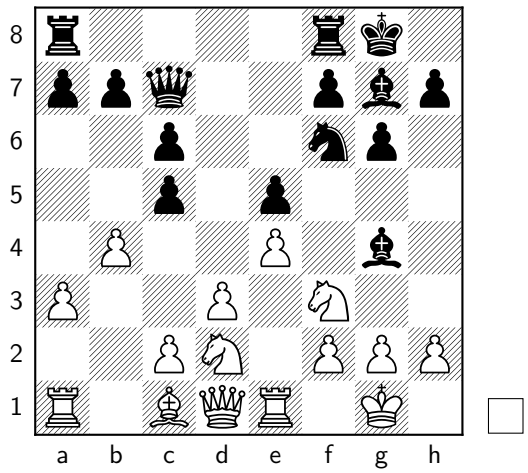
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 10...Bg4 ★

Biggest evaluation swing(s) (42cp)

Evaluation: +0.74



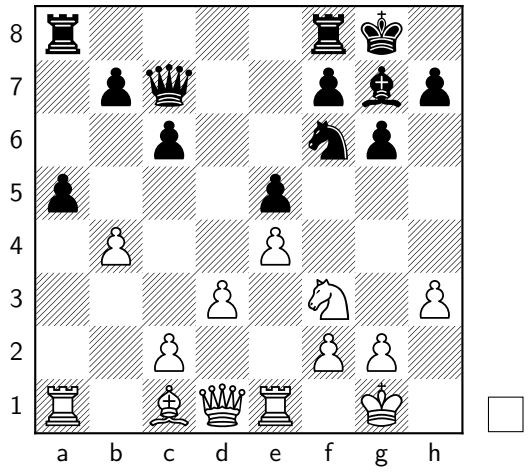
Metric	White	Black
Space	0.50	0.40
Mobility	4.30	4.90
King Safety	1.10	1.00
Threats	3.10	1.10

Instead of Bg4: cxb4, Nd7, Be6 Best continuation: bxc5 Nd7 a4

Position 2: After 13...a5 ★

Biggest evaluation swing(s) (43cp)

Evaluation: +0.75



Metric	White	Black
Space	0.50	0.40
Mobility	4.30	4.40
King Safety	1.00	0.80
Threats	1.40	3.10

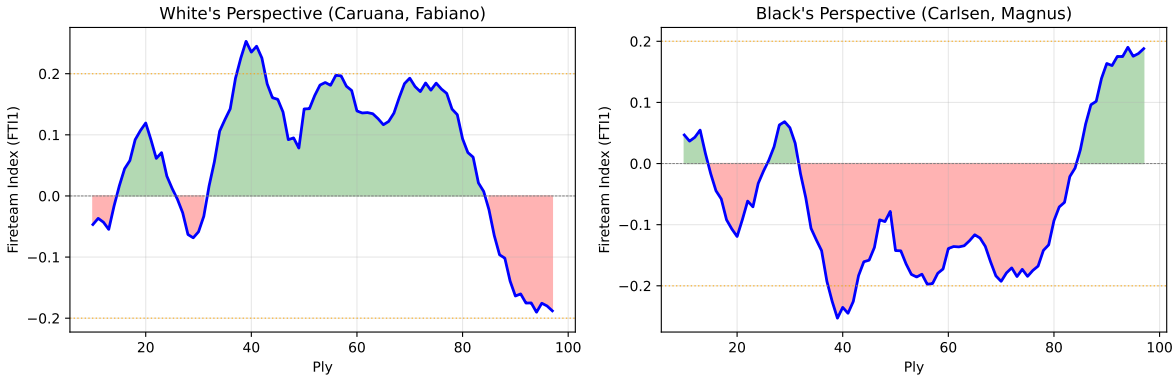
Instead of a5: Nd7, Rfd8, a6 Best continuation: bxa5 Rxa5 Rxa5

3.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

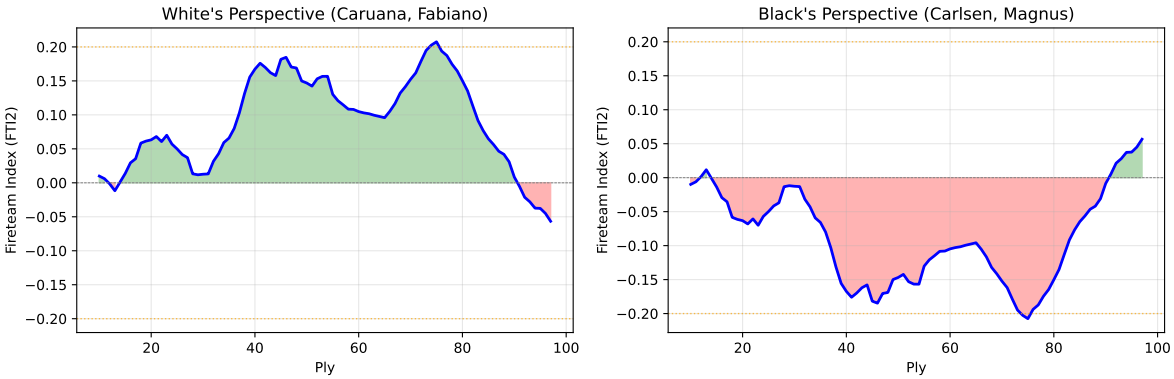
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	5 plies
Caruana, Fabiano	Windowed	DRAW	5 plies
Carlsen, Magnus	Per-ply	DRAW	2 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



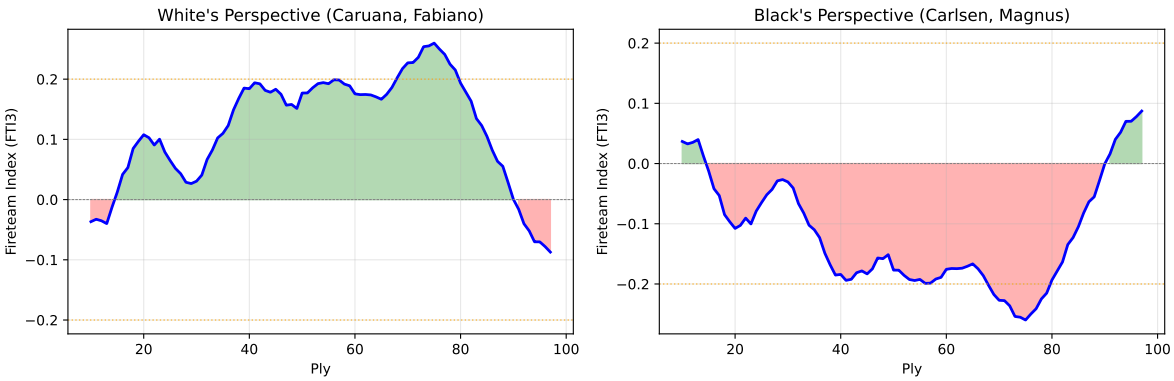
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	2 plies
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



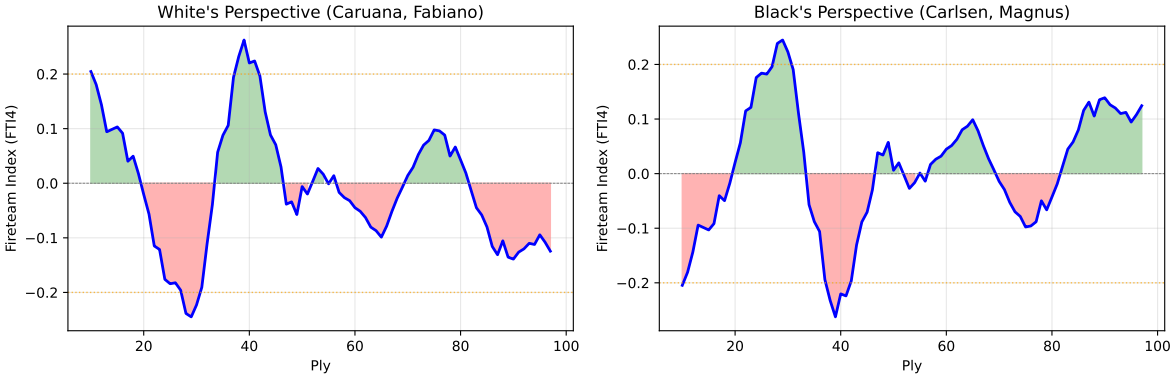
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	5 plies
Caruana, Fabiano	Windowed	WIN	12 plies
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



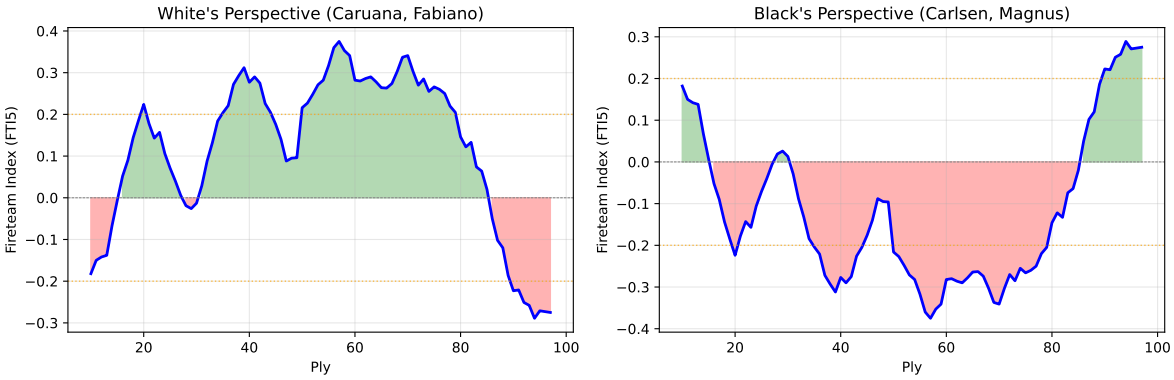
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	4 plies
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	3 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	11 plies
Caruana, Fabiano	Windowed	WIN	30 plies
Carlsen, Magnus	Per-ply	DRAW	6 plies
Carlsen, Magnus	Windowed	DRAW	8 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

3.7 Analysis Information

- Engine: Stockfish 16

- Depth: 20
- Analysis time: 137.7 seconds

Chapter 4

Carlsen, Magnus vs Caruana, Fabiano

4.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.13
Round	4
White	Carlsen, Magnus
Black	Caruana, Fabiano
Result	1/2-1/2
Opening	??? – Unknown

4.2 Player Statistics

4.2.1 Carlsen, Magnus (White)

- Total moves: 34
- Accuracy: 96.8%
- Average centipawn loss: 6.5
- Best/Excellent moves: 12 / 14
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

4.2.2 Caruana, Fabiano (Black)

- Total moves: 33
- Accuracy: 97.9%
- Average centipawn loss: 4.3

- Best/Excellent moves: 19 / 6
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

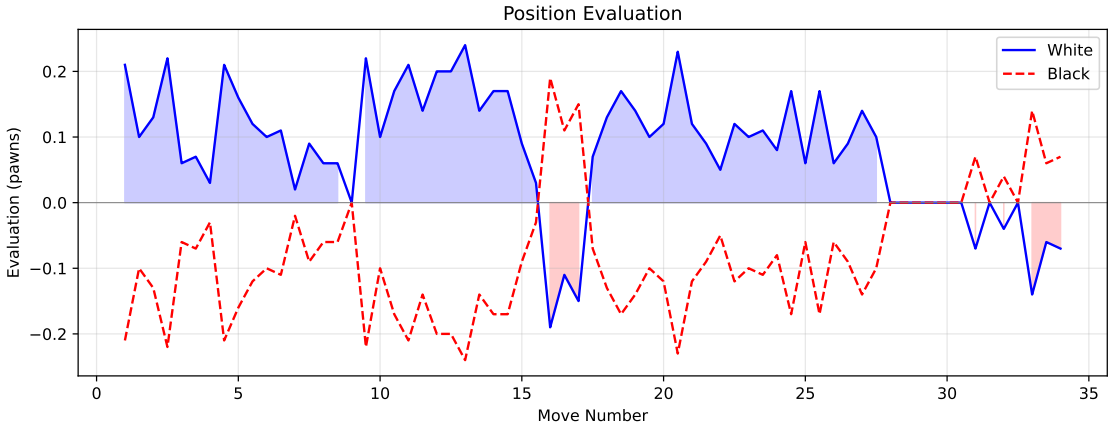
4.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

4.3.1 Position Evaluation Over Time

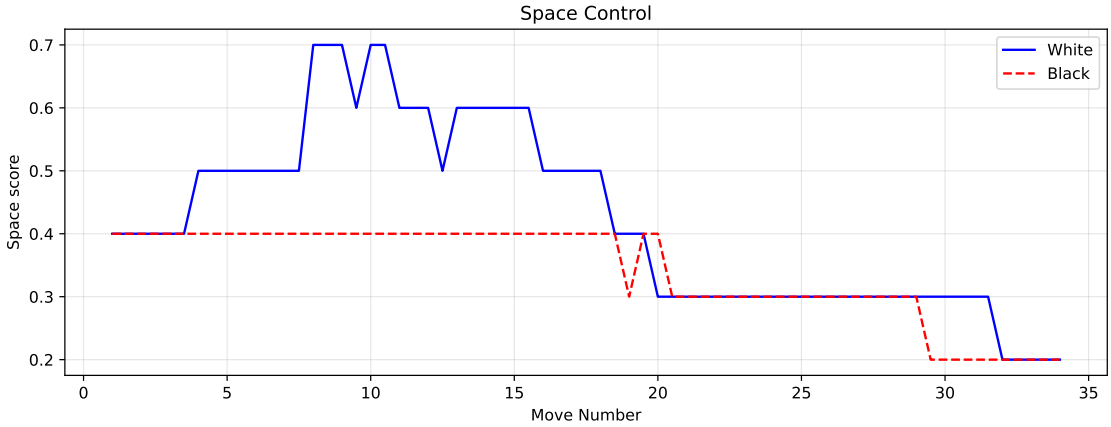
Game Character Classification. Evaluation range: minimum $m_1 = -0.19$, maximum $m_2 = +0.24$, spread $d = m_2 - m_1 = 0.43$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



4.3.2 Space Control

- White average: 0.42 (range: 0.20 to 0.70)
- Black average: 0.34 (range: 0.20 to 0.40)



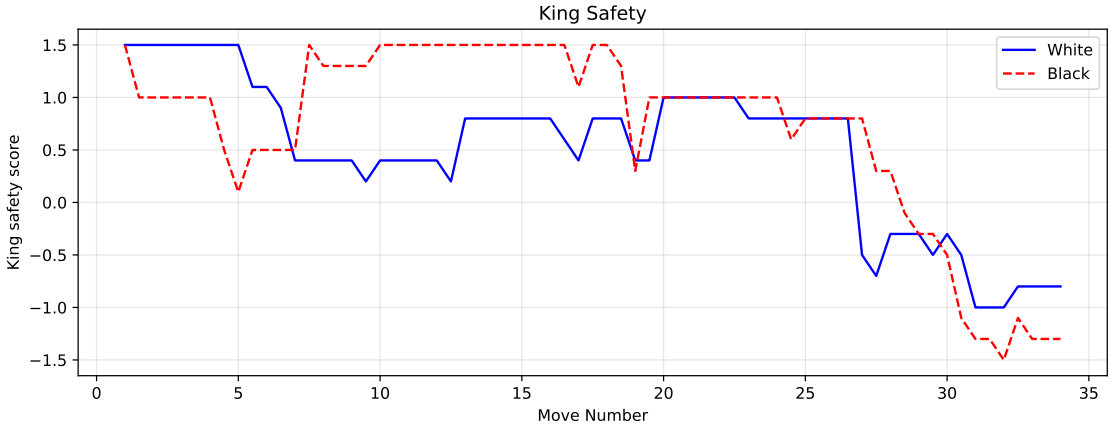
4.3.3 Mobility (Piece Activity)

- White average: 3.60 (range: 2.30 to 5.30)
- Black average: 3.83 (range: 2.30 to 5.90)



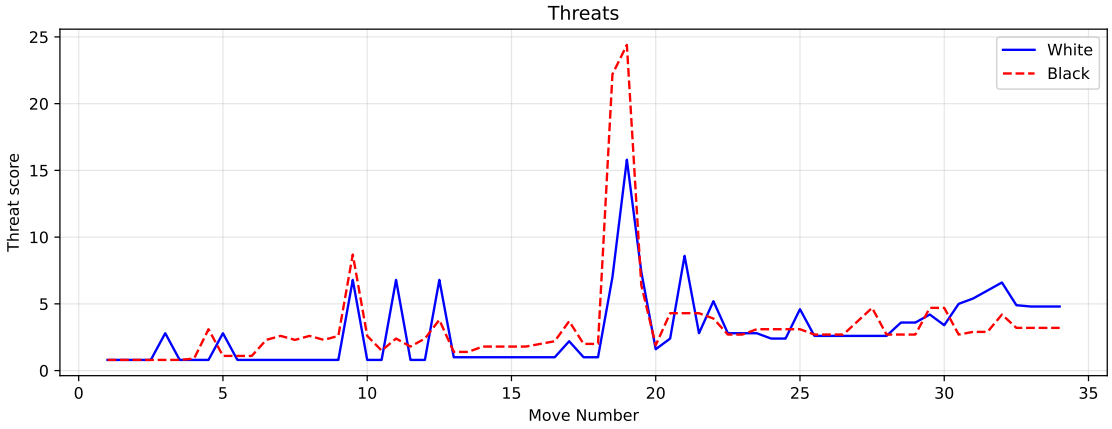
4.3.4 King Safety

- White average: 0.50
- Black average: 0.71



4.3.5 Threats

- White average: 2.91 (range: 0.80 to 15.80)
- Black average: 3.30 (range: 0.80 to 24.40)
- Average advantage: -0.39 (Black maintained more threats)



4.4 Annotated Game

1. c4 e5 2. Nc3 Nf6 3. Nf3 Nc6 4. g3 d5 5. cxd5 Nxd5 6. Bg2 Bc5 7. O-O O-O! 8. d3 Re8 9. Bd2 Nxc3 10. Bxc3! Nd4 11. b4! Bd6! 12. Rb1! Nxf3+! 13. Bxf3! a6 14. a4 c6! 15. Re1 Bd7! 16. e3 Qf6 17. Be4 Bf5 18. Qf3! Bxe4 19. Qxf6 gxf6! 20. dxe4! b5 21. Red1 Bf8! 22. axb5 axb5 23. Kg2 Red8! 24. Rdc1 Kg7 25. Be1 Rdc8 26. Rc2 Ra4 27. Kf3 h5 28. Ke2 Kg6! 29. h3! f5! 30. exf5+ Kxf5! 31. f3 Be7 32. e4+ Ke6 33. Bd2 Bd6 34. Rbc1

1/2-1/2

4.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

- **Instead of [move]**: The move(s) that should have been played instead.

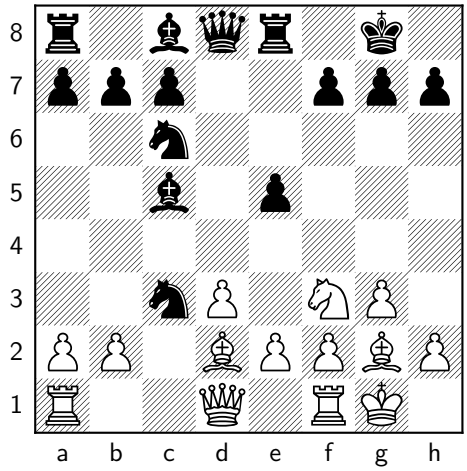
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 9...Nxc3 ★

Biggest evaluation swing(s) (22cp)

Evaluation: +0.22



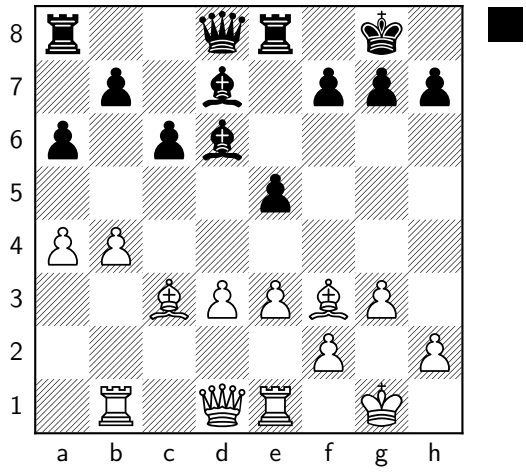
Metric	White	Black
Space	0.60	0.40
Mobility	4.50	5.90
King Safety	0.20	1.30
Threats	6.80	8.70

Instead of Nxc3: Nb6, a5 Best continuation: Bxc3 Nd4 b4

Position 2: After 16. e3 ★

Biggest evaluation swing(s) (22cp)

Evaluation: -0.19



Metric	White	Black
Space	0.50	0.40
Mobility	4.20	4.20
King Safety	0.80	1.50
Threats	1.00	2.00

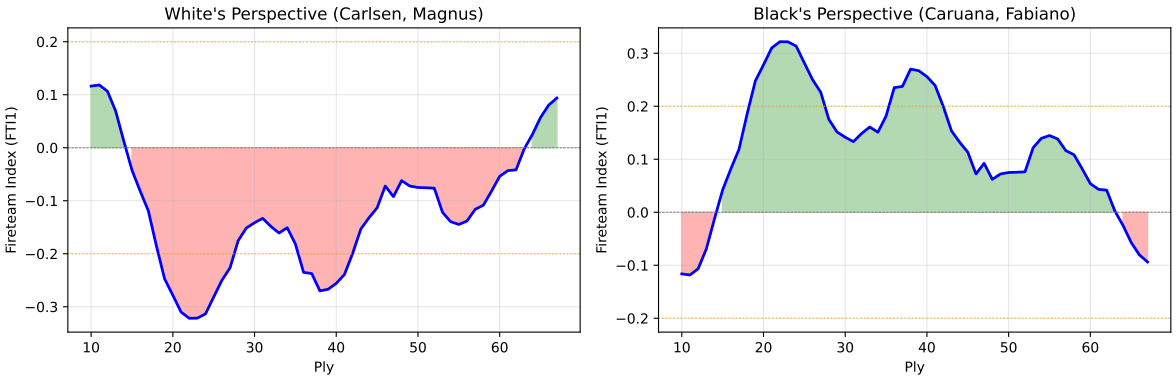
Instead of e3: Qc1, Qb3, Qc2 Best continuation: Qe7 Be4 Rad8

4.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

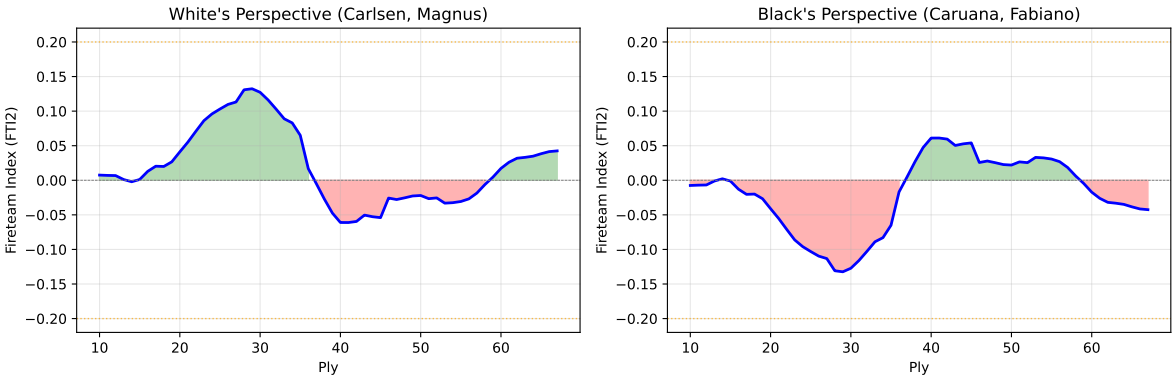
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	6 plies
Caruana, Fabiano	Windowed	DRAW	9 plies



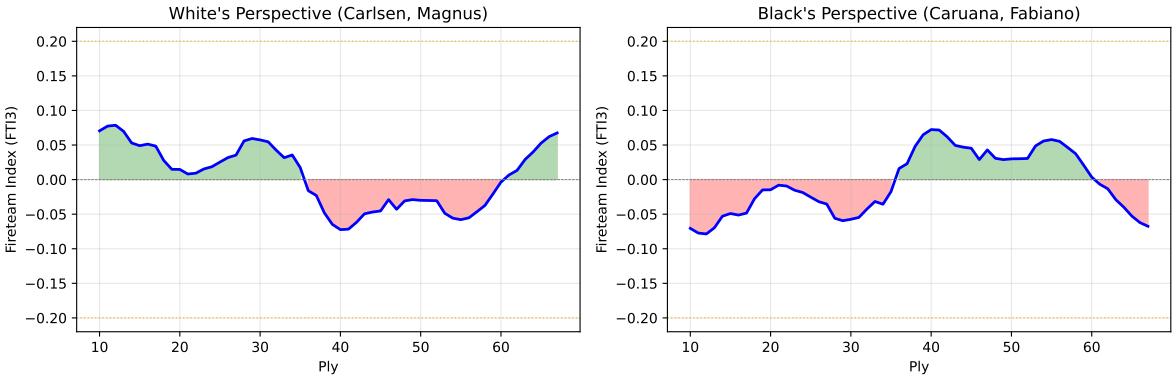
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



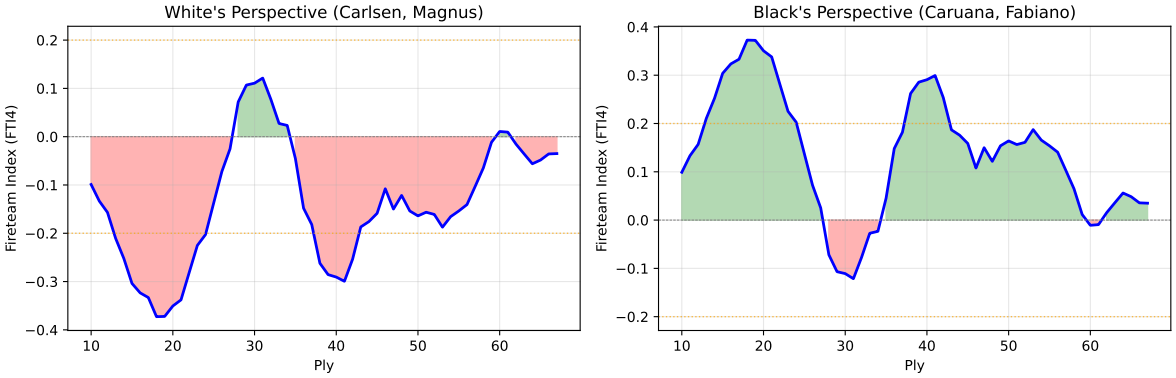
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



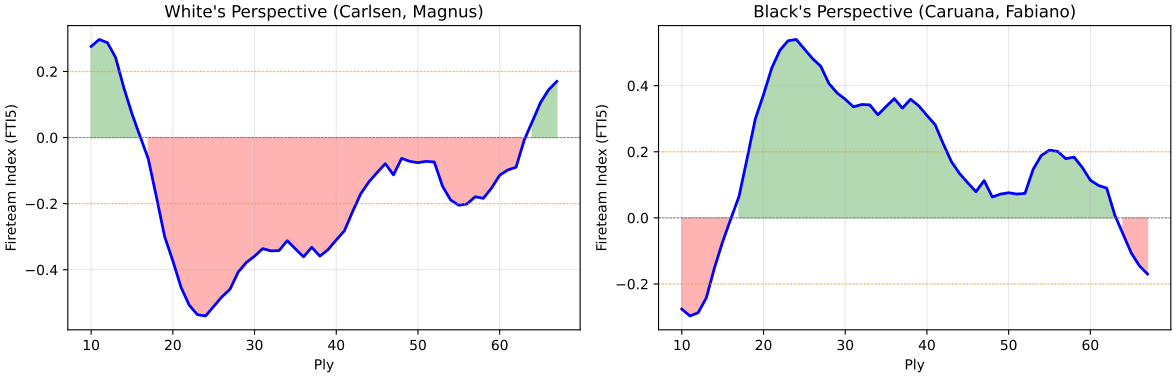
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	5 plies
Caruana, Fabiano	Windowed	DRAW	8 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	9 plies
Caruana, Fabiano	Windowed	WIN	24 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

4.7 Analysis Information

- Engine: Stockfish 16

- Depth: 20
- Analysis time: 124.5 seconds

Chapter 5

Caruana, Fabiano vs Carlsen, Magnus

5.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.15
Round	5
White	Caruana, Fabiano
Black	Carlsen, Magnus
Result	1/2-1/2
Opening	??? – Unknown

5.2 Player Statistics

5.2.1 Caruana, Fabiano (White)

- Total moves: 34
- Accuracy: 98.4%
- Average centipawn loss: 3.2
- Best/Excellent moves: 21 / 7
- Good moves: 6
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

5.2.2 Carlsen, Magnus (Black)

- Total moves: 33
- Accuracy: 98.2%
- Average centipawn loss: 3.6

- Best/Excellent moves: 22 / 4
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

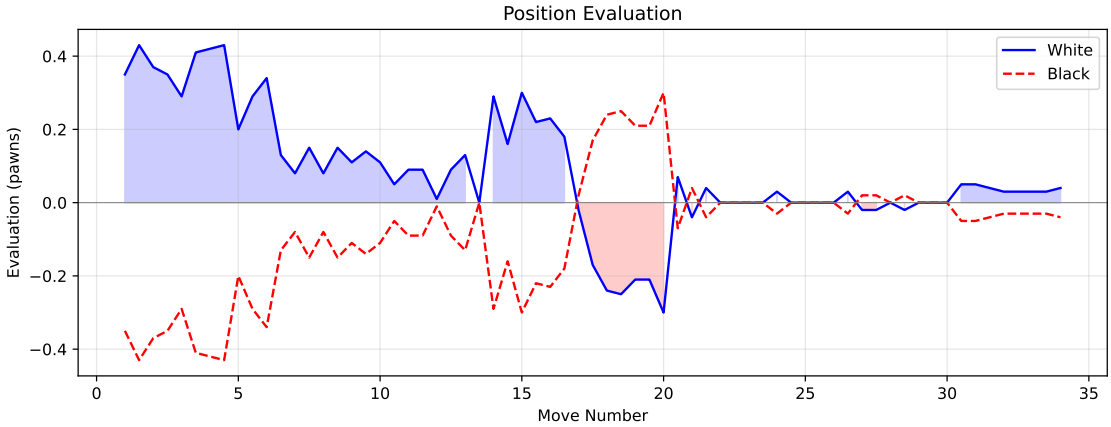
5.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

5.3.1 Position Evaluation Over Time

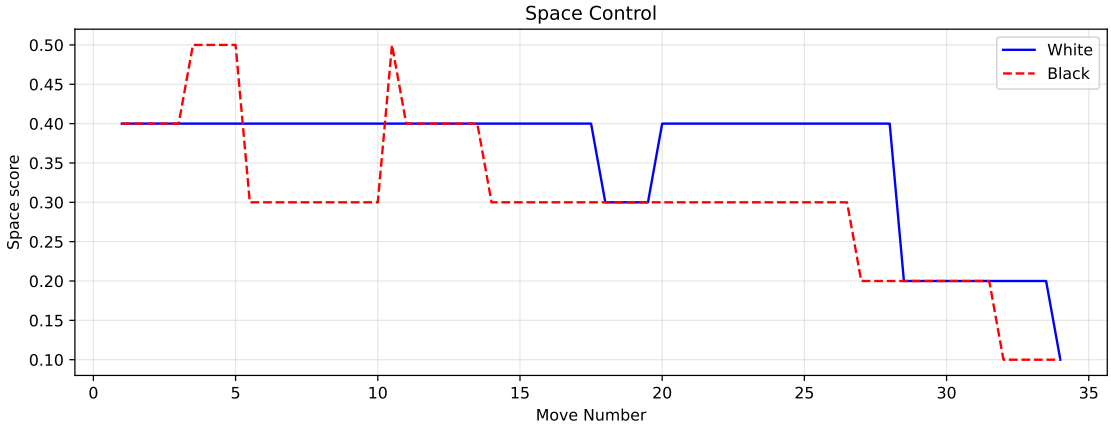
Game Character Classification. Evaluation range: minimum $m_1 = -0.30$, maximum $m_2 = +0.43$, spread $d = m_2 - m_1 = 0.73$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



5.3.2 Space Control

- White average: 0.36 (range: 0.10 to 0.40)
- Black average: 0.30 (range: 0.10 to 0.50)



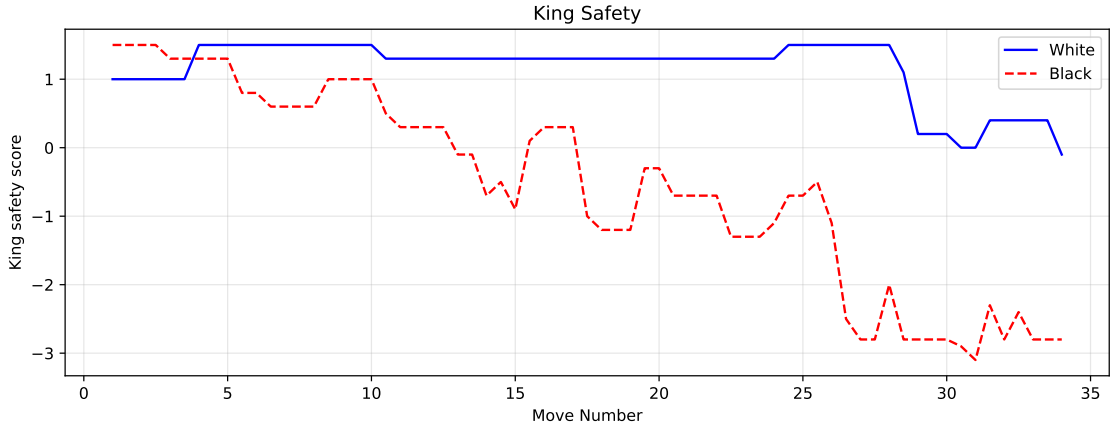
5.3.3 Mobility (Piece Activity)

- White average: 3.56 (range: 1.90 to 4.70)
- Black average: 3.45 (range: 2.00 to 4.60)



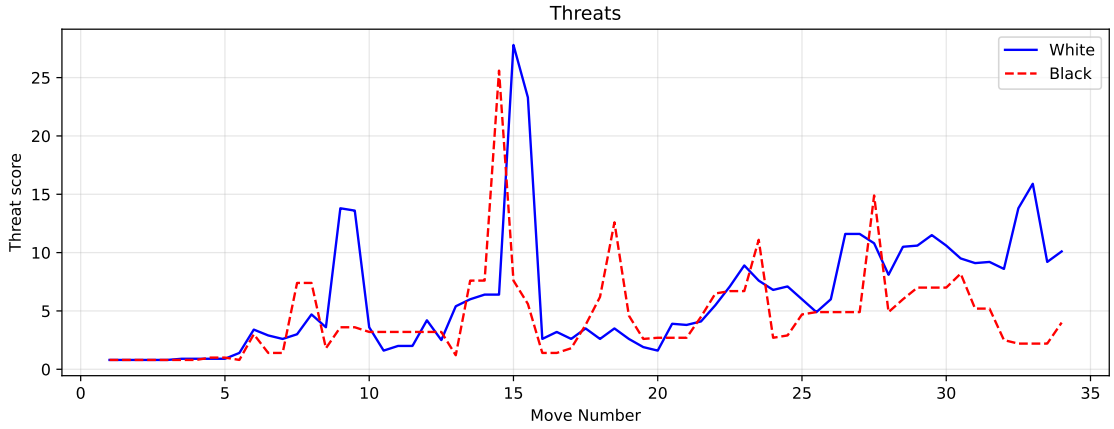
5.3.4 King Safety

- White average: 1.16
- Black average: -0.57



5.3.5 Threats

- White average: 6.18 (range: 0.80 to 27.80)
- Black average: 4.46 (range: 0.80 to 25.60)
- Average advantage: 1.72 (White maintained more threats)



5.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. Bb5 g6 4. O-O Bg7 5. Re1 e5 6. b4 Nxb4 7. Bb2 a6! 8. a3! axb5! 9. axb4 Rxa1! 10. Bxa1! d6! 11. bxc5! Ne7! 12. Qe2 b4 13. Qc4 Qa5! 14. cxd6 Be6! 15. Qc7! Qxc7! 16. dxc7! Nc6! 17. c3 Kd7! 18. cxb4! Ra8 19. Bc3! Kxc7! 20. d3 Kb6?! 21. Bd2! Rd8 22. Be3+ Kb5! 23. Nc3+! Kxb4! 24. Nd5+! Bxd5! 25. exd5! Rxd5! 26. Rb1+! Kc3! 27. Rxb7! Nd8! 28. Rc7+! Kxd3! 29. Kf1! h5 30. h3 Ke4 31. Ng5+! Kf5! 32. Nxf7! Nxf7 33. Rxf7+! Bf6! 34. g4+!

1/2-1/2

5.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

- **Instead of [move]**: The move(s) that should have been played instead.

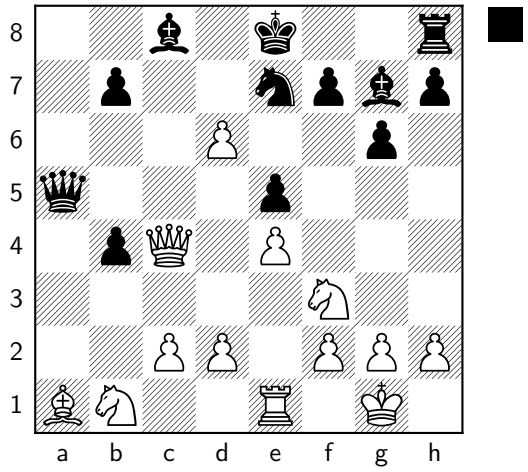
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 14. cxd6 ★

Biggest evaluation swing(s) (29cp)

Evaluation: +0.29



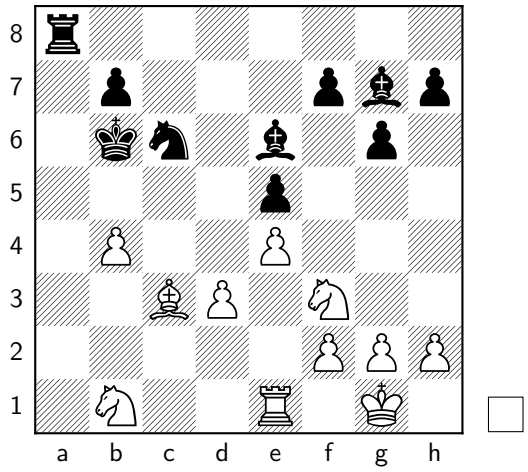
Metric	White	Black
Space	0.40	0.30
Mobility	4.30	3.90
King Safety	1.30	-0.70
Threats	6.40	7.60

Best continuation: Be6 Qc7 Qxc7

Position 2: After 20...Kb6 ★

Biggest evaluation swing(s) (37cp)

Evaluation: +0.07



Metric	White	Black
Space	0.40	0.30
Mobility	3.40	3.90
King Safety	1.30	-0.70
Threats	3.90	2.70

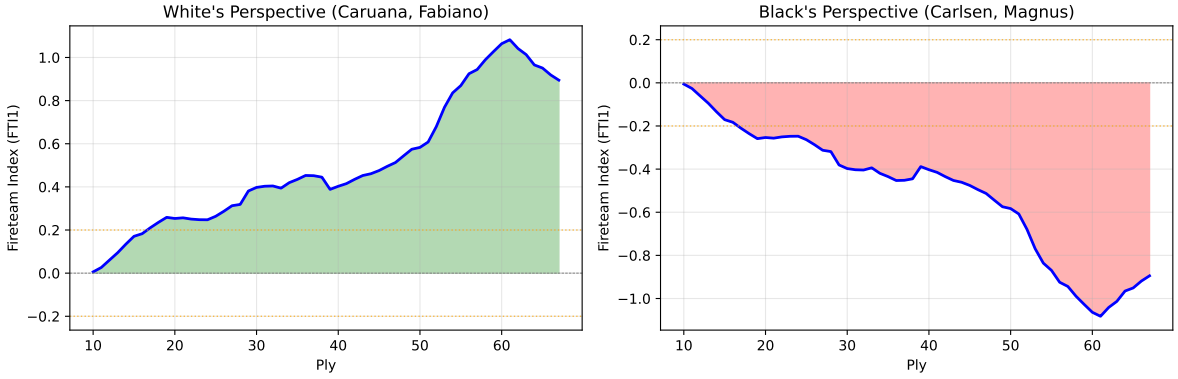
Instead of Kb6: b5, Kd7, Ra4 Best continuation: Bd2 Kc7 Kf1

5.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

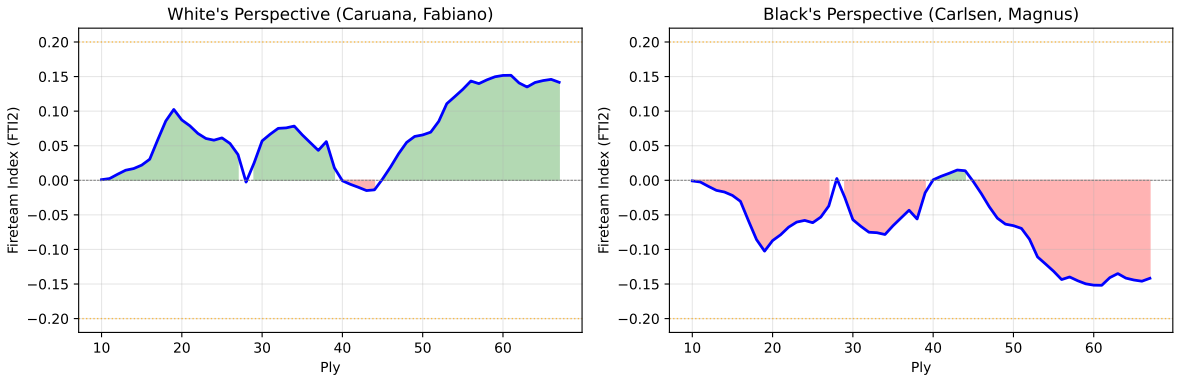
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	47 plies
Caruana, Fabiano	Windowed	WIN	51 plies
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



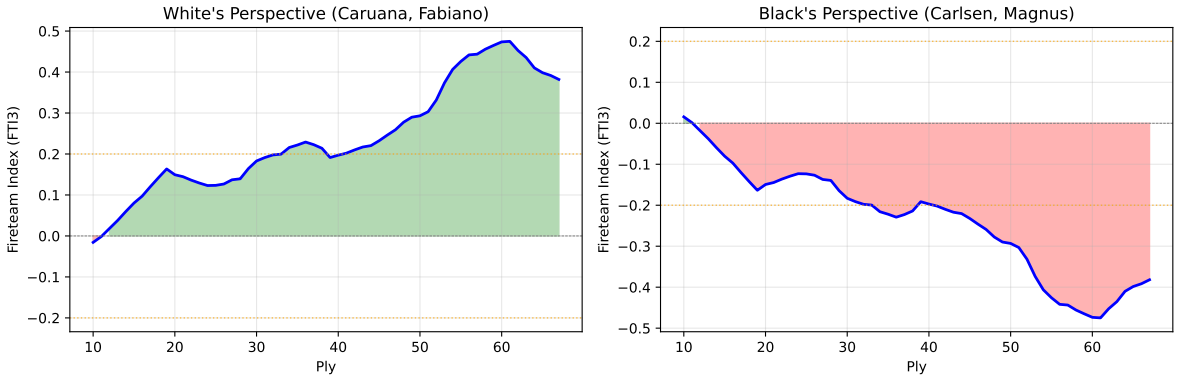
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	0 plies
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



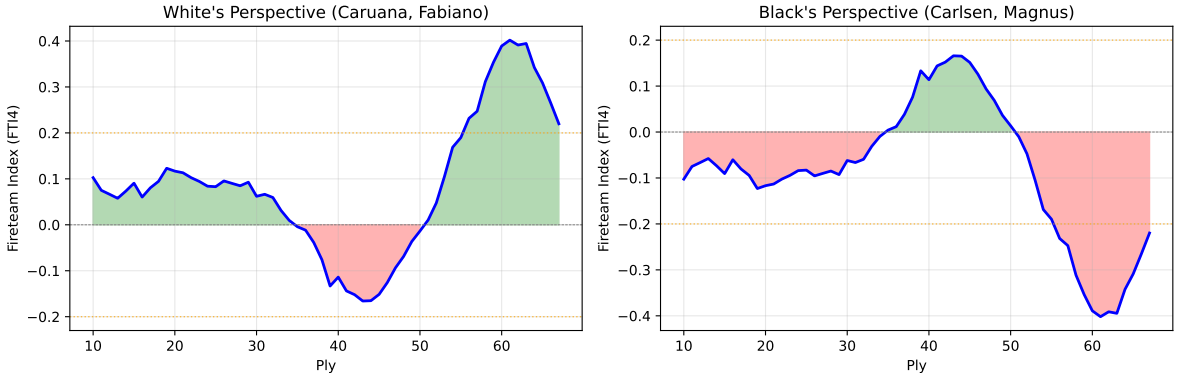
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	28 plies
Caruana, Fabiano	Windowed	WIN	27 plies
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



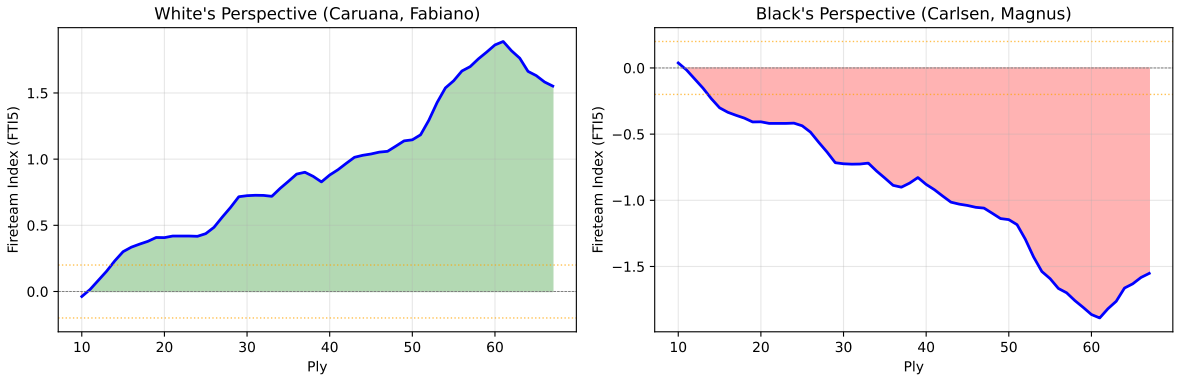
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	9 plies
Caruana, Fabiano	Windowed	WIN	12 plies
Carlsen, Magnus	Per-ply	DRAW	4 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	51 plies
Caruana, Fabiano	Windowed	WIN	51 plies
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



Interpretation. White (Caruana, Fabiano) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

5.7 Analysis Information

- Engine: Stockfish 16

- Depth: 20
- Analysis time: 131.5 seconds

Chapter 6

Carlsen, Magnus vs Caruana, Fabiano

6.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.16
Round	6
White	Carlsen, Magnus
Black	Caruana, Fabiano
Result	1/2-1/2
Opening	??? – Unknown

6.2 Player Statistics

6.2.1 Carlsen, Magnus (White)

- Total moves: 80
- Accuracy: 96.1%
- Average centipawn loss: 7.9
- Best/Excellent moves: 42 / 22
- Good moves: 9
- Inaccuracies: 6
- Mistakes: 1
- Blunders: 0

6.2.2 Caruana, Fabiano (Black)

- Total moves: 80
- Accuracy: 95.7%
- Average centipawn loss: 8.7

- Best/Excellent moves: 43 / 21
- Good moves: 11
- Inaccuracies: 4
- Mistakes: 0
- Blunders: 1

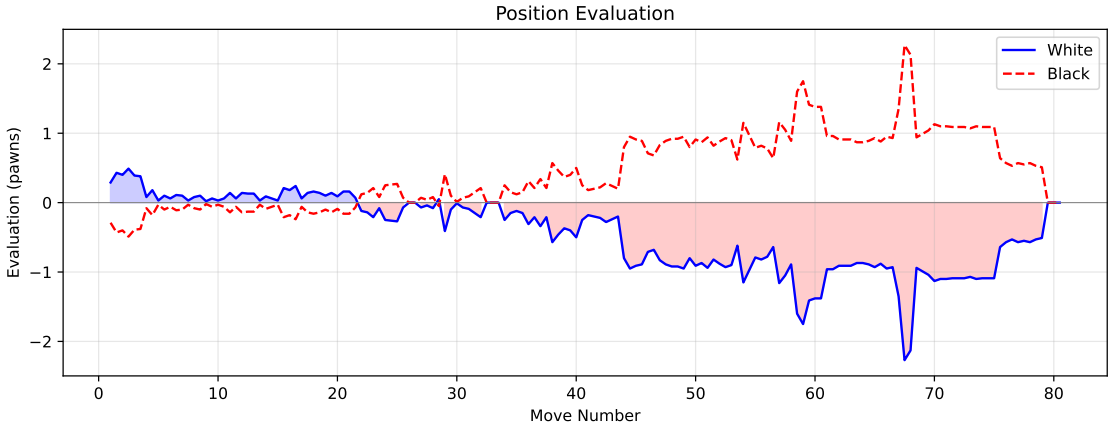
6.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

6.3.1 Position Evaluation Over Time

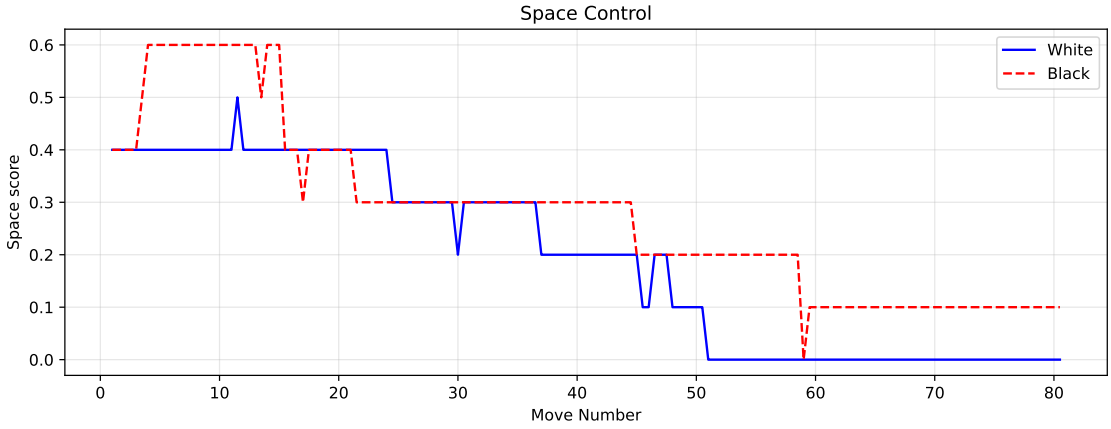
Game Character Classification. Evaluation range: minimum $m_1 = -2.27$, maximum $m_2 = +0.49$, spread $d = m_2 - m_1 = 2.76$.

This game is classified as **tense** (one-sided). A tense, one-sided game. Black maintained the advantage throughout.



6.3.2 Space Control

- White average: 0.19 (range: 0.00 to 0.50)
- Black average: 0.28 (range: 0.00 to 0.60)



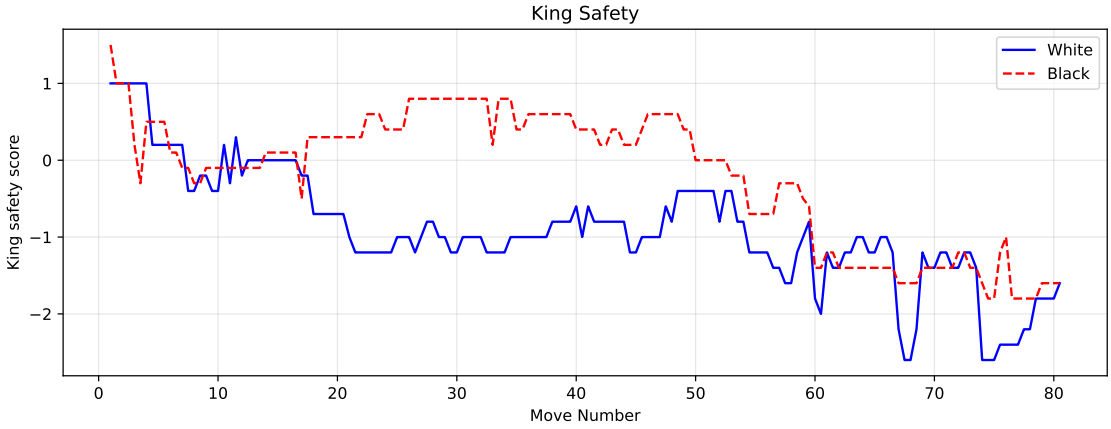
6.3.3 Mobility (Piece Activity)

- White average: 2.40 (range: 1.00 to 4.30)
- Black average: 2.55 (range: 1.00 to 5.20)



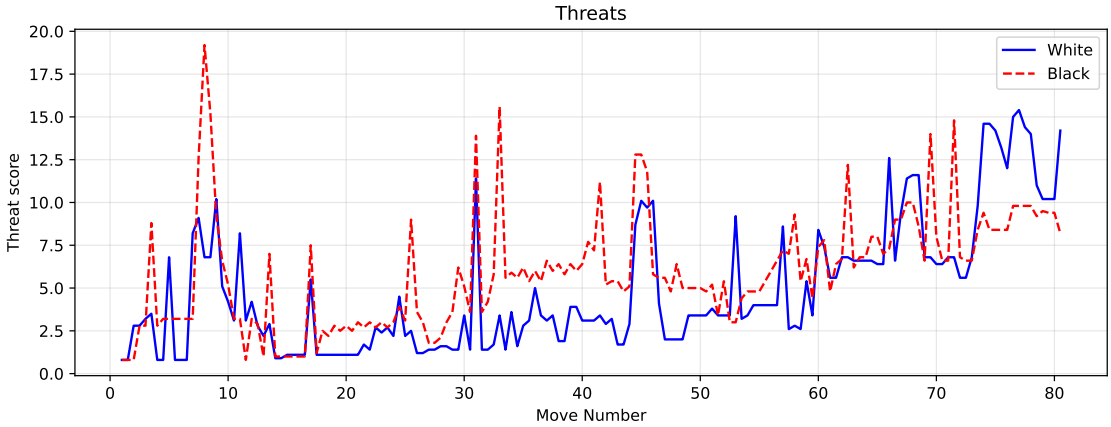
6.3.4 King Safety

- White average: -0.90
- Black average: -0.18



6.3.5 Threats

- White average: 4.86 (range: 0.80 to 15.40)
- Black average: 5.93 (range: 0.80 to 19.20)
- Average advantage: -1.07 (Black maintained more threats)



6.4 Annotated Game

1. e4 e5 2. Nf3 Nf6 3. Nxe5 d6 4. Nd3?! Nxe4 5. Qe2 Qe7 6. Nf4 Nc6 7. Nd5! Nd4! 8. Nxe7 Nxe2! 9. Nd5! Nd4! 10. Na3 Ne6! 11. f3! N4c5 12. d4! Nd7 13. c3 c6! 14. Nf4! Nb6! 15. Bd3 d5 16. Nc2! Bd6 17. Nxe6 Bxe6 18. Kf2! h5! 19. h4! Nc8 20. Ne3 Ne7 21. g3! c5! 22. Bc2 O-O 23. Rd1 Rfd8 24. Ng2 cxd4! 25. cxd4 Rac8 26. Bb3! Nc6 27. Bf4 Na5! 28. Rdc1! Bb4 29. Bd1?! Nc4 30. b3! Na3 31. Rxc8! Rxc8! 32. Rc1! Nb5 33. Rxc8+! Bxc8! 34. Ne3 Nc3! 35. Bc2! Ba3! 36. Bb8 a6 37. f4 Bd7 38. f5?! Bc6! 39. Bd1! Bb2 40. Bxh5 Ne4+ 41. Kg2! Bxd4! 42. Bf4 Bc5! 43. Bf3! Nd2 44. Bxd5? *White loses 62cp. Consider instead: Nf1, Nd1.*

44...Bxe3! 45. Bxc6! Bxf4! 46. Bxb7! Bd6 47. Bxa6 Ne4! 48. g4! Ba3! 49. Bc4 Kf8 50. g5 Nc3! 51. b4 Bxb4 52. Kf3 Na4! 53. Bb5 Nc5?! 54. a4?! f6 55. Kg4! Ne4 56. Kh5 Be1 57. Bd3?! Nd6 58. a5 Bxa5! 59. gxf6! gxf6! 60. Kg6! Bd8 61. Kh7 Nf7?! 62.

Bc4! Ne5 63. Bd5! Ba5 64. h5! Bd2 65. Ba2 Nf3 66. Bd5 Nd4 67. Kg6?! Bg5! 68. Bc4 Nf3?? *Black loses 151cp. Consider instead: Bh4.*

69. Kh7 Ne5 70. Bb3 Ng4 71. Bc4 Ne3! 72. Bd3! Ng4 73. Bc4 Nh6 74. Kg6 Ke7 75. Bb3! Kd6?! 76. Bc2! Ke5 77. Bd3 Kf4 78. Bc2 Ng4 79. Bb3 Ne3?! 80. h6! Bxh6!

1/2-1/2

6.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

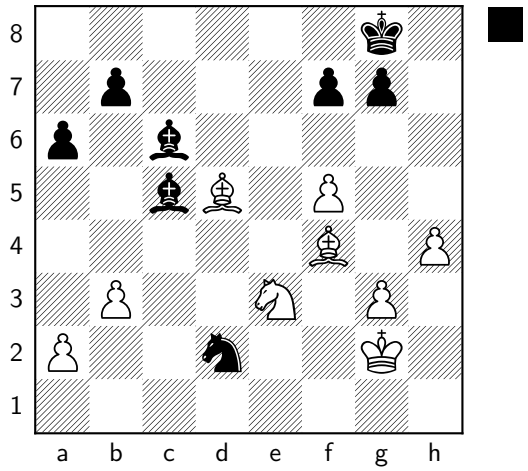
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 44. Bxd5 ★

Biggest evaluation swing(s) (60cp)

Evaluation: -0.80



Metric	White	Black
Space	0.20	0.30
Mobility	2.70	2.30
King Safety	-0.80	0.20
Threats	2.90	5.10

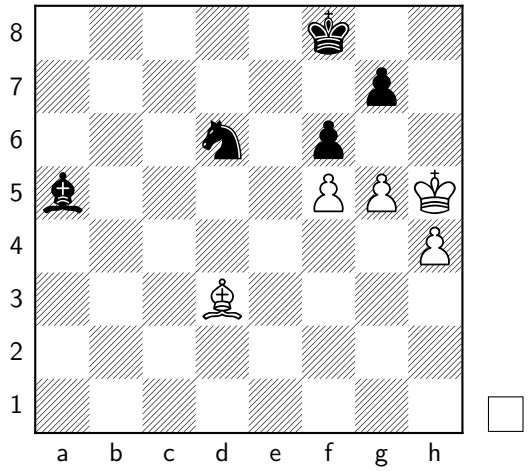
Instead of Bxd5: Nf1, Nd1

Best continuation: Bxe3 Bxc6 Bxf4

Position 2: After 58...Bxa5 ★

Biggest evaluation swing(s) (71cp)

Evaluation: -1.60



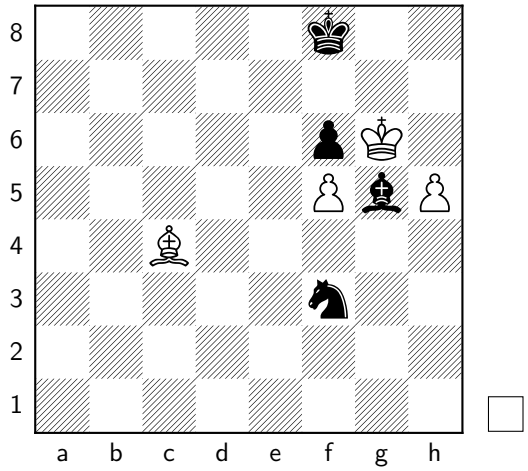
Metric	White	Black
Space	0.00	0.20
Mobility	1.10	1.60
King Safety	-1.20	-0.30
Threats	2.60	5.40

Best continuation: gxf6 gxf6 Kg6

Position 3: After 68...Nf3

Black blunders (+119cp)

Evaluation: -0.94



Metric	White	Black
Space	0.00	0.10
Mobility	1.40	1.70
King Safety	-2.20	-1.60
Threats	11.60	8.60

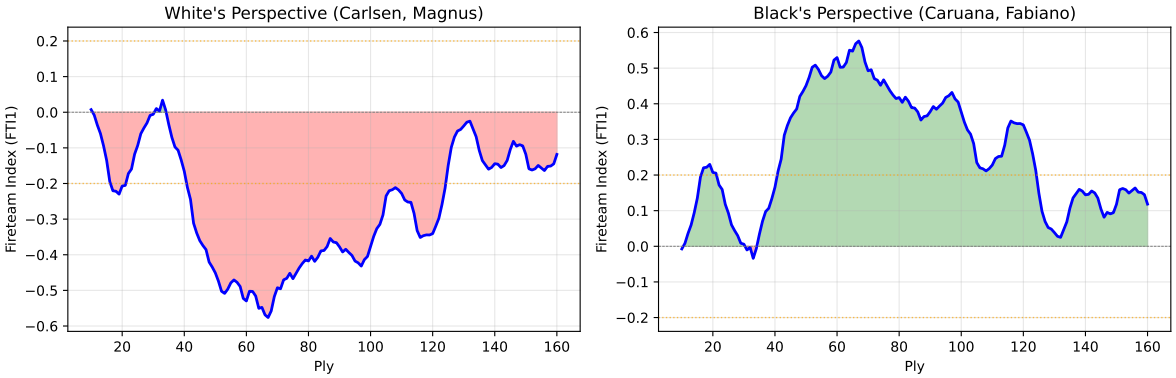
Instead of Nf3: Bh4 Best continuation: Kh7 Bd2 Bd3

6.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

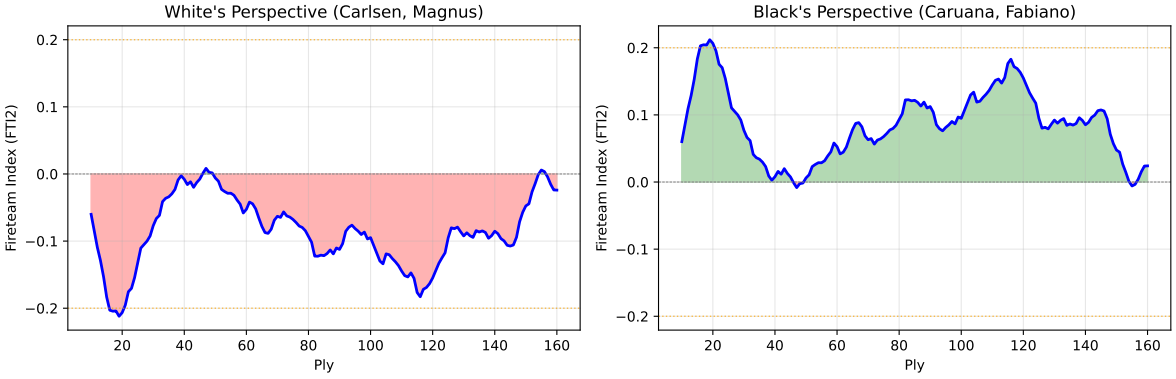
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	65 plies
Caruana, Fabiano	Windowed	WIN	84 plies



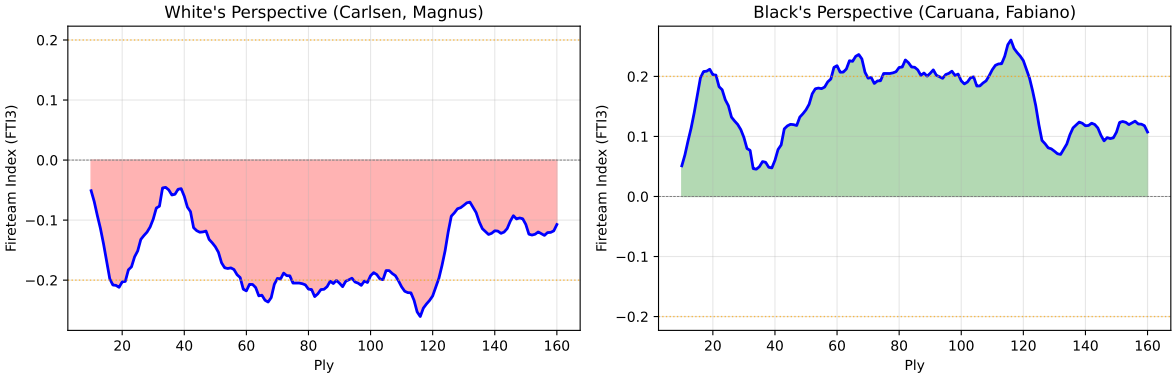
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	4 plies



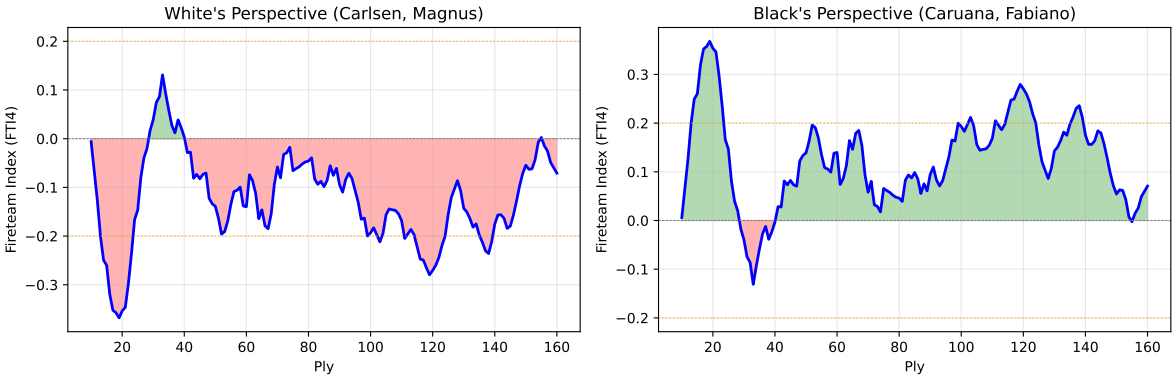
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	11 plies
Caruana, Fabiano	Windowed	WIN	18 plies



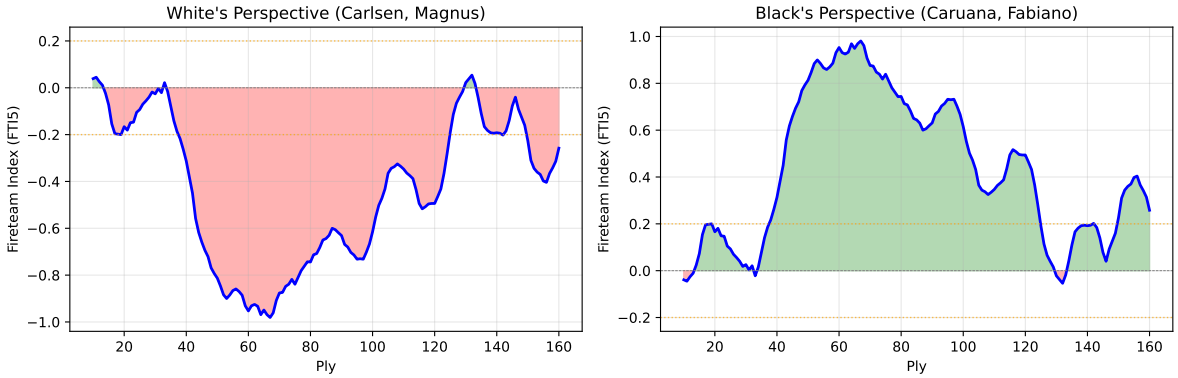
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	WIN	10 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	87 plies
Caruana, Fabiano	Windowed	WIN	87 plies



Interpretation. Black (Caruana, Fabiano) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting White successfully defended or Black failed to convert the advantage.

6.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 213.4 seconds

Chapter 7

Carlsen, Magnus vs Caruana, Fabiano

7.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.18
Round	7
White	Carlsen, Magnus
Black	Caruana, Fabiano
Result	1/2-1/2
Opening	??? – Unknown

7.2 Player Statistics

7.2.1 Carlsen, Magnus (White)

- Total moves: 40
- Accuracy: 98.1%
- Average centipawn loss: 3.9
- Best/Excellent moves: 26 / 5
- Good moves: 9
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

7.2.2 Caruana, Fabiano (Black)

- Total moves: 39
- Accuracy: 98.8%
- Average centipawn loss: 2.5

- Best/Excellent moves: 26 / 7
- Good moves: 6
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

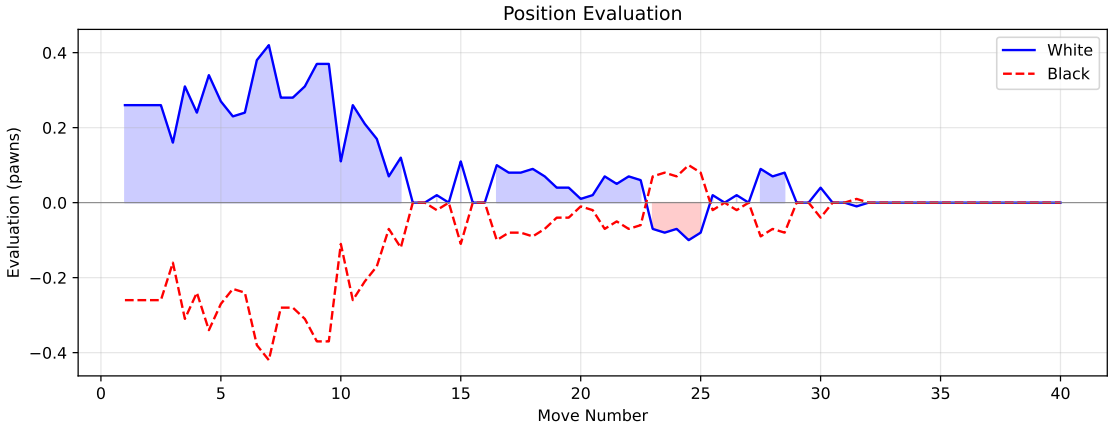
7.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

7.3.1 Position Evaluation Over Time

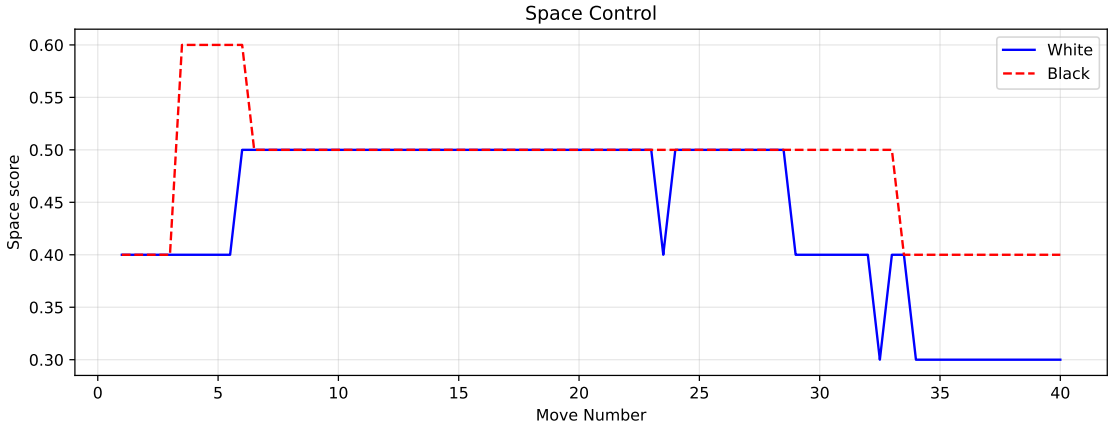
Game Character Classification. Evaluation range: minimum $m_1 = -0.10$, maximum $m_2 = +0.42$, spread $d = m_2 - m_1 = 0.52$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



7.3.2 Space Control

- White average: 0.44 (range: 0.30 to 0.50)
- Black average: 0.48 (range: 0.40 to 0.60)



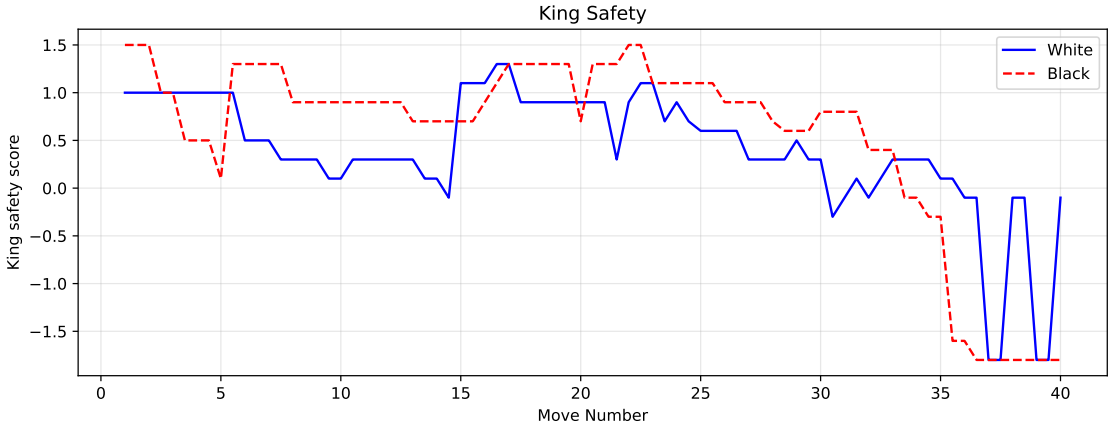
7.3.3 Mobility (Piece Activity)

- White average: 4.19 (range: 1.50 to 6.90)
- Black average: 3.92 (range: 1.40 to 6.30)



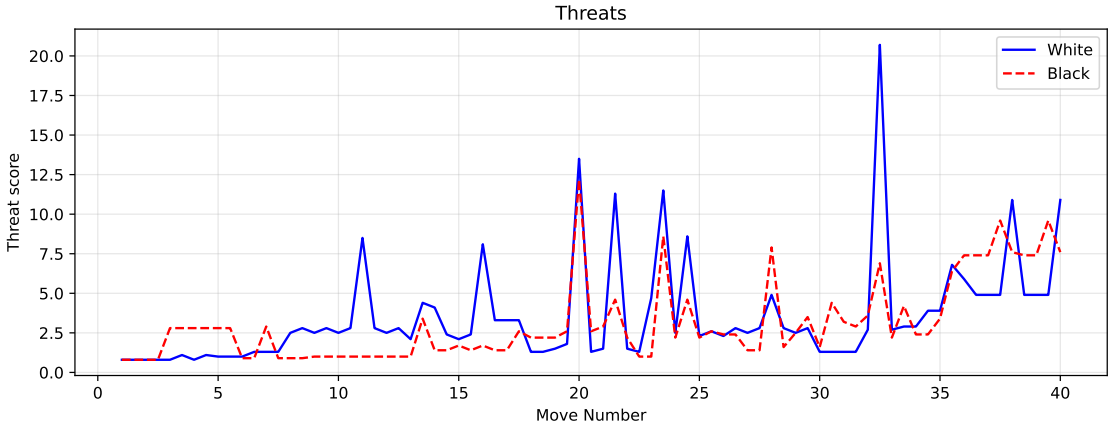
7.3.4 King Safety

- White average: 0.40
- Black average: 0.56



7.3.5 Threats

- White average: 3.58 (range: 0.80 to 20.70)
- Black average: 3.12 (range: 0.80 to 12.20)
- Average advantage: 0.45 (White maintained more threats)



7.4 Annotated Game

1. d4 Nf6 2. Nf3 d5 3. c4 e6 4. Nc3 Be7 5. Bf4 O-O 6. e3 c5 7. dxc5! Bxc5! 8. Qc2! Nc6 9. a3! Qa5 10. Nd2 Qd8 11. Nb3 Bb6! 12. Be2 Qe7 13. Bg5 dxc4! 14. Nd2! Ne5! 15. O-O! Bd7! 16. Bf4! Ng6! 17. Bg3! Bc6! 18. Nxc4! Bc7! 19. Rfd1! Rfd8! 20. Rxd8+ Rxd8! 21. Rd1 Rxd1+ 22. Qxd1! Nd5! 23. Qd4 Nxc3 24. Qxc3! Bxg3! 25. hxg3! Qd7 26. Bd3 b6 27. f3 Bb7 28. Bxg6! hxg6! 29. e4 Qc7! 30. e5 Qc5+! 31. Kh2 Ba6! 32. Nd6! Qxc3! 33. bxc3! f6 34. f4! Kf8! 35. Kg1 Ke7! 36. Kf2! Kd7! 37. Ke3 Bf1 38. Kf2! Ba6! 39. Ke3! Bf1 40. Kf2!

1/2-1/2

7.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

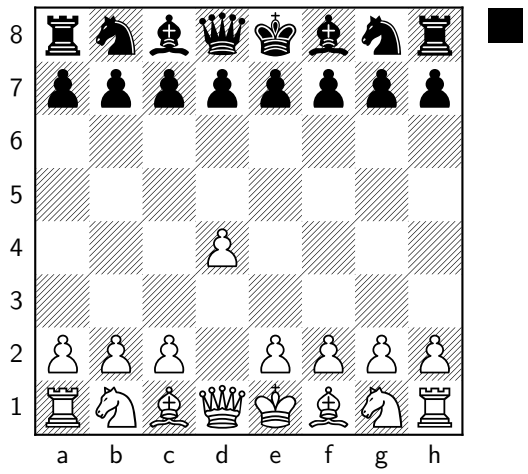
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 1. d4 ★

Biggest evaluation swing(s) (27cp)

Evaluation: +0.26



Metric	White	Black
Space	0.40	0.40
Mobility	3.80	3.30
King Safety	1.00	1.50
Threats	0.80	0.80

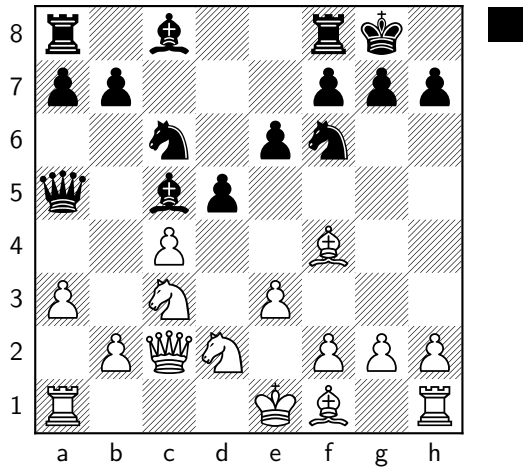
Instead of d4: e4, Nf3

Best continuation: Nf6 c4 e6

Position 2: After 10. Nd2 ★

Biggest evaluation swing(s) (26cp)

Evaluation: +0.11



Metric	White	Black
Space	0.50	0.50
Mobility	5.10	4.70
King Safety	0.10	0.90
Threats	2.50	1.00

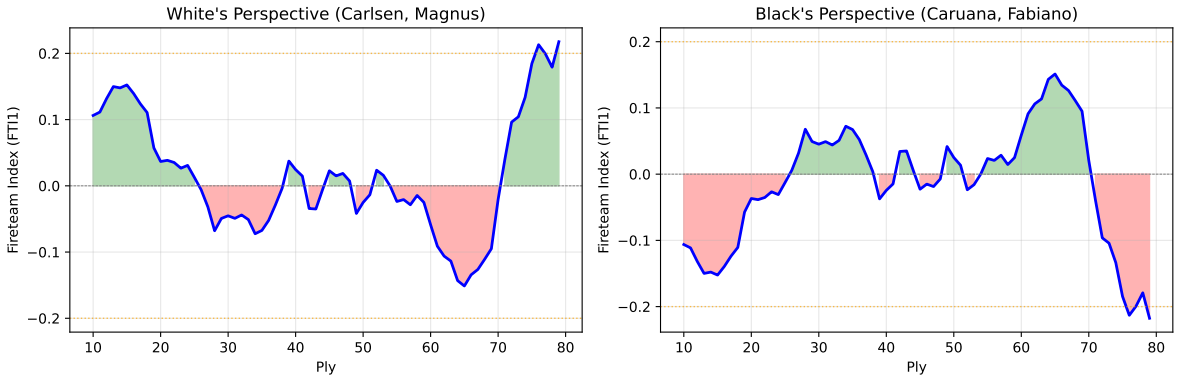
Instead of Nd2: Rd1, Be2 Best continuation: Bb4 cxd5 Nxd5

7.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

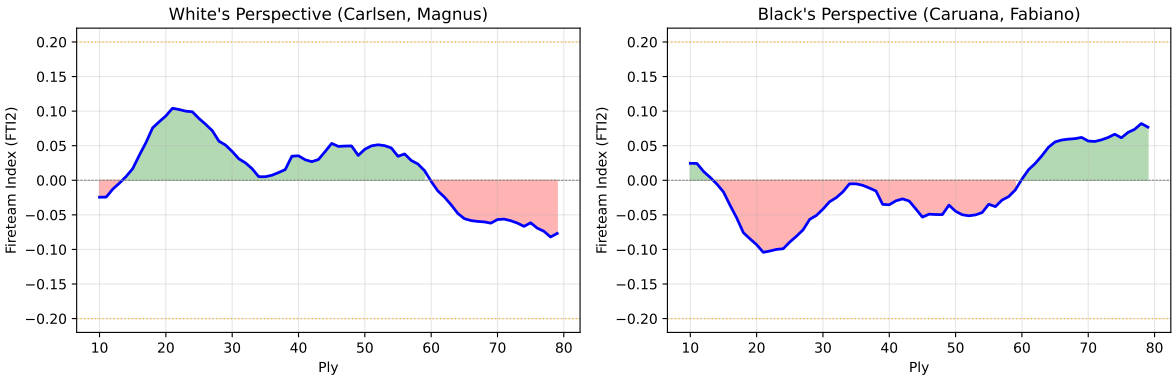
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	1 plies
Caruana, Fabiano	Per-ply	DRAW	2 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



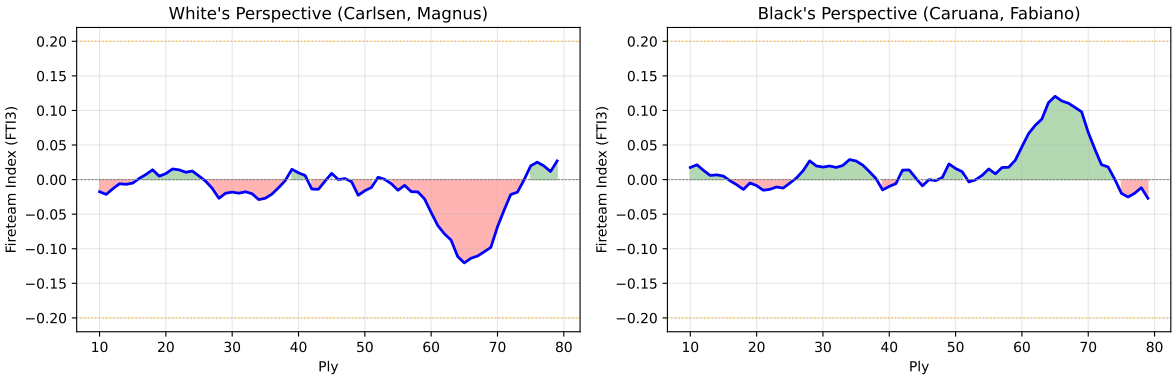
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



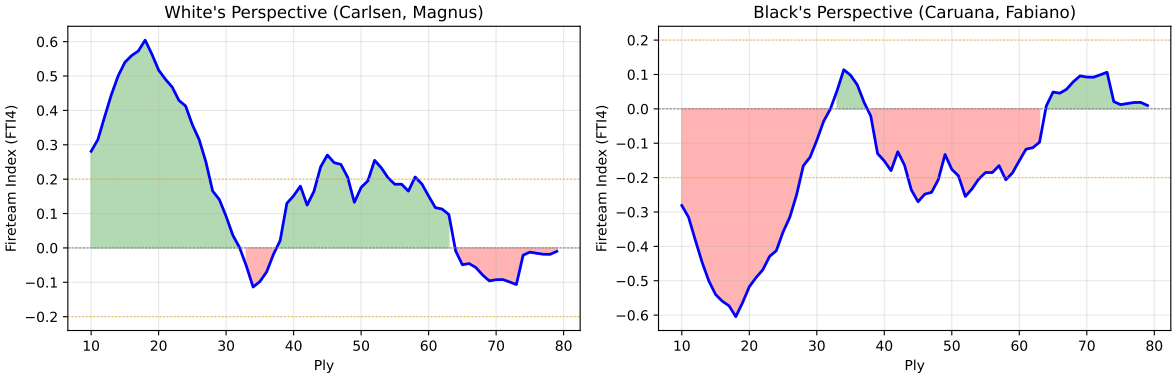
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



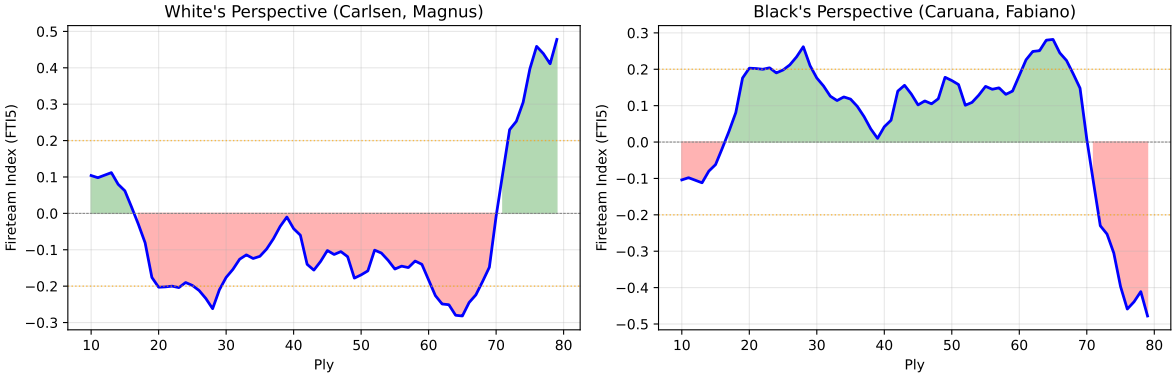
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	4 plies
Carlsen, Magnus	Windowed	WIN	11 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	8 plies
Caruana, Fabiano	Per-ply	DRAW	6 plies
Caruana, Fabiano	Windowed	DRAW	7 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

7.7 Analysis Information

- Engine: Stockfish 16

- Depth: 20
- Analysis time: 140.4 seconds

Chapter 8

Caruana, Fabiano vs Carlsen, Magnus

8.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.19
Round	8
White	Caruana, Fabiano
Black	Carlsen, Magnus
Result	1/2-1/2
Opening	??? – Unknown

8.2 Player Statistics

8.2.1 Caruana, Fabiano (White)

- Total moves: 38
- Accuracy: 96.4%
- Average centipawn loss: 7.3
- Best/Excellent moves: 22 / 8
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 1

8.2.2 Carlsen, Magnus (Black)

- Total moves: 37
- Accuracy: 96.4%
- Average centipawn loss: 7.3

- Best/Excellent moves: 20 / 7
- Good moves: 7
- Inaccuracies: 3
- Mistakes: 0
- Blunders: 0

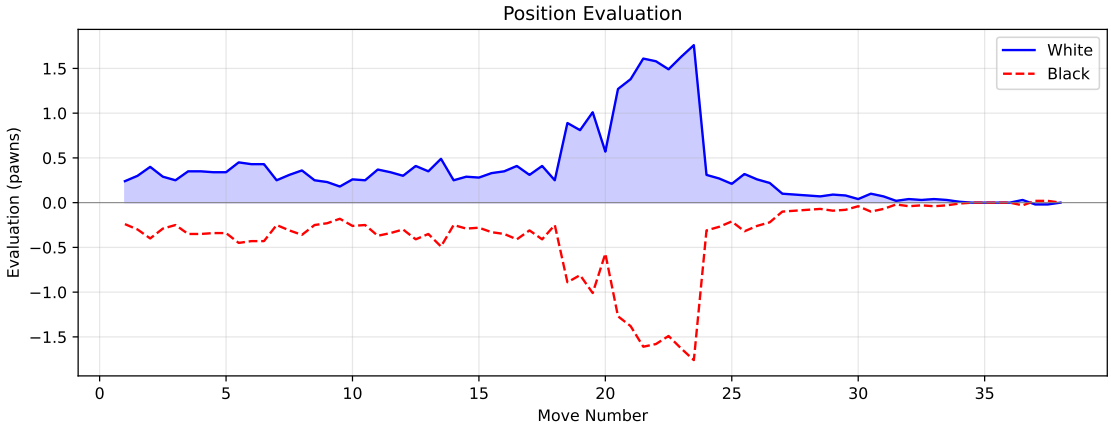
8.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

8.3.1 Position Evaluation Over Time

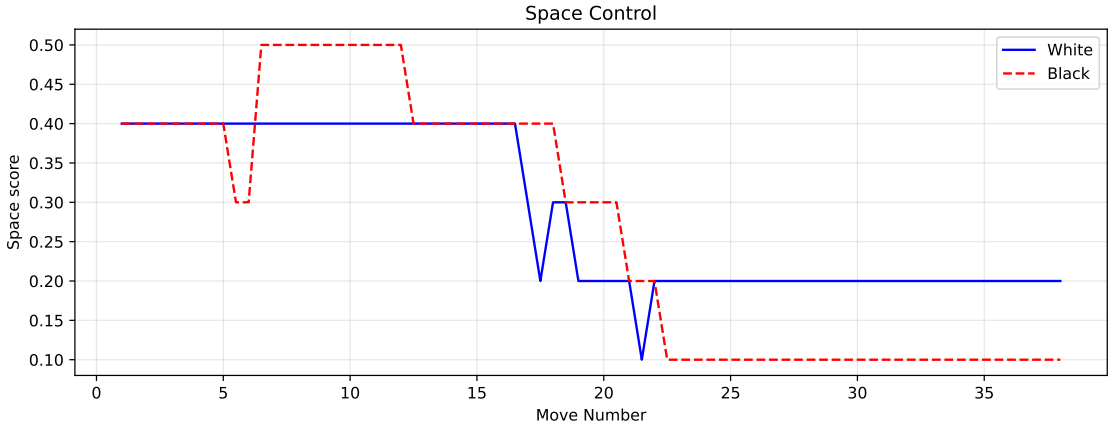
Game Character Classification. Evaluation range: minimum $m_1 = -0.02$, maximum $m_2 = +1.76$, spread $d = m_2 - m_1 = 1.78$.

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.



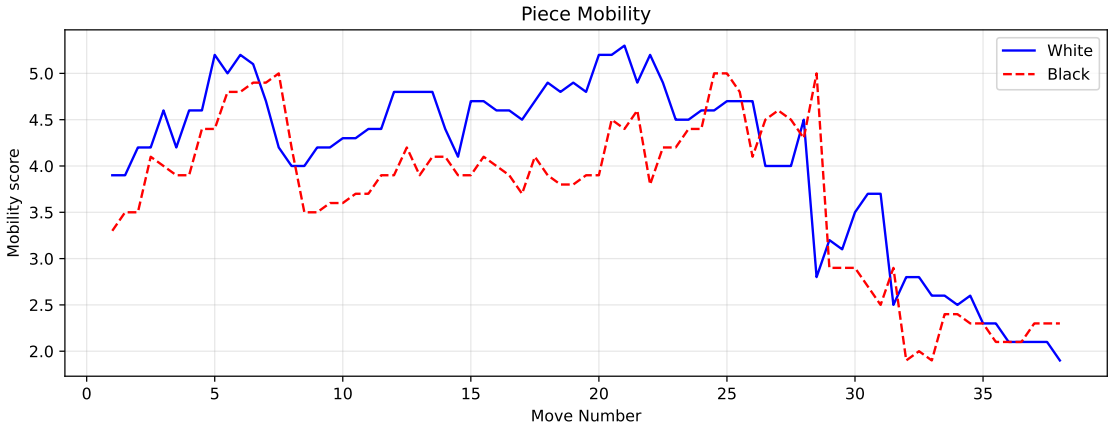
8.3.2 Space Control

- White average: 0.29 (range: 0.10 to 0.40)
- Black average: 0.27 (range: 0.10 to 0.50)



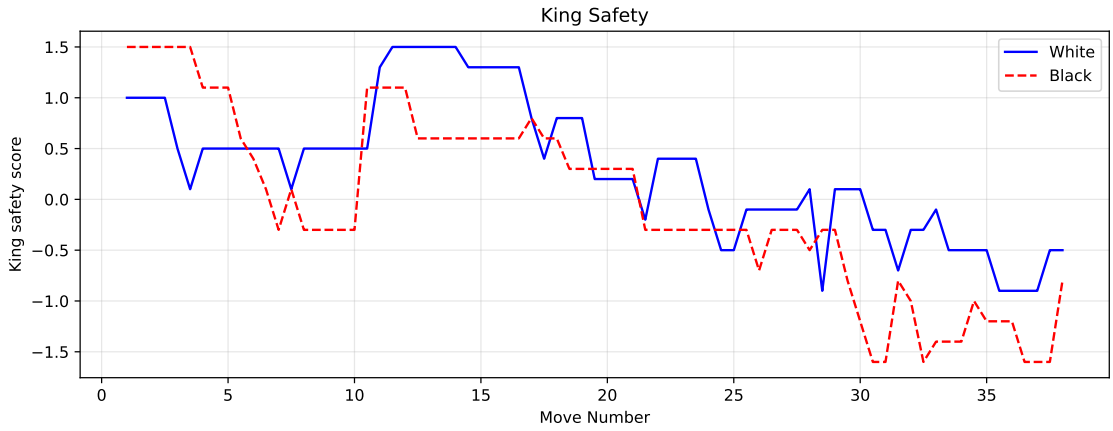
8.3.3 Mobility (Piece Activity)

- White average: 4.07 (range: 1.90 to 5.30)
- Black average: 3.70 (range: 1.90 to 5.00)



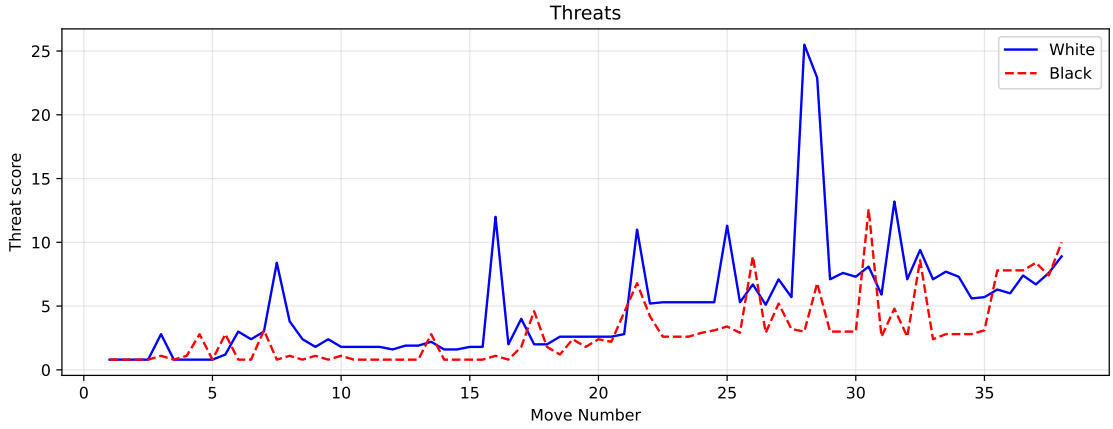
8.3.4 King Safety

- White average: 0.31
- Black average: -0.04



8.3.5 Threats

- White average: 4.95 (range: 0.80 to 25.50)
- Black average: 2.97 (range: 0.80 to 12.60)
- Average advantage: 1.98 (White maintained more threats)



8.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 e5 6. Ndb5 d6 7. Nd5 Nxd5! 8. exd5! Nb8! 9. a4! Be7! 10. Be2 O-O 11. O-O! Nd7 12. Bd2! f5 13. a5! a6 14. Na3 e4 15. Nc4! Ne5! 16. Nb6 Rb8! 17. f4 exf3 18. Bxf3 g5?! 19. c4! f4 20. Bc3?! Bf5?! 21. c5! Nxf3+?! 22. Qxf3! dxc5! 23. Rad1 Bd6 24. h3?? *White loses 135cp. Consider instead: Qh5, Nc4.*

24...Qe8! 25. Nc4! Qg6! 26. Nxd6 Qxd6 27. h4! gxh4! 28. Qxf4 Qxf4! 29. Rxf4! h5! 30. Re1! Bg4! 31. Rf6! Rxf6! 32. Bxf6! Kf7 33. Bxh4! Re8! 34. Rf1+ Kg8! 35. Rf6 Re2 36. Rg6+ Kf8 37. d6 Rd2! 38. Rg5!

1/2-1/2

8.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

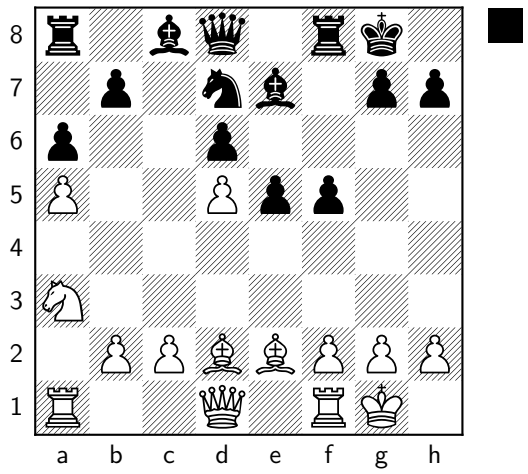
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 14. Na3 ★

Biggest evaluation swing(s) (24cp)

Evaluation: +0.25



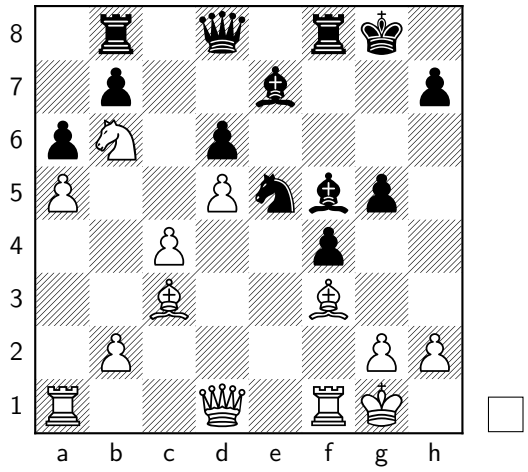
Metric	White	Black
Space	0.40	0.40
Mobility	4.40	4.10
King Safety	1.50	0.60
Threats	1.60	0.80

Instead of Na3: Nc3 Best continuation: e4 Nc4 Ne5

Position 2: After 20...Bf5 ★

Biggest evaluation swing(s) (70cp)

Evaluation: +1.27

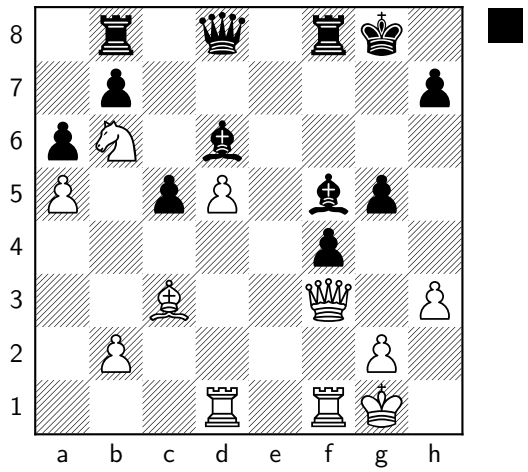


Metric	White	Black
Space	0.20	0.30
Mobility	5.20	4.50
King Safety	0.20	0.30
Threats	2.60	2.20

Instead of Bf5: Qc7 Best continuation: c5 Bf6 c6

Position 3: After 24. h3

White blunders (-145cp)
Evaluation: +0.31



Metric	White	Black
Space	0.20	0.10
Mobility	4.60	4.40
King Safety	-0.10	-0.30
Threats	5.30	2.90

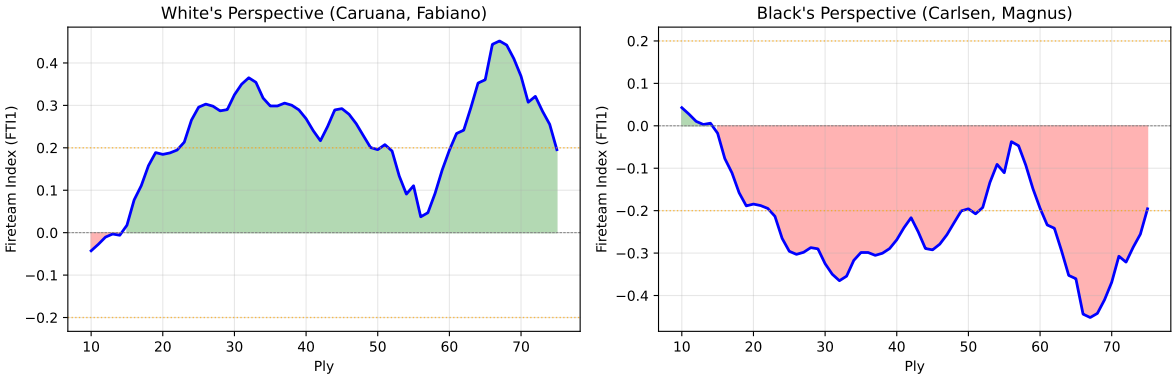
Instead of h3: Qh5, Nc4 Best continuation: Qe8 Nc4 Qg6

8.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

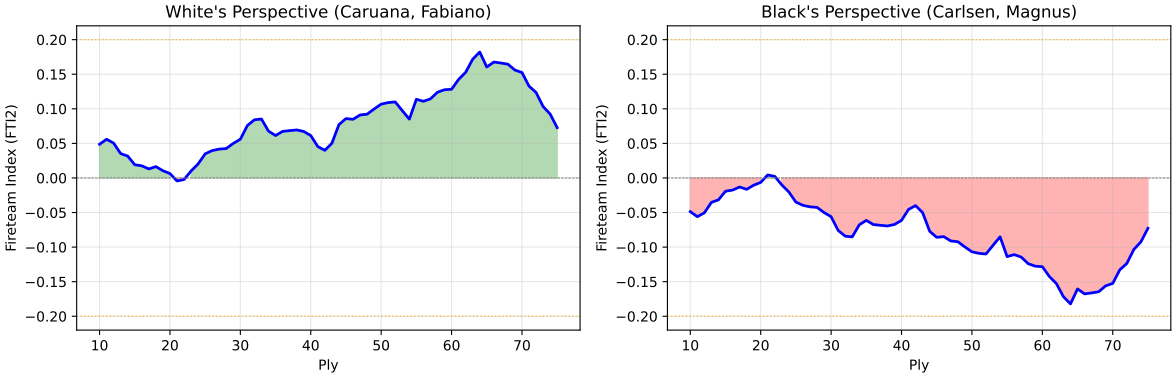
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	13 plies
Caruana, Fabiano	Windowed	WIN	27 plies
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



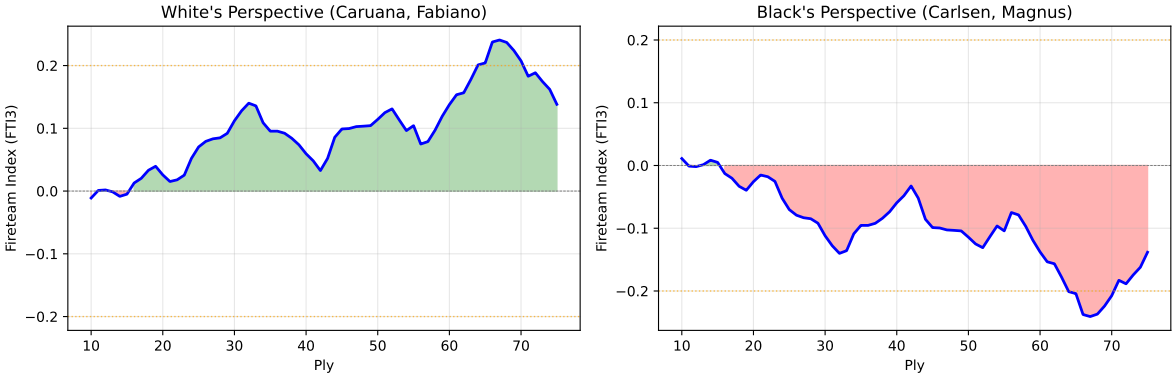
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



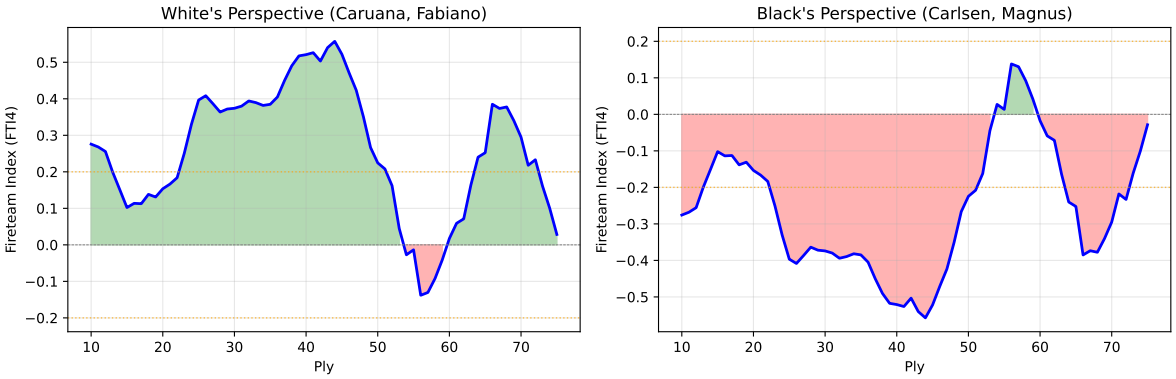
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	7 plies
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



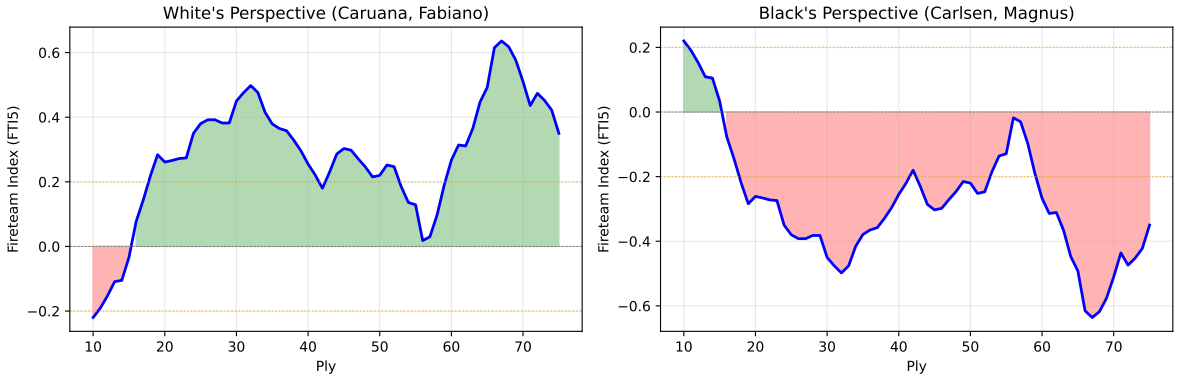
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	17 plies
Caruana, Fabiano	Windowed	WIN	29 plies
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	12 plies
Caruana, Fabiano	Windowed	WIN	24 plies
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



Interpretation. White (Caruana, Fabiano) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

8.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 155.0 seconds

Chapter 9

Carlsen, Magnus vs Caruana, Fabiano

9.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.21
Round	9
White	Carlsen, Magnus
Black	Caruana, Fabiano
Result	1/2-1/2
Opening	??? – Unknown

9.2 Player Statistics

9.2.1 Carlsen, Magnus (White)

- Total moves: 56
- Accuracy: 97.8%
- Average centipawn loss: 4.4
- Best/Excellent moves: 35 / 12
- Good moves: 9
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

9.2.2 Caruana, Fabiano (Black)

- Total moves: 56
- Accuracy: 97.8%
- Average centipawn loss: 4.5

- Best/Excellent moves: 37 / 8
- Good moves: 10
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

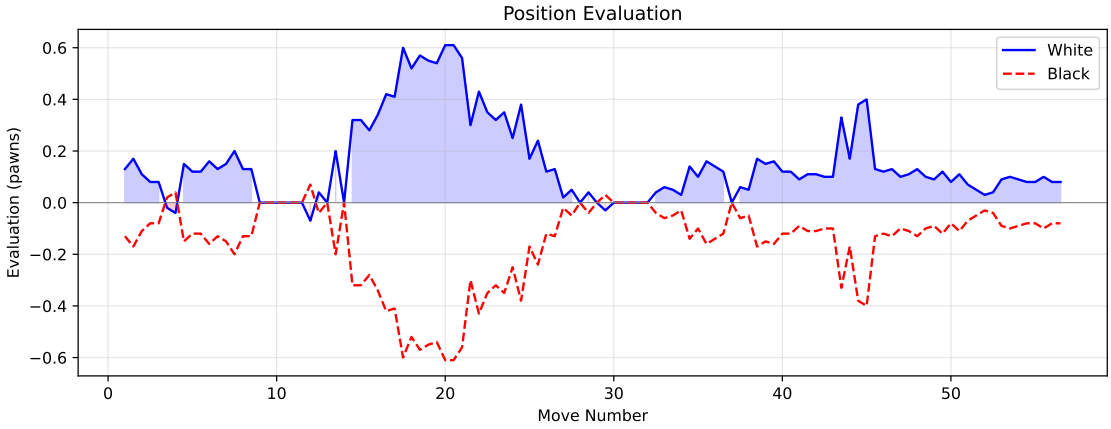
9.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

9.3.1 Position Evaluation Over Time

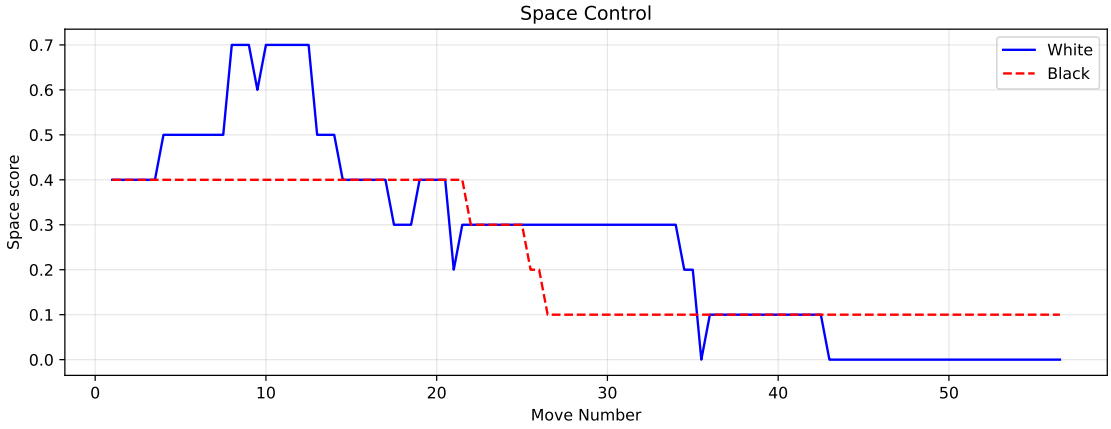
Game Character Classification. Evaluation range: minimum $m_1 = -0.07$, maximum $m_2 = +0.61$, spread $d = m_2 - m_1 = 0.68$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



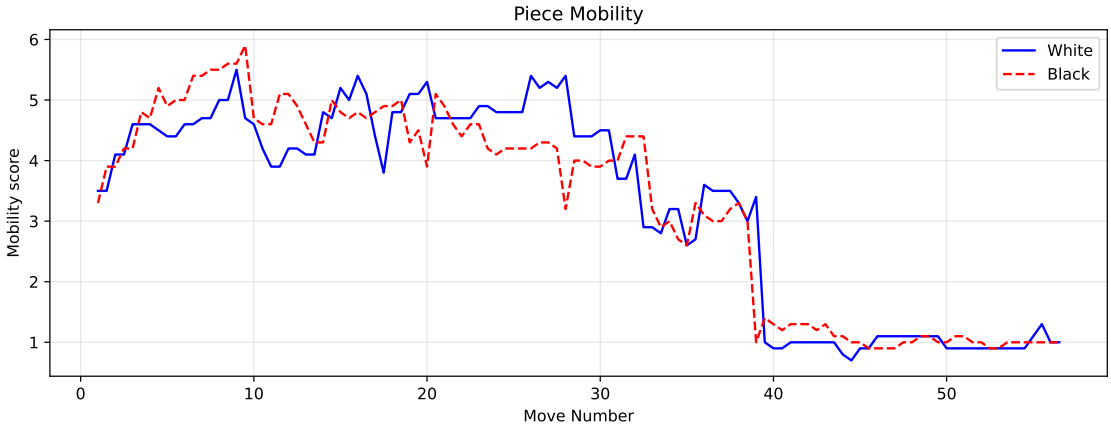
9.3.2 Space Control

- White average: 0.26 (range: 0.00 to 0.70)
- Black average: 0.23 (range: 0.10 to 0.40)



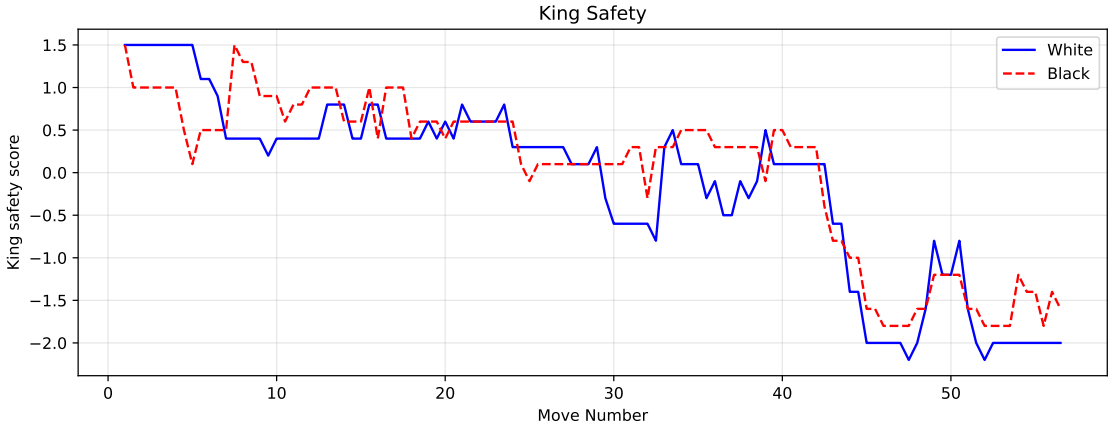
9.3.3 Mobility (Piece Activity)

- White average: 3.29 (range: 0.70 to 5.50)
- Black average: 3.28 (range: 0.90 to 5.90)



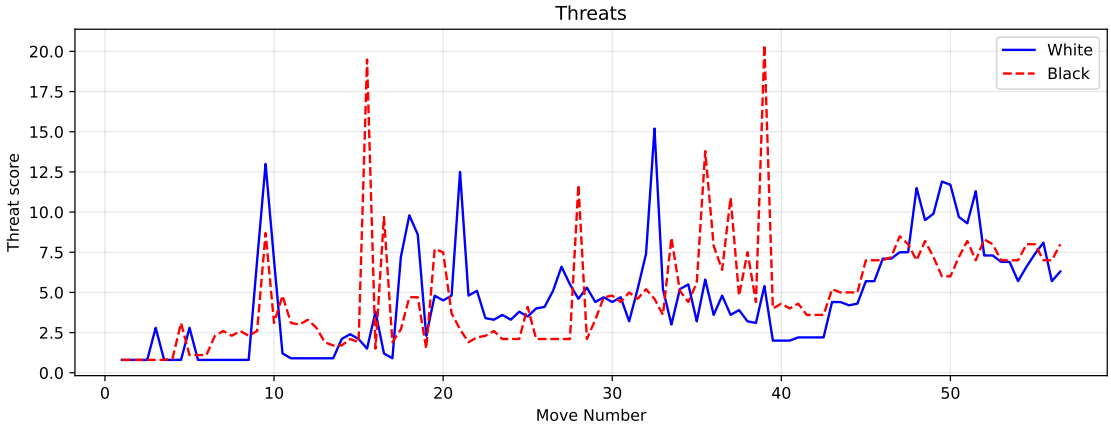
9.3.4 King Safety

- White average: -0.13
- Black average: 0.03



9.3.5 Threats

- White average: 4.60 (range: 0.80 to 15.20)
- Black average: 4.83 (range: 0.80 to 20.40)
- Average advantage: -0.23 (Black maintained more threats)



9.4 Annotated Game

1. c4 e5 2. Nc3 Nf6 3. Nf3 Nc6 4. g3 d5 5. cxd5 Nxd5 6. Bg2 Bc5 7. O-O! O-O! 8. d3! Re8 9. Bg5 Nxc3! 10. bxc3! f6 11. Bc1 Be6! 12. Bb2 Bb6 13. d4 Bd5 14. Qc2 exd4?! 15. cxd4! Be4! 16. Qb3+ Bd5 17. Qd1! Bxf3 18. Qb3+ Kh8 19. Bxf3 Nxd4! 20. Bxd4! Qxd4! 21. e3! Qe5! 22. Bxb7! Rad8! 23. Rad1 Qe7 24. h4 g6 25. h5 gxh5! 26. Qc4 f5! 27. Bf3 h4! 28. Rxd8 Rxd8! 29. gxh4 Rg8+ 30. Kh1! Qf6 31. Qf4! Bc5! 32. Rg1! Rxg1+ 33. Kxg1! Bd6! 34. Qa4! f4 35. Qxa7! fxe3! 36. Qxe3 Qxh4! 37. a4 Qf6 38. Bd1! Qe5 39. Qxe5+! Bxe5! 40. a5 Kg7! 41. a6! Bd4! 42. Kg2! Kf6 43. f4! Bb6 44. Kf3 h6 45. Ke4! Ba7 46. Bg4 Bg1 47. Kd5 Bb6 48. Kc6! Be3 49. Kb7 Bb6 50. Bh3 Be3! 51. Kc6 Bb6 52. Kd5 Ba7 53. Ke4! Bb6 54. Bf1 Ke6 55. Bc4+! Kf6! 56. Bd3 Ke6

1/2-1/2

9.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

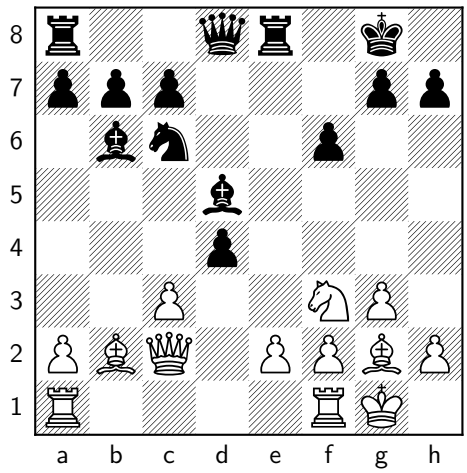
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 14...exd4 ★

Biggest evaluation swing(s) (32cp)

Evaluation: +0.32



Metric	White	Black
Space	0.40	0.40
Mobility	4.70	5.00
King Safety	0.40	0.60
Threats	2.40	2.10

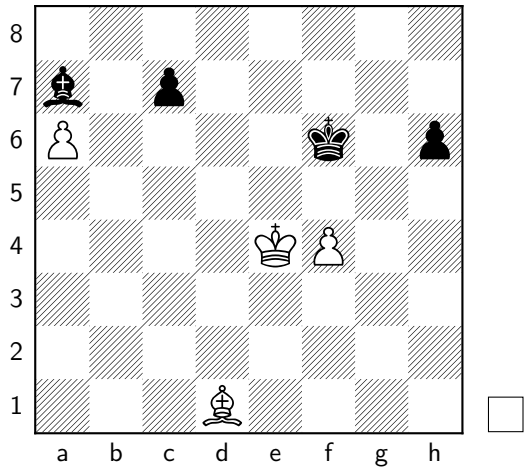
Instead of exd4: e4, Bc4

Best continuation: cxd4 Be4 Qb3+

Position 2: After 45...Ba7 ★

Biggest evaluation swing(s) (27cp)

Evaluation: +0.13



Metric	White	Black
Space	0.00	0.10
Mobility	0.90	0.90
King Safety	-2.00	-1.60
Threats	5.70	7.00

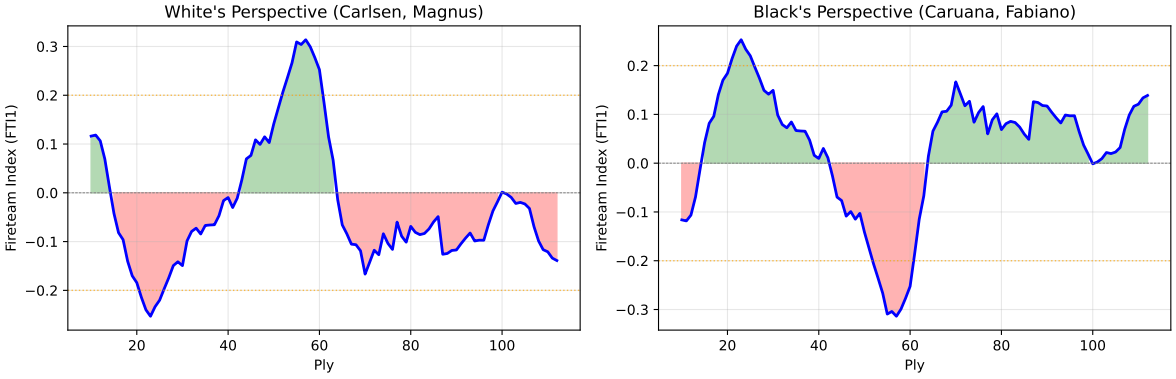
Instead of Ba7: Bc5, Bg1 Best continuation: Kf3 Bb8 Ke4

9.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

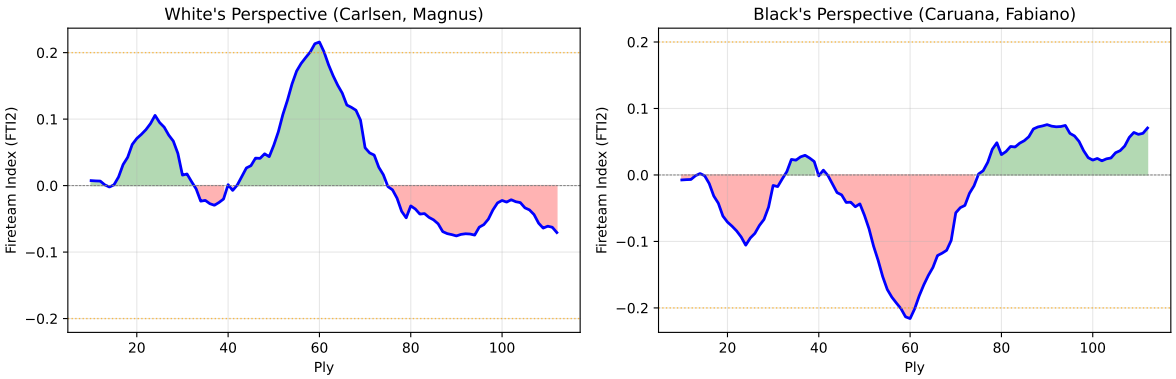
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	8 plies
Carlsen, Magnus	Windowed	DRAW	9 plies
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	5 plies



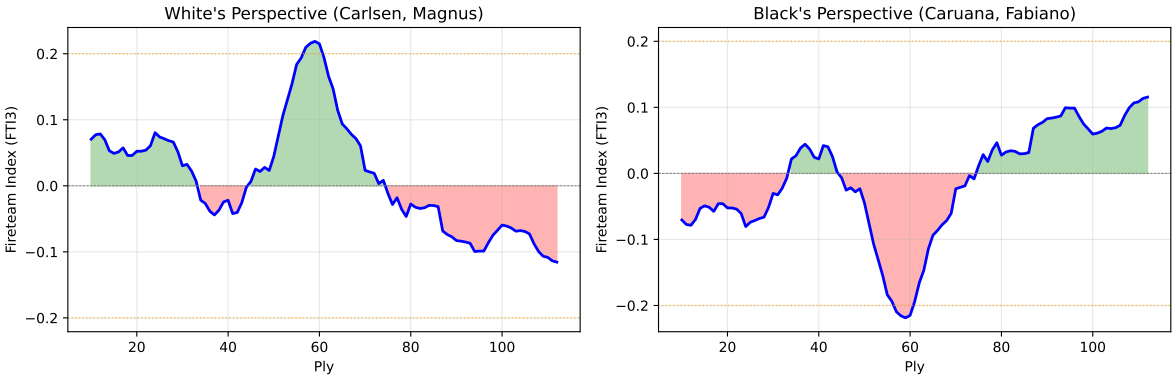
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	6 plies
Carlsen, Magnus	Windowed	DRAW	4 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



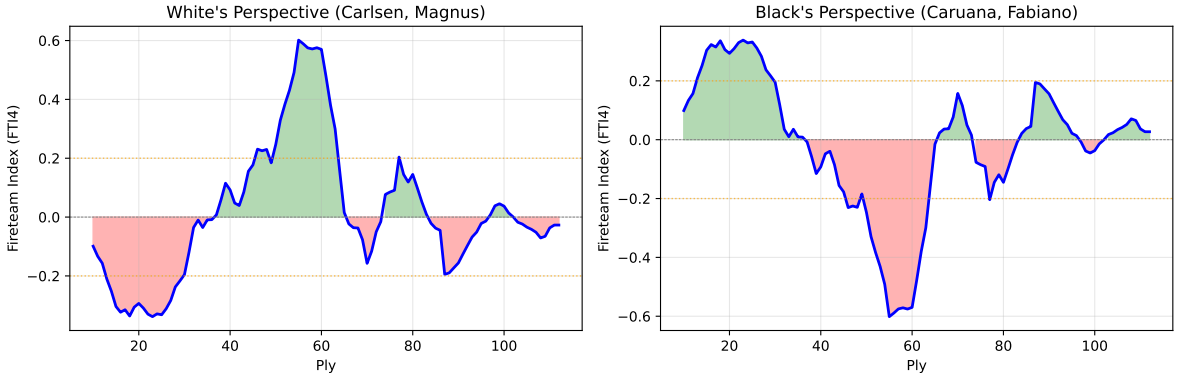
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	DRAW	4 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



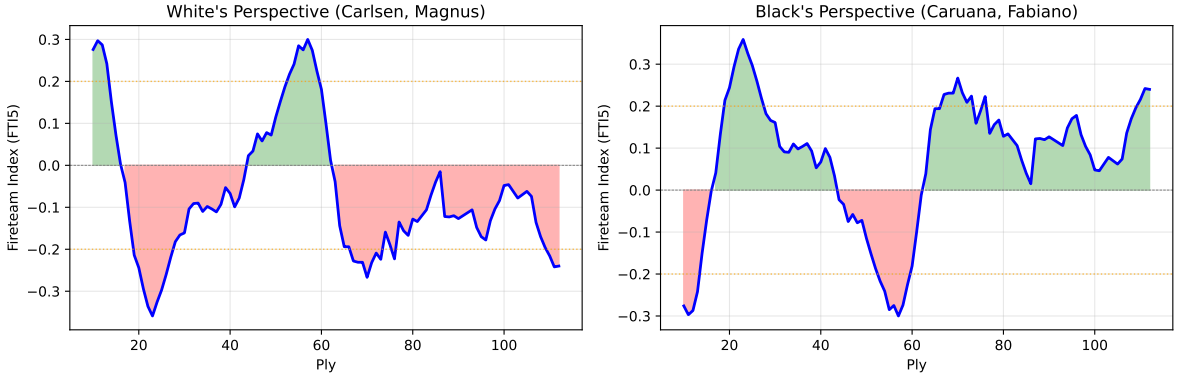
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	15 plies
Carlsen, Magnus	Windowed	WIN	14 plies
Caruana, Fabiano	Per-ply	DRAW	6 plies
Caruana, Fabiano	Windowed	WIN	13 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	8 plies
Carlsen, Magnus	Windowed	DRAW	7 plies
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	9 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

9.7 Analysis Information

- Engine: Stockfish 16

- Depth: 20
- Analysis time: 180.1 seconds

Chapter 10

Caruana, Fabiano vs Carlsen, Magnus

10.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.22
Round	10
White	Caruana, Fabiano
Black	Carlsen, Magnus
Result	1/2-1/2
Opening	??? – Unknown

10.2 Player Statistics

10.2.1 Caruana, Fabiano (White)

- Total moves: 54
- Accuracy: 96.1%
- Average centipawn loss: 7.9
- Best/Excellent moves: 33 / 8
- Good moves: 10
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 1

10.2.2 Carlsen, Magnus (Black)

- Total moves: 53
- Accuracy: 96.2%
- Average centipawn loss: 7.7

- Best/Excellent moves: 33 / 8
- Good moves: 9
- Inaccuracies: 1
- Mistakes: 2
- Blunders: 0

10.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

10.3.1 Position Evaluation Over Time

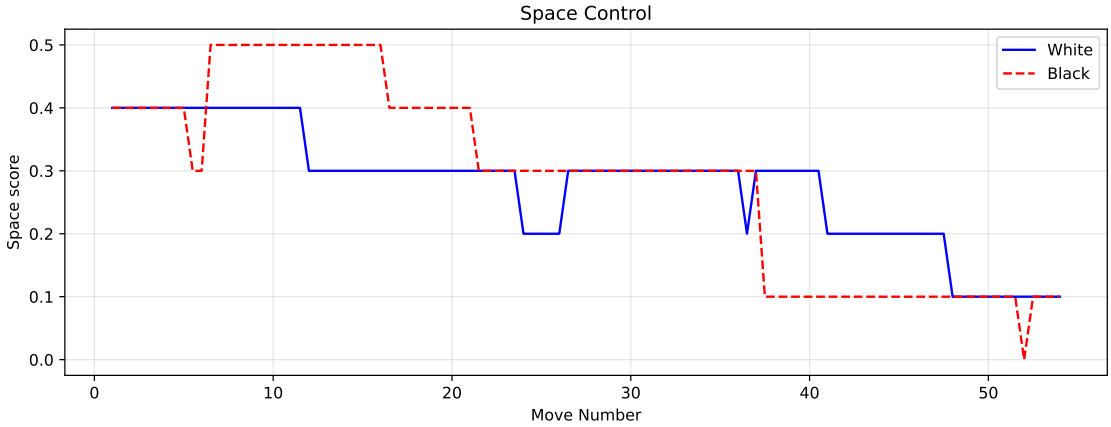
Game Character Classification. Evaluation range: minimum $m_1 = -0.21$, maximum $m_2 = +1.12$, spread $d = m_2 - m_1 = 1.33$.

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.



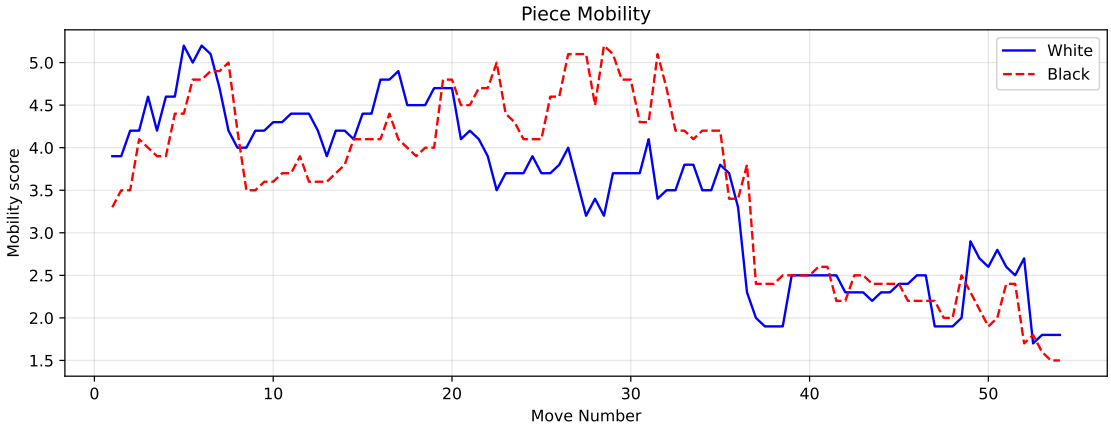
10.3.2 Space Control

- White average: 0.28 (range: 0.10 to 0.40)
- Black average: 0.29 (range: 0.00 to 0.50)



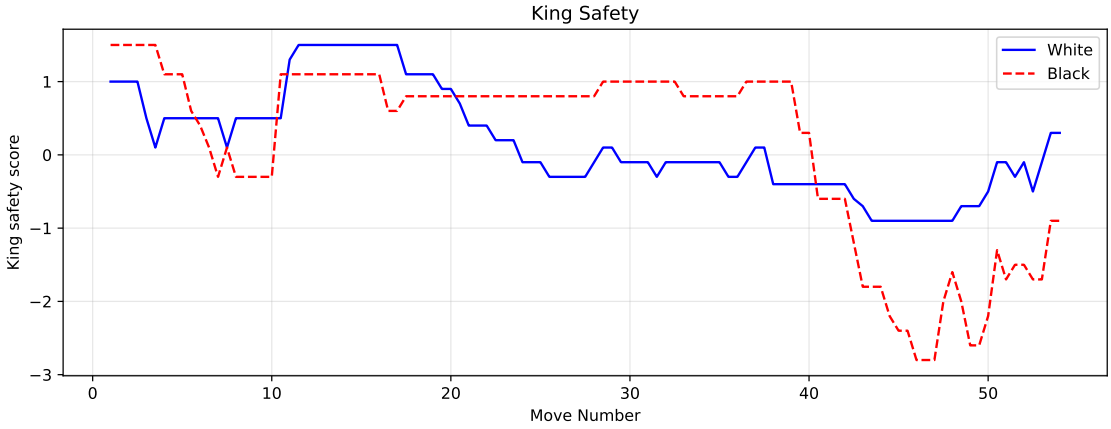
10.3.3 Mobility (Piece Activity)

- White average: 3.49 (range: 1.70 to 5.20)
- Black average: 3.58 (range: 1.50 to 5.20)



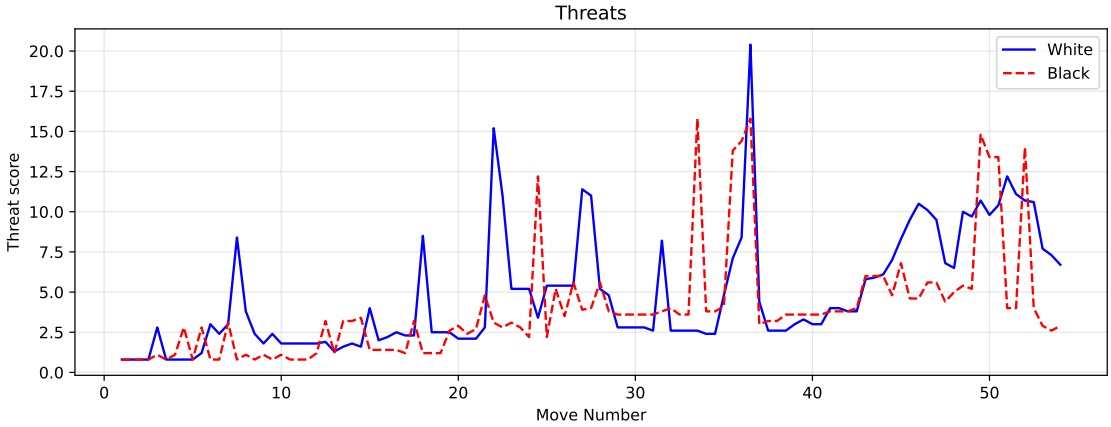
10.3.4 King Safety

- White average: 0.17
- Black average: 0.15



10.3.5 Threats

- White average: 4.82 (range: 0.80 to 20.40)
- Black average: 3.95 (range: 0.80 to 15.80)
- Average advantage: 0.86 (White maintained more threats)



10.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 e5 6. Ndb5 d6 7. Nd5! Nxd5 8. exd5! Nb8! 9. a4! Be7! 10. Be2! O-O! 11. O-O Nd7 12. b4 a6! 13. Na3! a5?! 14. bxa5! Rxa5 15. Nc4 Ra8! 16. Be3! f5 17. a5! f4 18. Bb6! Qe8! 19. Ra3?! Qg6 20. Bc7?! e4! 21. Kh1! b5 22. Nb6 Nxb6! 23. Bxb6 Qg5? *Black loses 118cp. Consider instead: b4, Bd7, Qh6.*

24. g3?? *White loses 130cp. Consider instead: Bxb5.*

24...b4 25. Rb3! Bh3 26. Rg1 f3 27. Bf1 Bxf1! 28. Qxf1! Qxd5! 29. Rxb4! Qe6! 30. Rb5 Bd8 31. Qe1! Bxb6 32. axb6! Rab8! 33. Qe3! Qc4 34. Rb2 Rb7 35. Rd1 Qe2 36. Re1 Qxe3! 37. Rxe3! d5 38. h4 Rc8 39. Ra3! Kf7 40. Kh2 Ke6! 41. g4! Rc6! 42. Ra6! Ke5! 43. Kg3! h6 44. h5! Kd4? *Black loses 60cp. Consider instead: g5, g6, Ke6.*

45. Rb5 Rd6! 46. Ra4+! Ke5! 47. Rab4 Ke6! 48. c4 dxc4! 49. Rxc4! Rdxb6! 50. Rxe4+! Kf7! 51. Rf5+! Rf6! 52. Rxf6+! Kxf6 53. Kxf3 Kf7! 54. Kg3

1/2-1/2

10.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

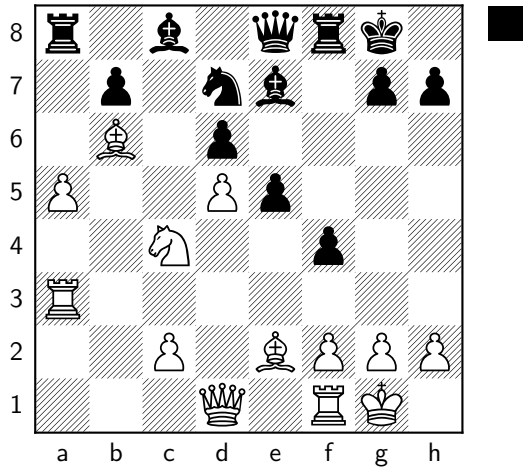
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 19. Ra3 ★

Biggest evaluation swing(s) (50cp)

Evaluation: +0.10



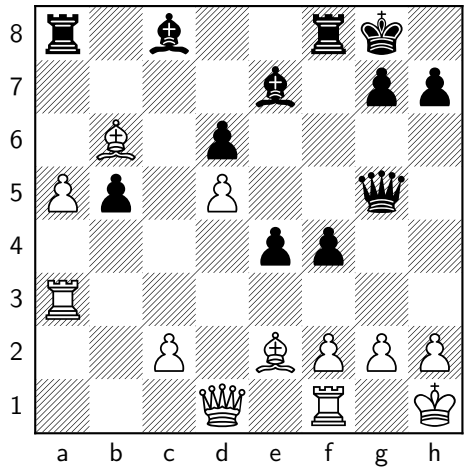
Metric	White	Black
Space	0.30	0.40
Mobility	4.70	4.00
King Safety	1.10	0.80
Threats	2.50	1.20

Instead of Ra3: Re1, Bc7, Kh1 Best continuation: Qg6 Kh1 Bd8

Position 2: After 23...Qg5

Critical moment

Evaluation: +1.12



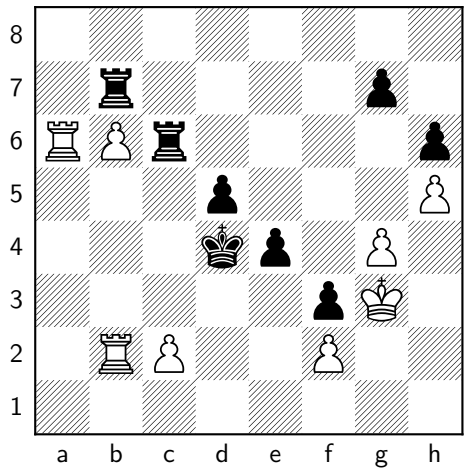
Metric	White	Black
Space	0.30	0.30
Mobility	3.70	4.30
King Safety	0.20	0.80
Threats	5.20	2.80

Instead of Qg5: b4, Bd7, Qh6 Best continuation: Bxb5 Rf6 Re1

Position 3: After 44...Kd4 ★

Biggest evaluation swing(s) (60cp)

Evaluation: +0.60



Metric	White	Black
Space	0.20	0.10
Mobility	2.30	2.40
King Safety	-0.90	-2.20
Threats	7.00	4.80

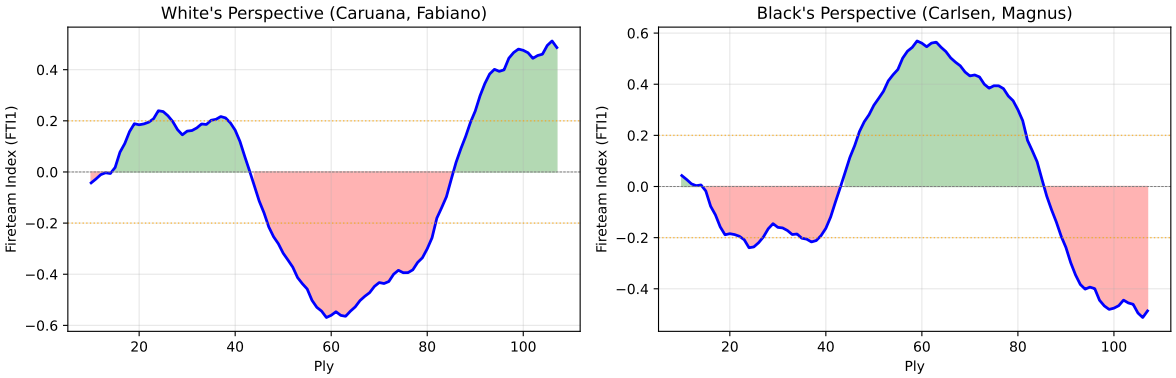
Instead of Kd4: g5, g6, Ke6 Best continuation: Rb5 Rd6 Ra4+

10.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

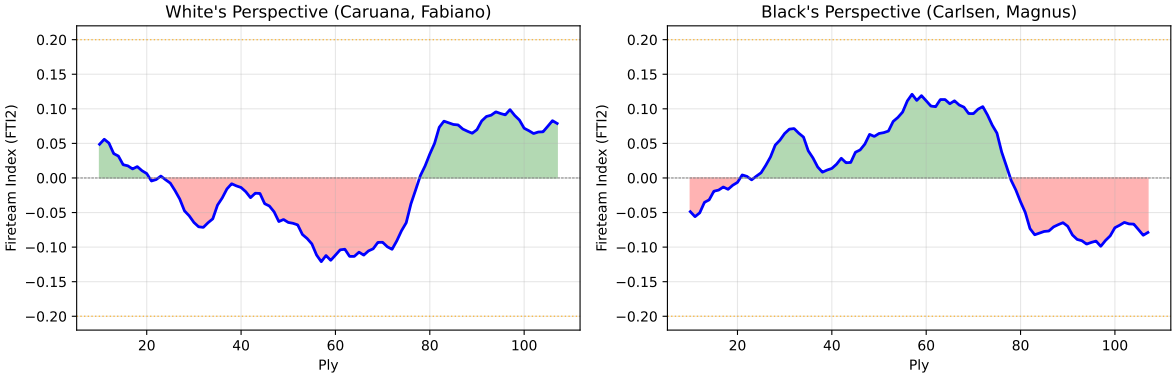
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	12 plies
Caruana, Fabiano	Windowed	WIN	18 plies
Carlsen, Magnus	Per-ply	WIN	34 plies
Carlsen, Magnus	Windowed	WIN	35 plies



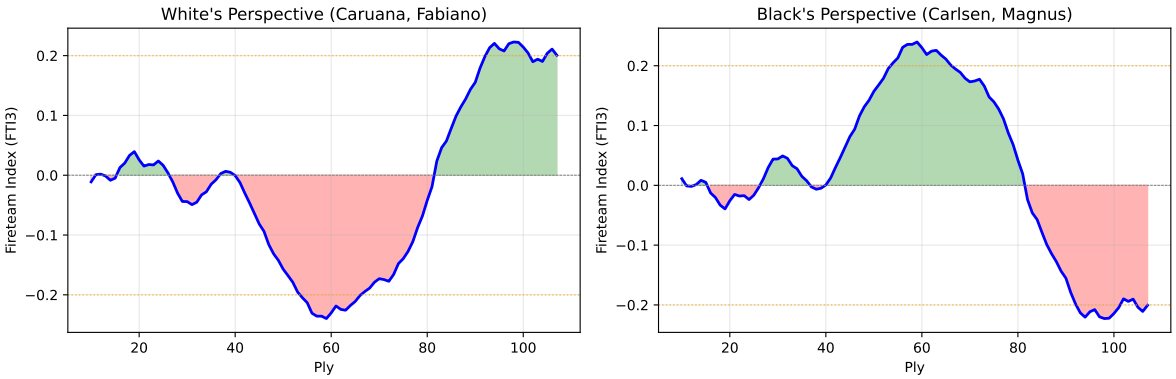
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



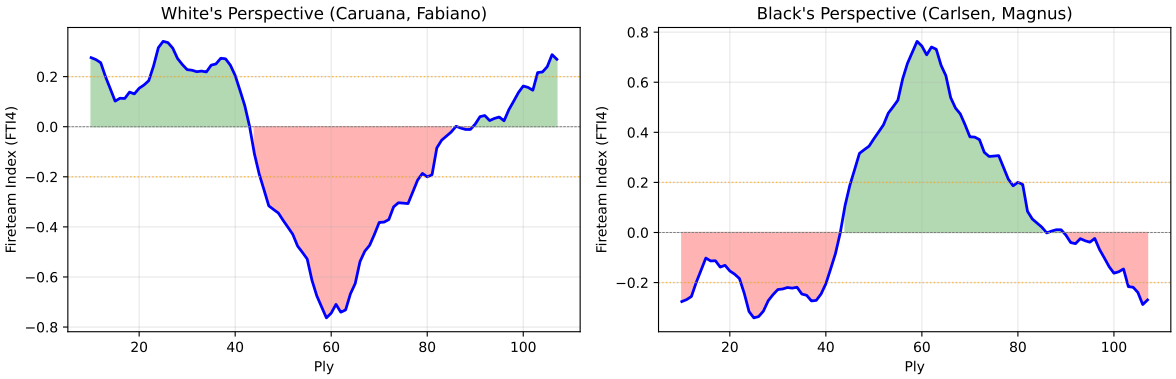
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	6 plies
Caruana, Fabiano	Windowed	DRAW	9 plies
Carlsen, Magnus	Per-ply	WIN	10 plies
Carlsen, Magnus	Windowed	WIN	13 plies



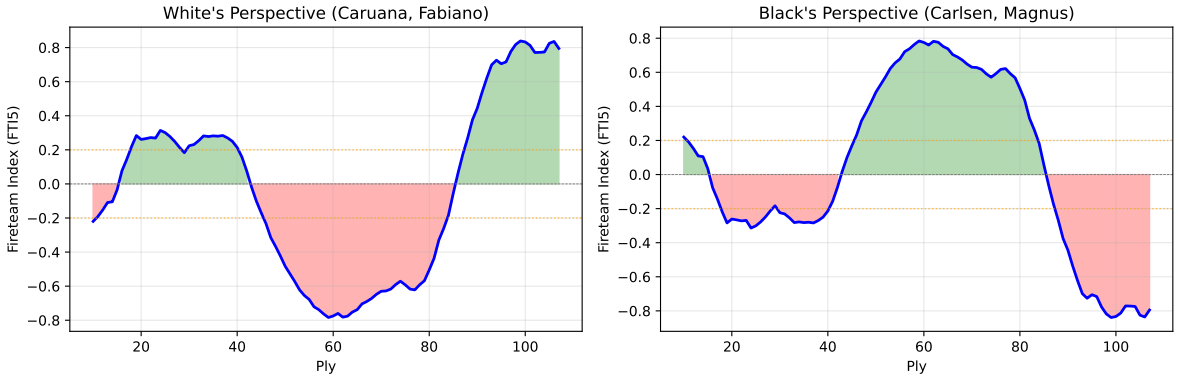
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	8 plies
Caruana, Fabiano	Windowed	WIN	18 plies
Carlsen, Magnus	Per-ply	WIN	21 plies
Carlsen, Magnus	Windowed	WIN	33 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	24 plies
Caruana, Fabiano	Windowed	WIN	20 plies
Carlsen, Magnus	Per-ply	WIN	39 plies
Carlsen, Magnus	Windowed	WIN	38 plies



Interpretation. Both players achieved sustained positional dominance at different points in the game. The FTI1 Index predicted wins for both sides, reflecting a hard-fought battle where momentum shifted. The draw is a reasonable outcome given the mutual winning chances.

10.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 212.0 seconds

Chapter 11

Carlsen, Magnus vs Caruana, Fabiano

11.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.24
Round	11
White	Carlsen, Magnus
Black	Caruana, Fabiano
Result	1/2-1/2
Opening	??? – Unknown

11.2 Player Statistics

11.2.1 Carlsen, Magnus (White)

- Total moves: 55
- Accuracy: 98.3%
- Average centipawn loss: 3.5
- Best/Excellent moves: 38 / 10
- Good moves: 7
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

11.2.2 Caruana, Fabiano (Black)

- Total moves: 55
- Accuracy: 98.8%
- Average centipawn loss: 2.3

- Best/Excellent moves: 43 / 5
- Good moves: 7
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

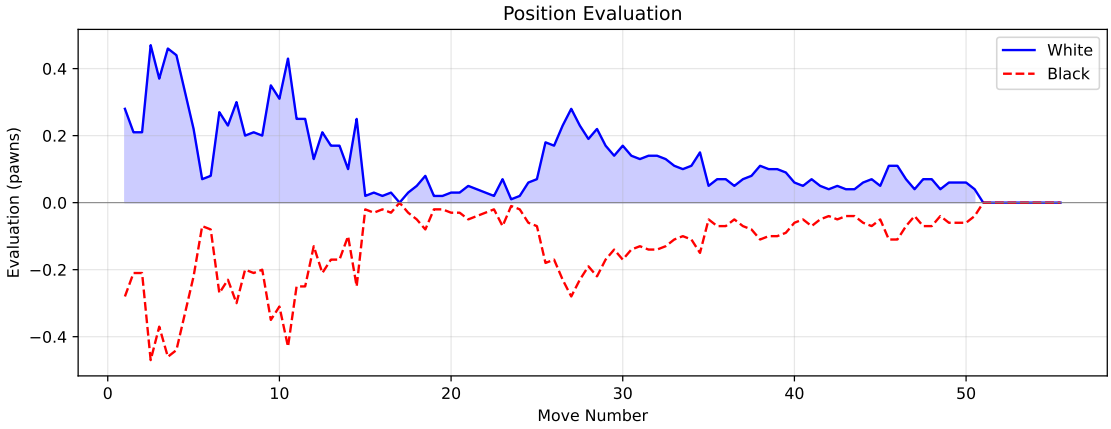
11.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

11.3.1 Position Evaluation Over Time

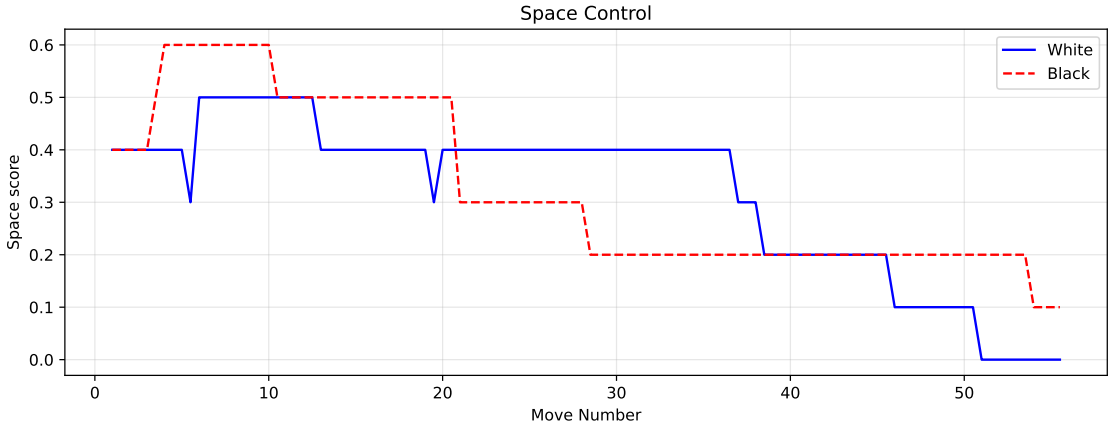
Game Character Classification. Evaluation range: minimum $m_1 = +0.00$, maximum $m_2 = +0.47$, spread $d = m_2 - m_1 = 0.47$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



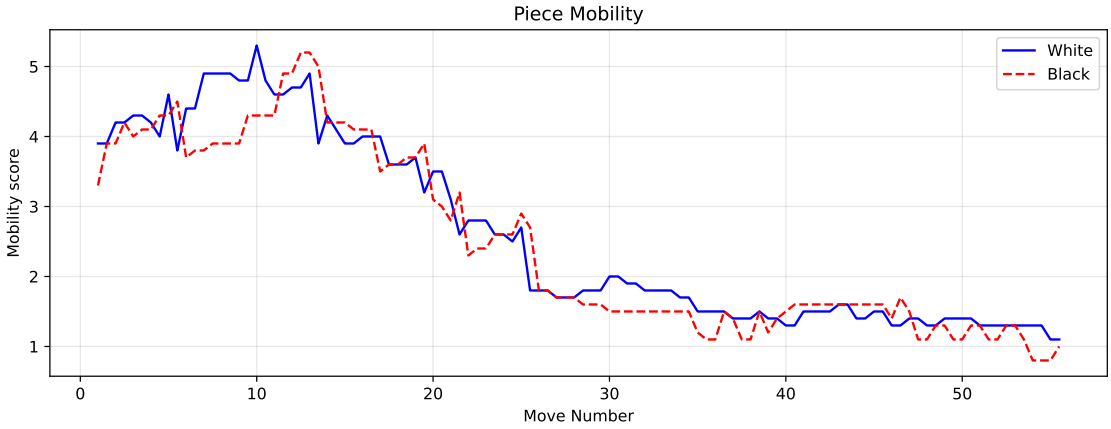
11.3.2 Space Control

- White average: 0.32 (range: 0.00 to 0.50)
- Black average: 0.33 (range: 0.10 to 0.60)



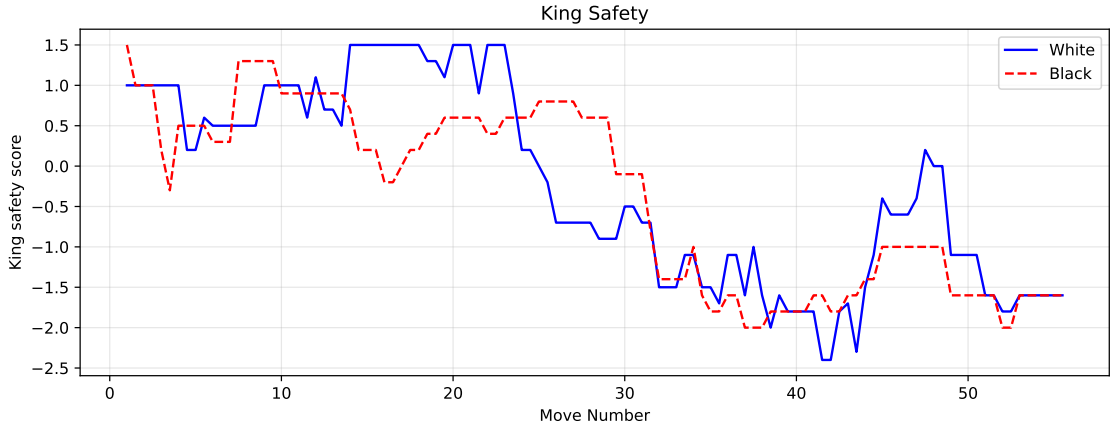
11.3.3 Mobility (Piece Activity)

- White average: 2.61 (range: 1.10 to 5.30)
- Black average: 2.48 (range: 0.80 to 5.20)



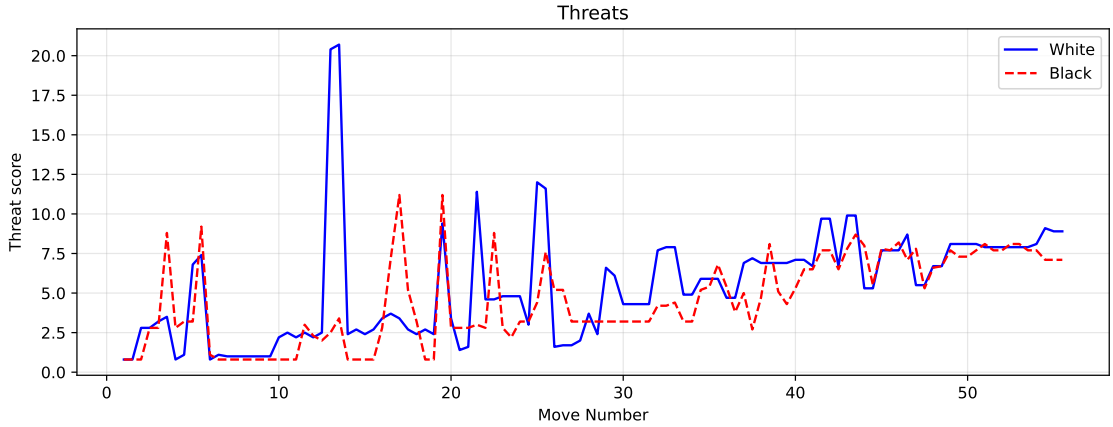
11.3.4 King Safety

- White average: -0.23
- Black average: -0.36



11.3.5 Threats

- White average: 5.41 (range: 0.80 to 20.70)
- Black average: 4.56 (range: 0.80 to 11.20)
- Average advantage: 0.85 (White maintained more threats)



11.4 Annotated Game

1. e4 e5 2. Nf3 Nf6 3. Nxe5 d6 4. Nf3 Nxe4 5. Nc3 Nxc3 6. dxc3 Be7 7. Be3 O-O 8. Qd2 Nd7 9. O-O-O Nf6 10. Bd3! c5 11. Rhe1 Be6! 12. Kb1 Qa5 13. c4 Qxd2! 14. Bxd2 h6 15. Nh4 Rfe8 16. Ng6 Ng4! 17. Nxe7+! Rxe7! 18. Re2! Ne5! 19. Bf4! Nxd3! 20. Rxd3! Rd7! 21. Rxd6 Rxd6! 22. Bxd6! Rd8 23. Rd2! Bxc4! 24. Kc1 b6 25. Bf4 Rxd2 26. Kxd2! a6 27. a3 Kf8 28. Bc7! b5! 29. Bd6+ Ke8! 30. Bxc5! h5! 31. Ke3! Kd7 32. Kd4 g6! 33. g3 Be2 34. Bf8 Kc6 35. b3 Bd1! 36. Kd3 Bg4! 37. c4! Be6! 38. Kd4! bxc4 39. bxc4! Bg4 40. c5 Be6! 41. Bh6 Bd5 42. Be3 Be6 43. Ke5 Bd5 44. Kf4 Be6 45. Kg5 Bd5 46. g4! hxg4! 47. Kxg4! Ba2 48. Kg5! Bb3 49. Kf6 Ba2! 50. h4 Bb3 51. f4 Ba2! 52. Ke7 Bb3! 53. Kf6! Ba2 54. f5 Bb1! 55. Bf2 Bc2!

1/2-1/2

11.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

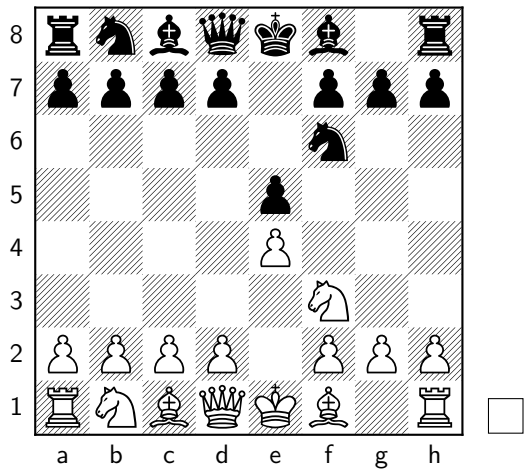
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 2...Nf6 ★

Biggest evaluation swing(s) (26cp)

Evaluation: +0.47



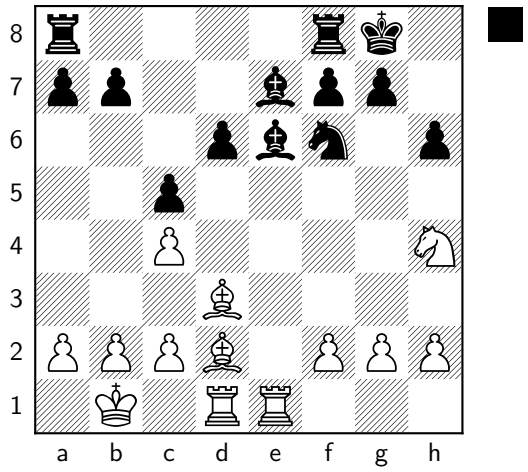
Metric	White	Black
Space	0.40	0.40
Mobility	4.20	4.20
King Safety	1.00	1.00
Threats	2.80	2.80

Instead of Nf6: Nc6, d6 Best continuation: Nxe5 d6 Nf3

Position 2: After 15. Nh4 ★

Biggest evaluation swing(s) (23cp)

Evaluation: +0.02



Metric	White	Black
Space	0.40	0.50
Mobility	3.90	4.20
King Safety	1.50	0.20
Threats	2.40	0.80

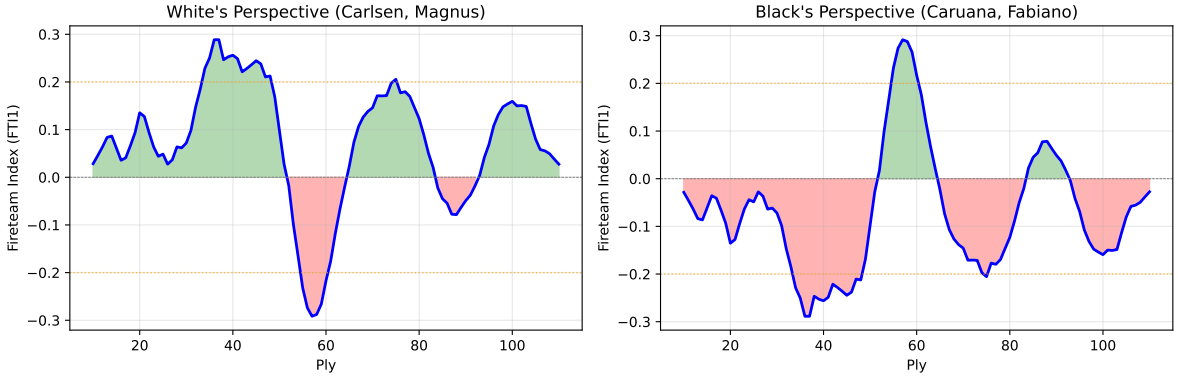
Instead of Nh4: h3, Re2, a3 Best continuation: Ng4 Ng6 Rfe8

11.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

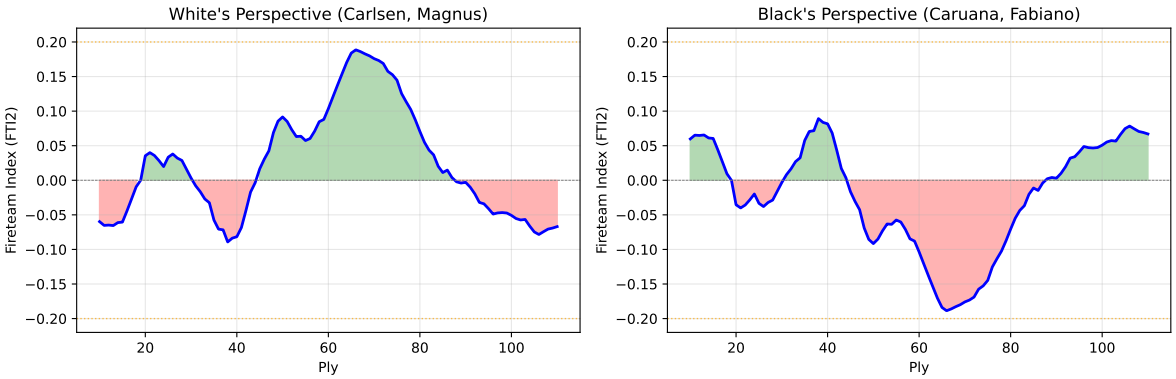
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	9 plies
Carlsen, Magnus	Windowed	WIN	15 plies
Caruana, Fabiano	Per-ply	DRAW	8 plies
Caruana, Fabiano	Windowed	DRAW	6 plies



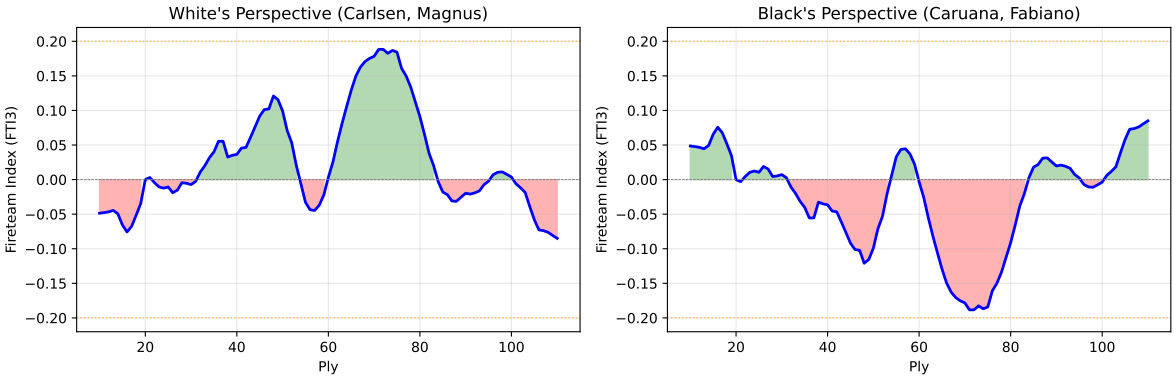
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



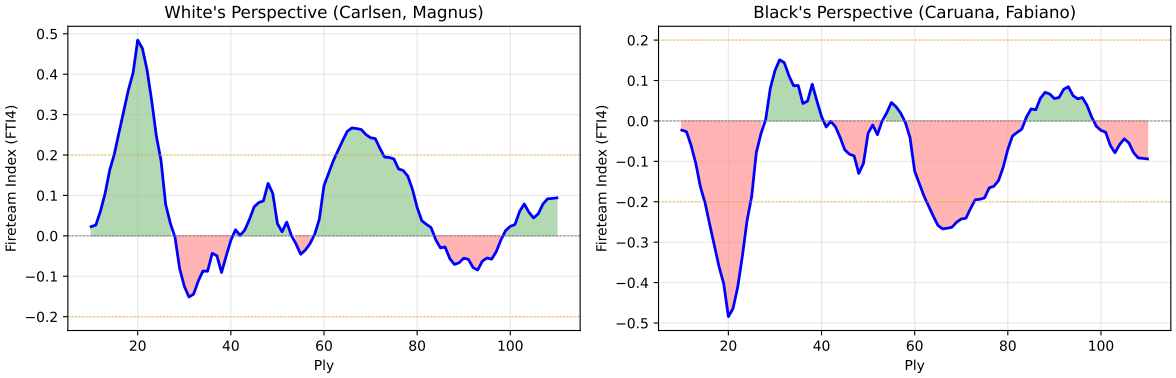
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



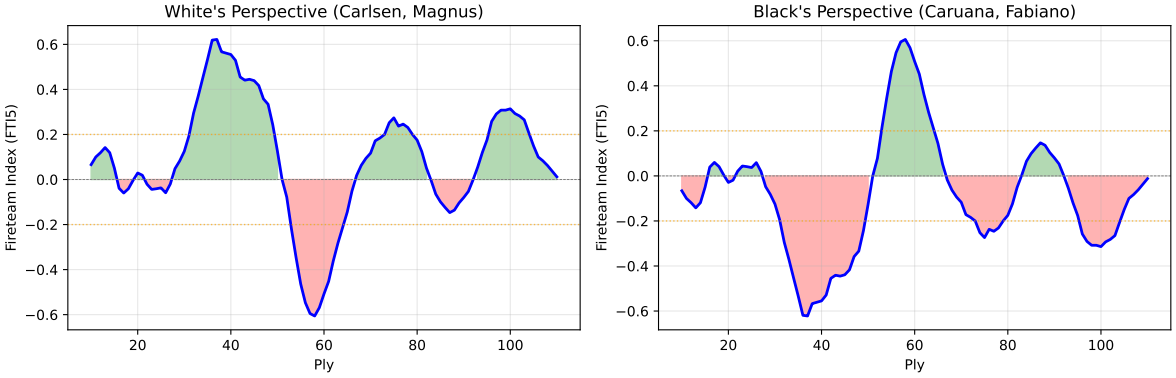
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	8 plies
Carlsen, Magnus	Windowed	WIN	10 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	11 plies
Carlsen, Magnus	Windowed	WIN	18 plies
Caruana, Fabiano	Per-ply	WIN	10 plies
Caruana, Fabiano	Windowed	WIN	12 plies



Interpretation. White (Carlsen, Magnus) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

11.7 Analysis Information

- Engine: Stockfish 16

- Depth: 20
- Analysis time: 133.4 seconds

Chapter 12

Caruana, Fabiano vs Carlsen, Magnus

12.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.26
Round	12
White	Caruana, Fabiano
Black	Carlsen, Magnus
Result	1/2-1/2
Opening	??? – Unknown

12.2 Player Statistics

12.2.1 Caruana, Fabiano (White)

- Total moves: 31
- Accuracy: 93.1%
- Average centipawn loss: 14.3
- Best/Excellent moves: 15 / 4
- Good moves: 7
- Inaccuracies: 2
- Mistakes: 3
- Blunders: 0

12.2.2 Carlsen, Magnus (Black)

- Total moves: 31
- Accuracy: 94.8%
- Average centipawn loss: 10.8

- Best/Excellent moves: 9 / 11
- Good moves: 9
- Inaccuracies: 0
- Mistakes: 2
- Blunders: 0

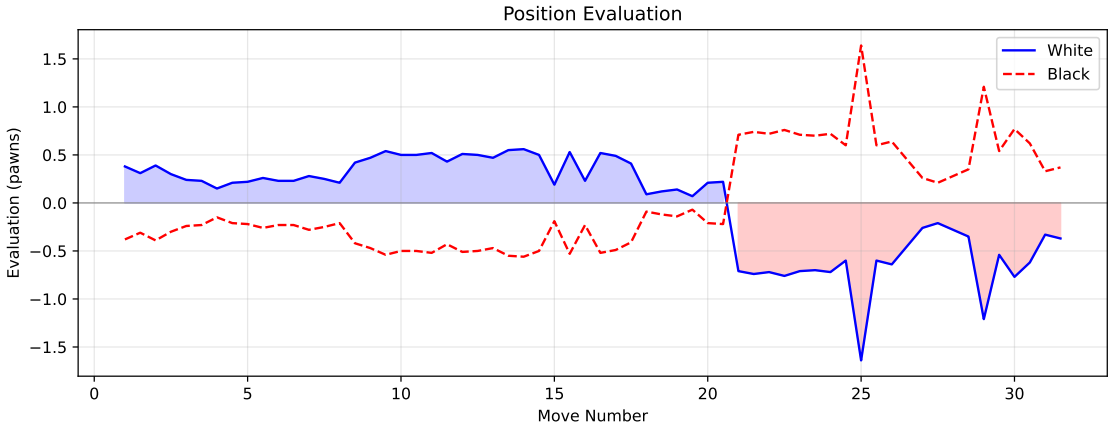
12.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

12.3.1 Position Evaluation Over Time

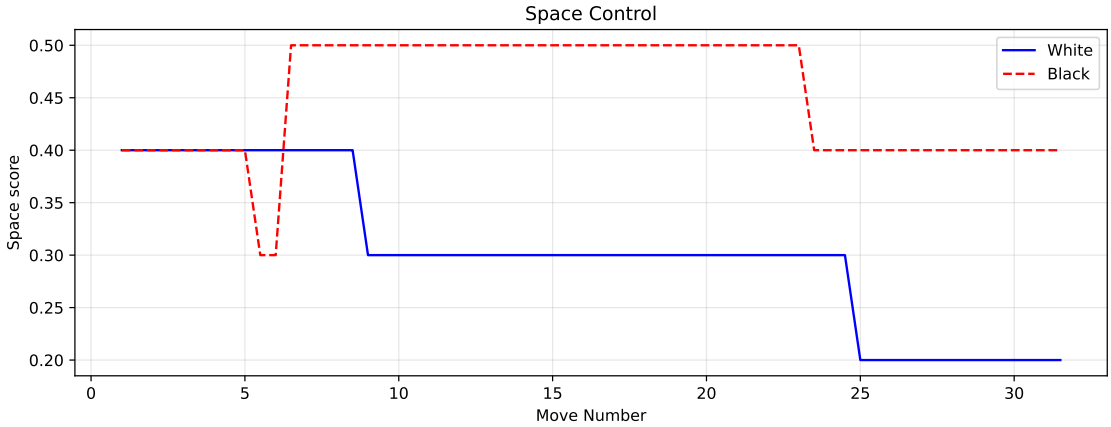
Game Character Classification. Evaluation range: minimum $m_1 = -1.64$, maximum $m_2 = +0.56$, spread $d = m_2 - m_1 = 2.20$.

This game is classified as **tense** (normal). A tense game with moderate swings.



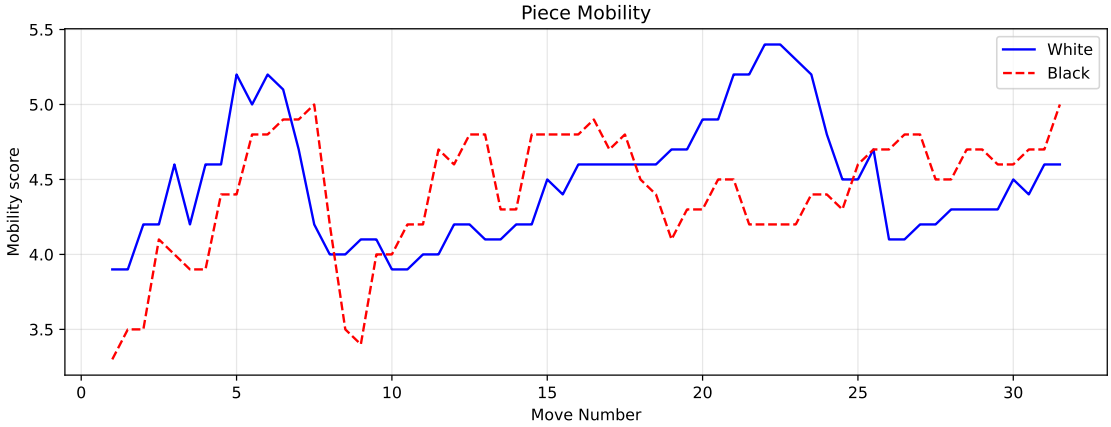
12.3.2 Space Control

- White average: 0.30 (range: 0.20 to 0.40)
- Black average: 0.45 (range: 0.30 to 0.50)



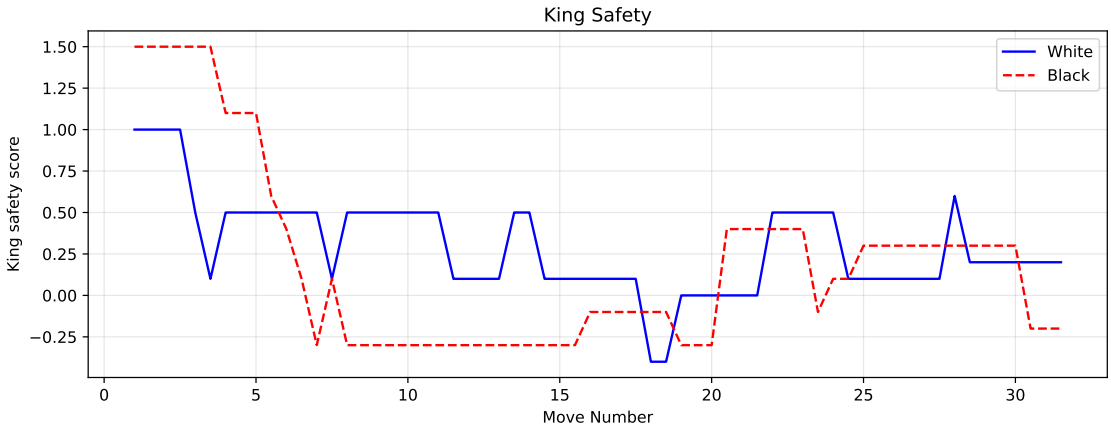
12.3.3 Mobility (Piece Activity)

- White average: 4.49 (range: 3.90 to 5.40)
- Black average: 4.42 (range: 3.30 to 5.00)



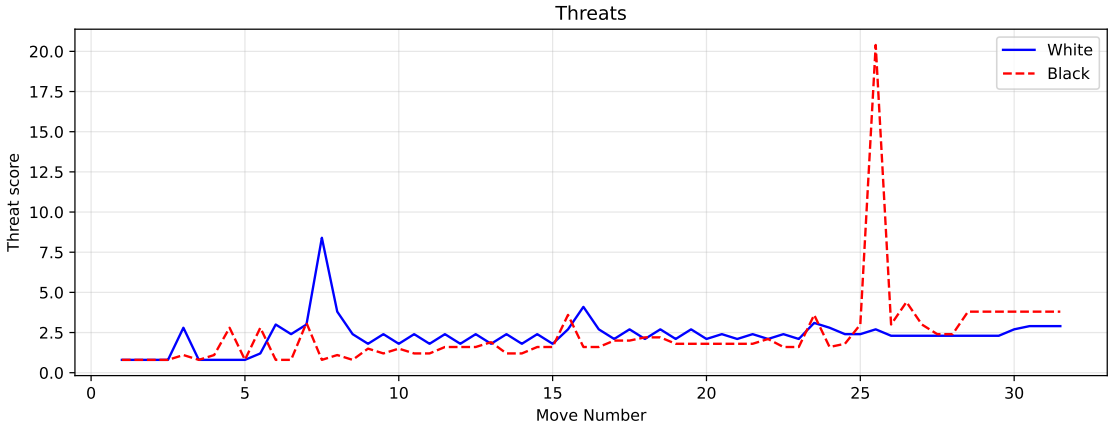
12.3.4 King Safety

- White average: 0.29
- Black average: 0.20



12.3.5 Threats

- White average: 2.31 (range: 0.80 to 8.40)
- Black average: 2.30 (range: 0.80 to 20.40)
- Average advantage: 0.01 (threats were balanced)



12.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 e5 6. Ndb5 d6 7. Nd5! Nxd5! 8. exd5! Ne7 9. c4! Ng6 10. Qa4! Bd7 11. Qb4! Bf5! 12. h4! h5 13. Qa4! Bd7 14. Qb4! Bf5 15. Be3?! a6 16. Nc3 Qc7 17. g3! Be7! 18. f3?! Nf8 19. Ne4 Nd7! 20. Bd3! O-O 21. Rh2? *White loses 97cp. Consider instead: O-O, Qd2, Qc3.*

21...Rac8 22. O-O-O Bg6! 23. Rc2 f5! 24. Nf2 Nc5 25. f4? *White loses 102cp. Consider instead: Bxc5, Qa3.*

25...a5? *Black loses 98cp. Consider instead: exf4, b5.*

26. Qd2 e4 27. Be2! Be8 28. Kb1! Bf6! 29. Re1? *White loses 85cp. Consider instead: b3, Nh3, Rdc1.*

29...a4? *Black loses 76cp. Consider instead: Ba4, Rb8.*

30. Qb4 g6 31. Rd1 Ra8

1/2-1/2

12.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

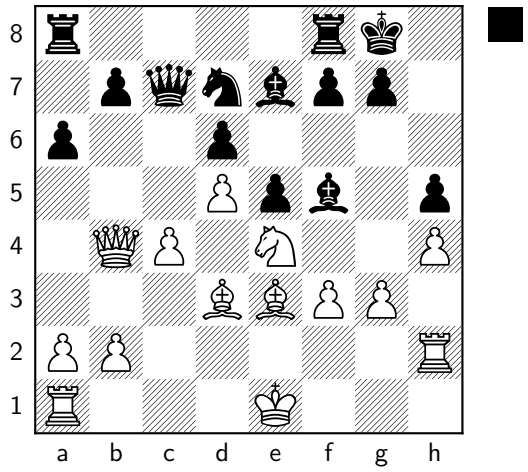
- **Instead of [move]**: The move(s) that should have been played instead.
- **Best continuation**: The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 21. Rh2 ★

Biggest evaluation swing(s) (93cp)

Evaluation: -0.71



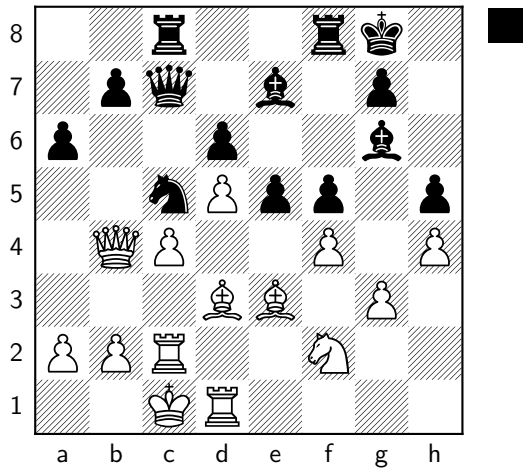
Metric	White	Black
Space	0.30	0.50
Mobility	5.20	4.50
King Safety	0.00	0.40
Threats	2.10	1.80

Instead of Rh2: O-O, Qd2, Qc3 Best continuation: Rac8 Rc1 Bg6

Position 2: After 25. f4

Critical moment

Evaluation: -1.64



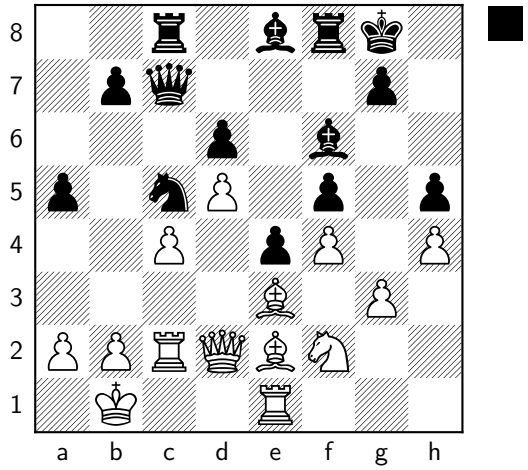
Metric	White	Black
Space	0.20	0.40
Mobility	4.50	4.60
King Safety	0.10	0.30
Threats	2.40	3.00

Instead of f4: Bxc5, Qa3 Best continuation: exf4 gxf4 Bxh4

Position 3: After 29. Re1 ★

Biggest evaluation swing(s) (86cp)

Evaluation: -1.21



Metric	White	Black
Space	0.20	0.40
Mobility	4.30	4.70
King Safety	0.20	0.30
Threats	2.30	3.80

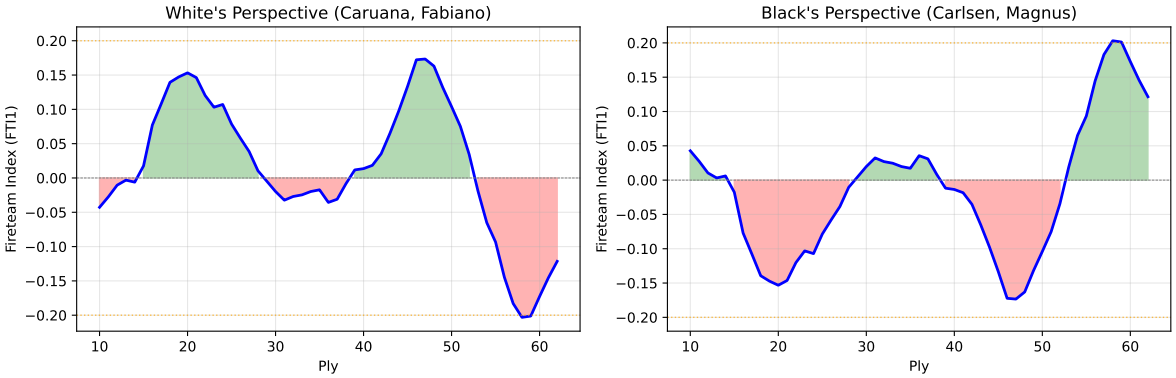
Instead of Re1: b3, Nh3, Rdc1 Best continuation: Ba4 Rcc1 b5

12.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

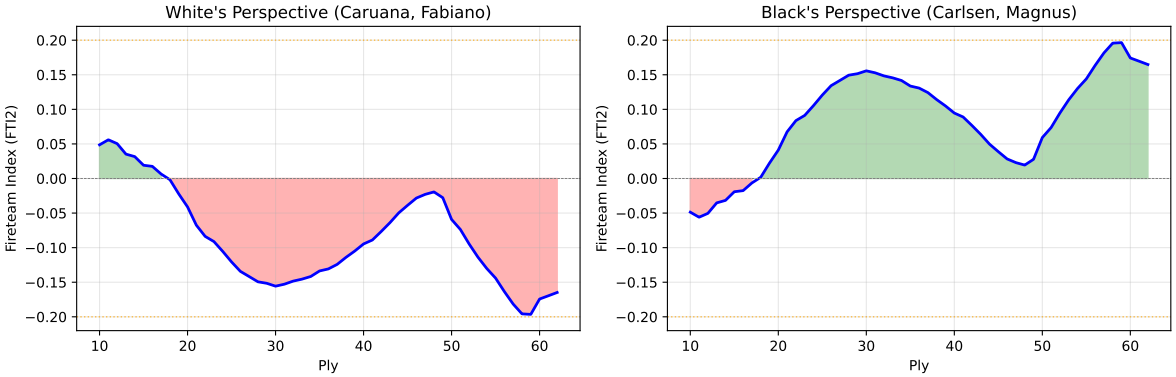
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	0 plies
Carlsen, Magnus	Per-ply	DRAW	4 plies
Carlsen, Magnus	Windowed	DRAW	2 plies



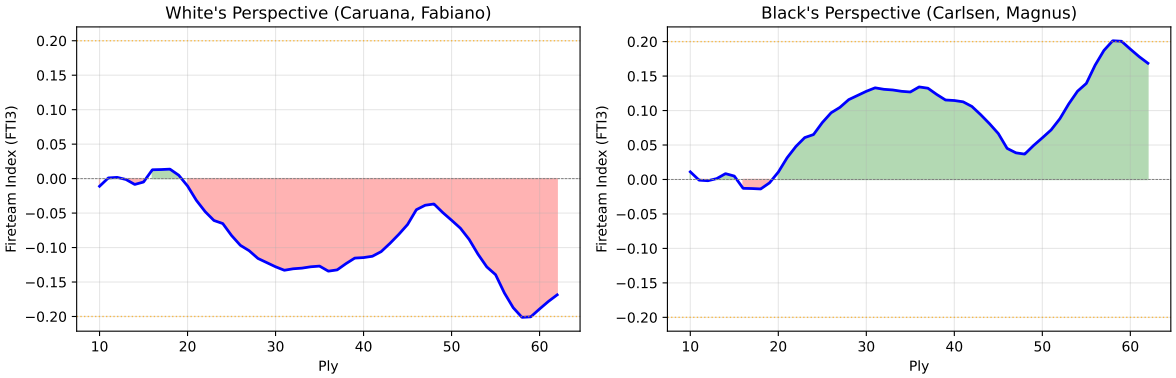
FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



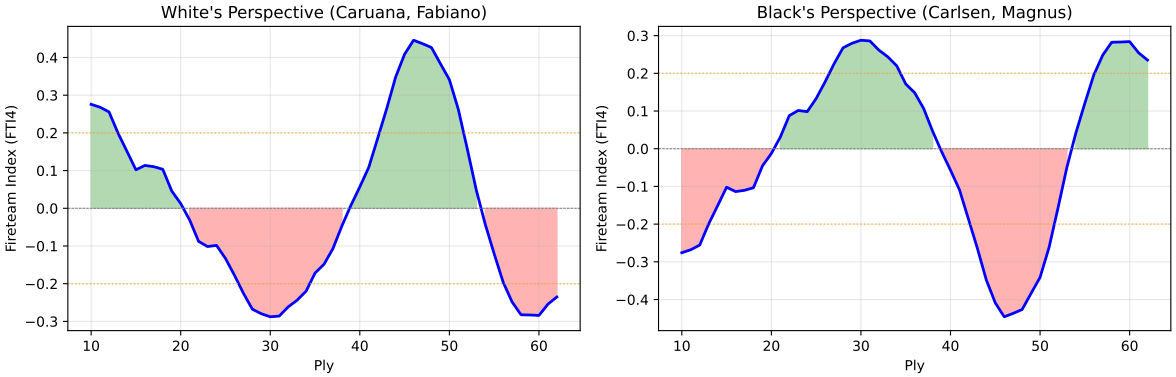
FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies
Carlsen, Magnus	Per-ply	DRAW	4 plies
Carlsen, Magnus	Windowed	DRAW	2 plies



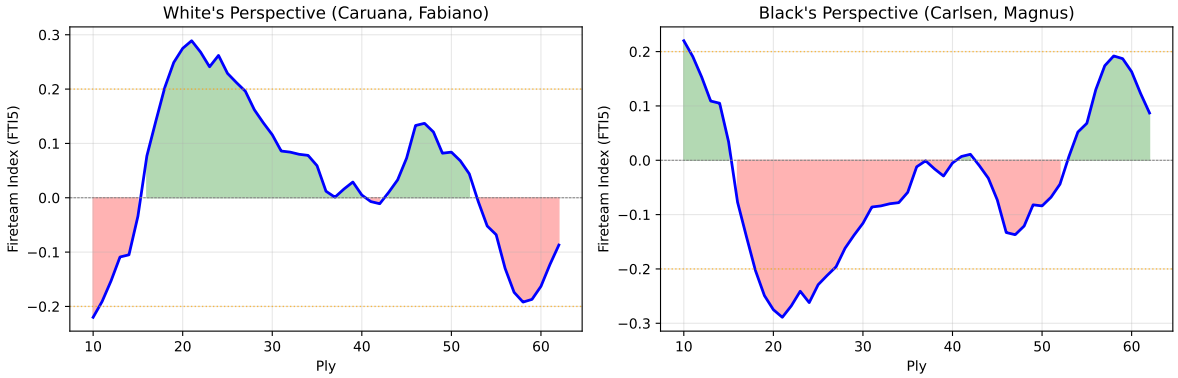
FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	WIN	11 plies
Caruana, Fabiano	Windowed	DRAW	9 plies
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	DRAW	8 plies



FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Player	Algorithm	Prediction	Max Streak
Caruana, Fabiano	Per-ply	DRAW	5 plies
Caruana, Fabiano	Windowed	DRAW	9 plies
Carlsen, Magnus	Per-ply	DRAW	4 plies
Carlsen, Magnus	Windowed	DRAW	0 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

12.7 Analysis Information

- Engine: Stockfish 16
- Depth: 20
- Analysis time: 204.3 seconds

Chapter 13

Fireteam Index Prediction Summary

This chapter summarizes the accuracy of the Fireteam Index prediction algorithms across all games in this collection.

13.1 Overall Results

Metric	Value
Total games	12
Decisive games	0
Draws (both sides analyzed)	12
Games with prediction	12

13.1.1 FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Algorithm	Correct	Accuracy
Per-ply (raw)	5/12	41.7%
Windowed (smoothed)	5/12	41.7%
Balance	0/12	0.0%

13.1.2 FTI2: Tactical $(0.6, 0.1, 0, 0.3)$

Algorithm	Correct	Accuracy
Per-ply (raw)	5/12	41.7%
Windowed (smoothed)	5/12	41.7%
Balance	0/12	0.0%

13.1.3 FTI3: Strategic $(0.7, 0.1, 0.1, 0.1)$

Algorithm	Correct	Accuracy
Per-ply (raw)	4/12	33.3%
Windowed (smoothed)	4/12	33.3%
Balance	0/12	0.0%

13.1.4 FTI4: Dynamic $(0.2, 0.6, 0, 0.2)$

Algorithm	Correct	Accuracy
Per-ply (raw)	2/12	16.7%
Windowed (smoothed)	2/12	16.7%
Balance	0/12	0.0%

13.1.5 FTI5: Solid $(0.3, 0.2, 0.5, 0)$

Algorithm	Correct	Accuracy
Per-ply (raw)	3/12	25.0%
Windowed (smoothed)	3/12	25.0%
Balance	0/12	0.0%

13.1.6 Comparison

FTI1 (Harmonious) (41.7%) and **FTI2 (Tactical)** (41.7%) show similar predictive accuracy (within 2%).

13.1.7 Held-out Tests of the Fireteam Index

Earlier versions of CGA printed "empirically optimal" FTI weights fitted to all of the games at once. Those weights and their R^2 values were not meaningful: the fit had no constant term and its weights were clipped and rescaled afterwards. This section instead tests the FTI the honest way: every prediction below is for a game the model was *not* fitted on (leave-one-game-out), and it is compared with simple baselines.

1. Predicting results. All 12 games ended in a draw for the tracked side, so there is no variation in results for any weights to learn from, and the FTI cannot be tested as a predictor of results on this collection.

2. Predicting the type of move a player chooses. In positions where Stockfish rates two or more moves within 50 centipawns of its best, which one does the player pick? If Space, Mobility, King Safety and Threats captured a personal style, adding them to Stockfish's evaluation would predict the player's choices better, and a model fitted to the player's own moves would beat one fitted to the other players' moves.

Player	Choices	Random	Eval only	Eval + metrics	Others' model
Carlsen, Magnus	385	35%	61%	60%	59%
Caruana, Fabiano	371	36%	62%	54%	56%

Each percentage is how often the model's most likely move was the move actually played, on games the model was not fitted on.

Adding the four metrics does not improve the predictions for any player, so on this collection they do not reveal a playing style. No player's own model predicts their choices better than the model fitted to the other players' moves.

13.2 Interpretation

These predictions tracked the eventual winner in each decisive game. Three prediction algorithms are compared:

- **Per-ply**: Predicts WIN if FTI $> \epsilon$ for 10 consecutive plies (raw values)
- **Windowed**: Same logic but applied to a 10-ply rolling average
- **Balance**: Uses normalized balance B_n (plies ahead − plies behind) and volatility V (sign changes), combined as confidence $C = |B_n| \cdot \max(0, 1 - V/V_{\text{thr}})$ with $V_{\text{thr}} = 0.2\times$(plies). Predicts DRAW if $C < 0.15$

The Balance algorithm explicitly models draws by detecting games where neither side maintains a consistent advantage, or where the advantage changes hands frequently.

The different FTI signatures may perform better or worse depending on the playing style of the players being analyzed. FTI2 (Tactical) tends to predict well for players who emphasize territorial control and tactical threats, while FTI3 (Strategic) may better capture the style of players who prioritize space and piece activity.

Appendix A

Methodology: How Metrics Are Computed

This appendix explains how the positional metrics are computed.

This section explains how Stockfish's classical evaluation computes each positional factor. Note that modern Stockfish primarily uses NNUE (neural network) evaluation, but the classical evaluation terms provide interpretable metrics for educational purposes.

A.1 Space

Space evaluation counts squares in the center (files c-f, ranks 2-4 for White, 5-7 for Black) that are:

1. Attacked by at least one friendly pawn, AND
2. Not occupied by any enemy piece

The computation is weighted by the number of pieces behind the pawn chain—this rewards positions where you control territory AND have pieces to exploit that space. The space term is scaled by total piece count; it matters more in closed positions with many pieces.

A.2 Mobility

Mobility counts legal moves for each piece type, with different weights:

- **Knights**: Safe squares attacked (excluding squares defended by enemy pawns)
- **Bishops**: Safe squares on diagonals (bonus for long, unobstructed diagonals)
- **Rooks**: Squares on files/ranks (bonus for open files, 7th rank penetration)
- **Queens**: Combination of bishop and rook mobility patterns

"Safe squares" exclude those defended by enemy pawns or where exchanges would lose material. Mobility in the opponent's territory is valued higher than mobility in your own.

A.3 King Safety

King safety combines several factors:

- **Pawn shield**: Pawns on 2nd/3rd rank directly in front of the castled king
- **King tropism**: Distance of enemy pieces to your king (closer = more dangerous)
- **Attack units**: Count of enemy pieces attacking squares adjacent to your king
- **Safe checks**: Number of safe checking squares available to the opponent

A.4 Threats

Threat evaluation considers:

- Hanging pieces (attacked but not adequately defended)
- Safe pawn advances that create threats
- Minor pieces attacking major pieces
- Weak squares in the pawn structure

A.5 Middlegame vs Endgame (MG/EG)

Each term has separate middlegame (MG) and endgame (EG) values. Stockfish blends these based on remaining material (the "phase"). With all pieces on the board, MG dominates; as pieces trade, EG values become more important. For example, space matters greatly in the middlegame but is nearly irrelevant in endgames.

A.6 Game Character Classification

Games are classified based on evaluation volatility and directionality. Let m_2 be the maximum evaluation for White during the game (in pawns) and m_1 be the minimum. The *spread* $d = m_2 - m_1$ measures the total evaluation range.

Spread Classifications. Based on the magnitude of d:

- **Balanced** ($d < 1$): Minimal advantage shifts; controlled, technical play with neither side gaining a significant edge.
- **Tense** ($1 \leq d < 3$): Normal competitive tension; typical of well-played games where small advantages are contested.
- **Tactical** ($3 \leq d < 6$): Significant evaluation swings; tactical complications, missed opportunities, or speculative play.
- **Chaotic** ($d \geq 6$): Wild swings in evaluation; likely includes major blunders, speculative sacrifices, or complex tactical melees.

Directionality Classifications. Based on whether the advantage changed hands:

- **One-sided**: Either $m_1 > -\frac{1}{2}$ (White always comfortable) or $m_2 < \frac{1}{2}$ (Black always comfortable). The leading side maintained control throughout.
- **Seesaw**: Both $m_1 < -1$ and $m_2 > 1$. The advantage genuinely changed hands, with each side holding a significant advantage at some point.
- **Normal**: Neither one-sided nor seesaw; moderate swings without complete reversals.

These classifications combine to describe the overall game character. For example, a "tactical seesaw battle" indicates significant evaluation swings ($3 \leq d < 6$) with the advantage changing hands multiple times.

A.7 Fireteam Index (FTI) and Win Prediction

The Fireteam Index is a composite metric inspired by the ancient mathematical board game *Rithmomachia*, where a "progression victory" is achieved when a coordinated group of pieces (a "fireteam") establishes control deep in enemy territory. Similarly, in chess, sustained positional dominance across multiple factors often precedes victory.

The FTI weights can be thought of as *signatures* that capture different playing styles. By adjusting which positional factors receive emphasis, the FTI can be tuned to better predict outcomes for players with different strategic preferences.

Threats Scaling. The raw Threats metric (T) produces values typically ranging from 5–20, while Space, Mobility, and King Safety range from approximately 0.1–0.5. To make these comparable, Threats is divided by a scaling factor of 20 before weighting. The scaled value $T' = T/20$ brings Threats into a similar range as the other metrics.

FTI1: Harmonious Formula. The harmonious Fireteam Index weights all four positional metrics equally (with Threats pre-scaled):

$$\text{FTI}_1 = \frac{1}{4}\Delta S + \frac{1}{4}\Delta M + \frac{1}{4}\Delta K + \frac{1}{4}\Delta T' \tag{A.1}$$

where ΔS = Space, ΔM = Mobility, ΔK = King Safety, and $\Delta T' = T/20$ = scaled Threats, each computed as (Player's value − Opponent's value). Positive FTI indicates the player has overall positional superiority; negative indicates the opponent is better.

FTI2: Tactical. This variant emphasizes territorial control and tactical pressure through Space and Threats:

$$\text{FTI}_2 = 0.6\Delta S + 0.1\Delta M + 0.3\Delta T' \tag{A.2}$$

FTI3: Strategic. This variant emphasizes long-term spatial dominance:

$$\text{FTI}_3 = 0.7\Delta S + 0.1\Delta M + 0.1\Delta K + 0.1\Delta T' \tag{A.3}$$

FTI4: Dynamic. This variant emphasizes piece activity and maneuvering:

$$\text{FTI}_4 = 0.2\Delta S + 0.6\Delta M + 0.2\Delta T' \tag{A.4}$$

FTI5: Solid. This variant emphasizes king safety and defensive play:

$$\text{FTI}_5 = 0.3\Delta S + 0.2\Delta M + 0.5\Delta K \tag{A.5}$$

Prediction Algorithms. Three algorithms are used to predict game outcomes based on FTI (applied independently to each of the five variants):

1. **Per-Ply (Raw)**: Examines the raw FTI value at each ply after the opening phase (ply 16, approximately 8 moves). If FTI $> \epsilon$ (default $\epsilon = 0.25$) for 10 consecutive plies (5 full moves), a WIN is predicted for that player.

2. **Windowed (Smoothed)**: Computes a 10-ply rolling average of FTI before applying the same streak detection. This smooths out tactical noise and captures sustained trends rather than momentary fluctuations.

3. **Balance (Confidence-based)**: Predicts outcomes using a confidence score that combines balance and volatility:

   - *Normalized Balance* $B_n = \frac{(\text{plies with FTI}>\epsilon) - (\text{plies with FTI}<-\epsilon)}{\text{total plies}}$
   - *Volatility* $V =$ number of times FTI crosses zero (sign changes)
   - *Volatility threshold* $V_{\max} = 0.2 \times$ total plies (scales with game length)
   - *Confidence* $C = |B_n| \times \max(0, 1 - V/V_{\max})$

   The prediction is: WIN if $C > 0.15$ and $B_n > 0$; LOSS if $C > 0.15$ and $B_n < 0$; DRAW otherwise. Low confidence (due to balanced play or high volatility) predicts a draw.

The Per-Ply and Windowed algorithms predict DRAW only when no streak is achieved, whereas the Balance algorithm explicitly detects drawish characteristics (balanced plies, back-and-forth play).

Interpretation. The FTI prediction measures *positional* dominance, which is a necessary but not sufficient condition for winning. A player may dominate positionally yet fail to convert due to:

- Tactical errors (blunders, missed combinations)
- Time pressure in rapid/blitz games
- Tenacious defense by the opponent
- Drawing tendencies in certain structures (opposite-colored bishops, fortresses)

Conversely, a player may win without ever achieving sustained FTI dominance through a single tactical strike or opponent error. The FTI metric is most predictive in games where positional understanding—rather than pure calculation—determines the outcome.

Choosing a Signature. Different FTI signatures may perform better for different players or playing styles. FTI2 (Tactical) tends to predict well for players who build up territorial control and convert through direct threats. FTI3 (Strategic) may better capture the style of players who emphasize piece activity and spatial domination. Comparing prediction accuracy across signatures can reveal which positional factors most strongly correlate with a player's wins.

For Draws. When analyzing drawn games, the FTI is computed from both players' perspectives. This reveals whether:

- Neither side achieved dominance (a "true draw" with balanced play)
- One side dominated but failed to convert (a "missed win")
- Both sides dominated at different points (a "mutual chances draw")